# DOT-Design

Version Next, 2026-10-23

# Overview

**Describe a tool once. Reuse it in different deployments and compositions.**

DOT-Design provides a portable description of software components and the `dotx` command-line tool to validate, deploy and connect them in the MARE 6G security ecosystem. A Dynamic Open Trust component (**DOT**) is a reusable definition, not a particular running container, VM or Kubernetes Pod.

For example, one Prometheus DOT can describe its API and supported deployment choices. A composition can then use it to scrape a selected metrics exporter, without creating a second Prometheus product definition for that use case.

## Choose your first step

- **Use an existing tool:** Getting started takes you from CLI installation to a running Prometheus service, a verified API response and cleanup. Java and contributor tooling are not required.
- **Watch or present the ecosystem:** the 10-15-minute guided demo connects the concepts to executable evidence.
- **Integrate your own component:** start with the model below, then use the developer tour and the authoring and composition guide.



The bottom-left version selector offers **Next**, the current development documentation, alongside immutable stable releases. Select **Next** to follow the latest walkthrough and examples.

## The model in one minute

| Question | Model | Plain-language meaning |
|---|---|---|
| What is this component? | `Dot` | Reusable identity and version, described by `dot.json`. |
| What can it do? | `Skill` | Optional high-level capability classification, not executable code or an API. |
| What interface does it implement or require? | `Contract` | An authoritative OpenAPI, AsyncAPI, Protobuf or GraphQL document. Complete declarations identify its raw HTTPS payload and SHA-256 integrity pin. |
| Which interaction can another DOT connect to? | `Anchor` | A stable `PROVIDES` or `REQUIRES` point, normally selecting an operation from a declared Contract rather than copying its interface metadata. |
| Where and how can it run? | `Runtime`, `Platform`, `Deployment`, `DeploymentManifest` | A platform is **where** it runs; a deployment selects **how** to launch it. Kubernetes may offer both manifest and Helm alternatives. |

| Question | Model | Plain-language meaning |
| --- | --- | --- |
| Which service implements the interface? | `DeploymentService` | Logical services, named ports, Anchor bindings and supported configuration inside a selected deployment. |
| How do components work together? | `Composition`, `DotInstance`, `Wire` | A separate domain aggregate selects pinned DOTs and connects their Anchors; it does not copy or embed the reusable DOT definitions. |

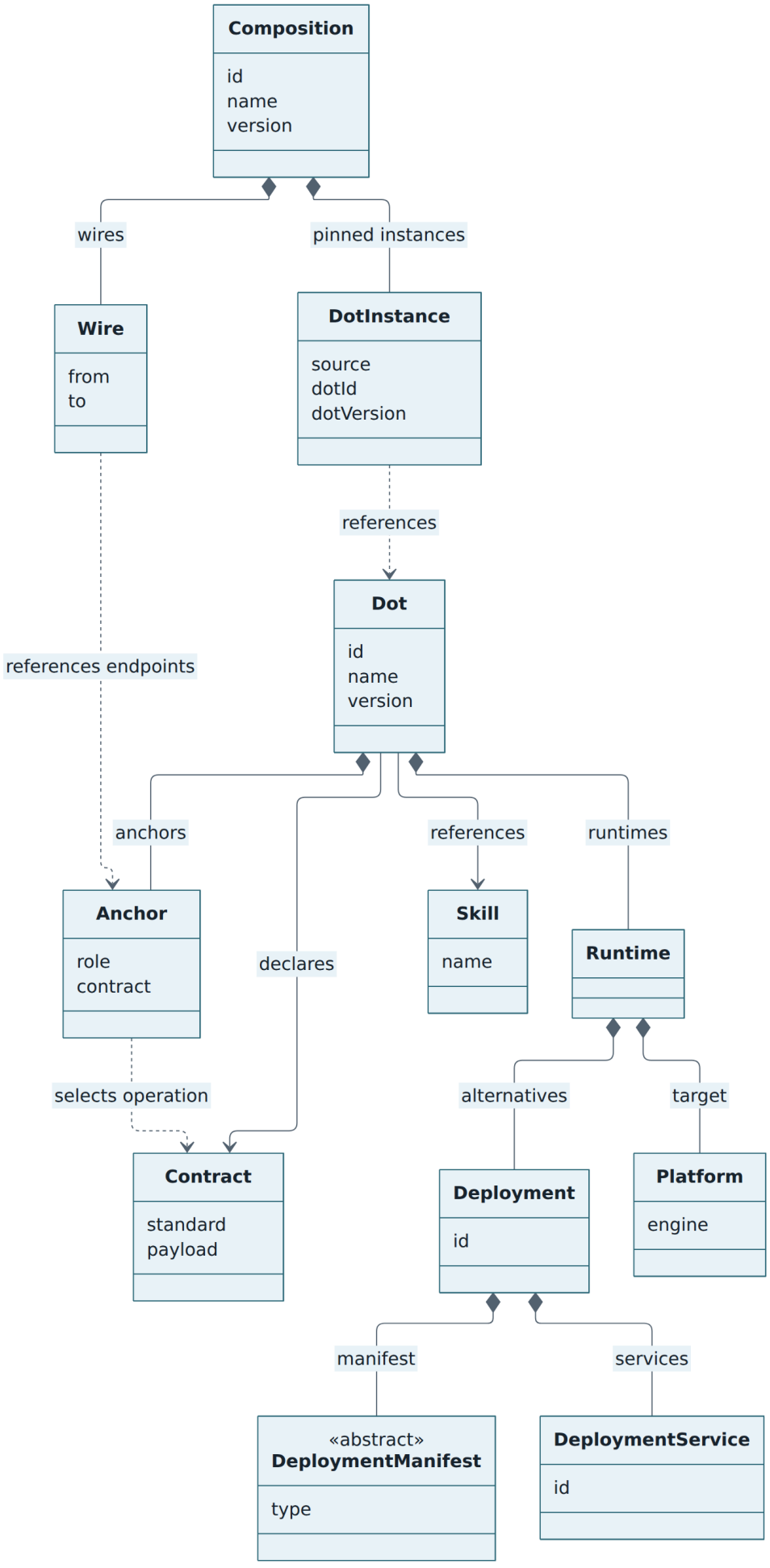



*Figure 1. Simplified domain model: reusable DOT definitions and a separate Composition*

Diamonds indicate embedded values; arrows indicate declarations or references. The dashed arrow applies to Contract-backed Anchors. Collection sizes, configuration, ports, binding helpers and audit metadata are omitted for readability; see the full generated class diagram for implementation detail. `Composition` is a separate DOT-Domain aggregate: its instances reference DOTs and its Wires reference their Anchors. Neither is embedded in the reusable Java `Dot` aggregate.

Read the runtime branch as **alternatives**, not a requirement to deploy everything: choose one supported deployment for each composition instance. An Anchor binding maps the logical Anchor to a named port of a deployment service; concrete reachable addresses are resolved for the running instance.

Groups, descriptions and metadata help classify or audit a definition. They do not change API semantics or grant runtime permissions. For field-level validation and the complete generated model, continue to DOT-Domain.

# From one DOT to a working composition

**A DOT describes a reusable component; a composition describes one use case.** A separate `composition.yaml`:

1. Selects named DOT instances with pinned identities and versions.
2. Chooses a supported deployment for each instance.
3. Connects a provider Anchor to a consumer Anchor with a **wire**.
4. Supplies use-case configuration and Secret references; a reviewed lock pins the resolved definitions, Contracts and inputs.

For example, the PoC10 walkthrough connects Keycloak, APISIX, OPA, Oxigraph and a machine client. Realm configuration, gateway routes, authorization policy and credentials belong to that use case, not to separate copies of those products' DOTs.

## Follow one API call

In the Prometheus quickstart, the official Contract describes the build-information operation. The `build-information` Anchor selects it, and the chosen deployment binds that Anchor to a service port. The verified response comes from the running Prometheus instance at `http://127.0.0.1:18909/api/v1/status/buildinfo`.

The Contract describes the interaction; the Anchor gives it a stable composable identity; the binding makes it reachable. None of these is the same thing as a successful live call.



`dot.json` is the deployable DOT definition and the only source inspected by `dotx deploy`. Connections between deployed DOT instances are described in a separate composition document. A wire does not create an API, translate protocols automatically or prove production readiness. Concrete addresses, credentials and live policy results never become Anchor fields.

# Documentation map

| Read | When you need to |
|---|---|
| Getting started | Install the released CLI, deploy Prometheus with Docker Compose, verify its API, and clean up. |
| Contributor guide | Build the project, run shared local/GitHub checks, and generate documentation. |
| Guided demo | Follow a 10-15-minute live presenter route, a five-minute read-only overview, or a deeper onboarding walkthrough from catalogue validation to live evidence. |
| Developer demo: reuse and extend | Inspect a real DOT, prove an invalid wire is rejected, and map PoC changes to shared definitions, composition inputs and executable acceptance. |
| Architecture | Understand repository boundaries, sources of truth, runtime/deployment separation, and the composition resolution path. |
| DOT Anchors and Composition | Author Anchors, bind them to services, connect DOT instances, materialize configuration, handle security inputs, and operate a composition. |
| Verification and examples | Choose conformance, dry-run, source-adapted, or live evidence and find the executable MARE PoC scenarios. |
| DOT-Domain | Work with the Java aggregate, value objects, validation, Contracts, Runtimes, and deployment manifests. |
| Integration handoff | Implement a catalogue, composer, or SFC adapter without coupling to a specific service, language, framework, or storage technology. |

# Repository components

| Component | Responsibility | Build ecosystem |
|---|---|---|
| `DOT-Domain` | Java domain model and validation library. It is not a web application. | Maven 3.10.0 / Java 27 |
| `DOT-CLI` | Independent `dotx` CLI for authoring support, validation, inspection, deployment, runtime lifecycle, and the Composition v1 reference executor. | Go 1.27.2 |
| `DOT-Repository` | Independently cloneable Git submodule containing reusable DOT definitions, reusable compositions, generated JSON Schema, and optional high-level Skill classifications. The filesystem is the catalogue index. | Git submodule |

| Component | Responsibility | Build ecosystem |
| --- | --- | --- |
| DOT-Testing | Repository-level conformance, composition, interoperability, and end-to-end evidence. It is test infrastructure, not a Maven module. | Maven, Go, shell, container tools |

# Supported Contract standards

| Standard | Anchor selector |
| --- | --- |
| OpenAPI | RFC 6901 URI-fragment JSON Pointer to an Operation Object |
| AsyncAPI | RFC 6901 URI-fragment JSON Pointer to an Operation Object |
| Protobuf | Fully qualified service method |
| GraphQL | Root-field schema coordinate |

Contractless Anchors remain available for interactions that cannot be described by one of these standards. They must declare direction, protocol, and media type explicitly, and they do not pass the strict `interoperable` interface profile. An explicit SPARQL or metrics interface is not an invented OpenAPI document; a runtime-only broker does not own its applications' message Contracts.

# Maturity and scope

The DOT-owned Anchor model, service-port binding, generated schema, repository resolution, conformance fixtures, `dot.mare6g.eu/v1` composition document, and `dotx compose` executor form the stable Composition v1 profile. The Java domain and the independently released CLI need not share a release number.

The reference executor is intentionally fail-closed. It does not claim to be a general multi-host orchestrator, trust-policy engine, active security scanner, or replacement for Kubernetes, Helm, Docker, Vagrant, or an SFC implementation. See the supported composition profile for the exact boundary.

# First steps

After installing the CLI and checking out the catalogue, these commands run from the `DOT-Design` checkout without deploying anything. Topology rendering requires DOT-CLI `0.2.5` or a newer pinned checkout.

```
# Validate the real catalogue using verified local Contract assets
dotx validate DOT-Repository --schema DOT-Repository/Schema/schema.json --offline

# Inspect the reviewed five-DOT PoC10 plan, not live traffic
composition=DOT-Repository/Compositions/keycloak-apisix-opa-sparql/composition.yaml
dotx compose validate "$composition" --locked
dotx compose topology "$composition" --locked --color auto
```

Use the guided demo for live execution and cleanup. Build toolchains, Java tests and the full local/GitHub CI route belong in the contributor guide.

# Project links

- MARE project
- MARE proof-of-concept catalogue
- DOT-Design source repository

# Getting started

Run your first DOT on Linux amd64 using the released CLI and the existing Prometheus definition. You need Bash, Git, GitHub CLI (`gh`), `curl`, `jq`, SHA-256 utilities, and Docker with Docker Compose. No Java, Go, Kubernetes, or Task is needed for this path.

This example runs Prometheus 3.15.0 on port `18909`. Allow time for initial image and Contract downloads. Use an isolated development host: the Compose manifest publishes unauthenticated HTTP on all host interfaces. Restrict access using your host firewall. Cleanup deletes the example's metrics data.

## Install the released CLI

Authenticate with an account granted access to the private MARE repositories and releases. From a workspace directory:

```
gh auth login
gh auth setup-git
mkdir -p dot-quickstart-download
cd dot-quickstart-download
gh release download v0.2.5 --repo MARE-EU-Project/DOT-CLI \
  --pattern dotx-0.2.5-linux-amd64 --pattern SHA256SUMS
sha256sum --check --ignore-missing SHA256SUMS
mkdir -p "$HOME/.local/bin"
install -m 0755 dotx-0.2.5-linux-amd64 "$HOME/.local/bin/dotx"
export PATH="$HOME/.local/bin:$PATH"
dotx version
cd ..
```

Stop if checksum verification fails. Expected version: `dotx 0.2.5`. Add the `PATH` export to your shell profile for future sessions. The release also provides an SBOM and build/acceptance revision manifest.

## Check out the catalogue

Only the independently cloneable catalogue is needed. Pin the revision used by this walkthrough, then check your Docker access:

```
gh repo clone MARE-EU-Project/DOT-Repository
cd DOT-Repository
git checkout --detach 49395d60a70a7f22d9109a3e5c4006c68f3218ad
docker info
docker compose version
```

Run the following blocks in the same Bash shell from the **DOT-Repository root**. Run each block only after the preceding block succeeds. The selected descriptor embeds its Compose manifest; no

neighbouring Compose file is inferred.

## Select and validate

Choose an unused runtime name. For a different host port, run `export DOTX_HTTP_PORT=18919` before the selection block.

```
DOT_REPOSITORY="${DOT_REPOSITORY:-$PWD}"
DOT_BUNDLE="$DOT_REPOSITORY/DOTs/external/Prometheus-DOT"
DOT_NAME="${DOT_NAME:-dot-quickstart}"
DOTX_HTTP_PORT="${DOTX_HTTP_PORT:-18909}"
DOT_PROJECT_FILTER="label=com.docker.compose.project=$DOT_NAME"
export DOTX_HTTP_PORT
```

```
dotx validate "$DOT_BUNDLE" --schema "$DOT_REPOSITORY/Schema/schema.json" \
  --profile reusable
dotx inspect "$DOT_BUNDLE" --json
dotx deploy "$DOT_BUNDLE" --deployment docker-compose --name "$DOT_NAME" --dry-run
```

Validation checks the schema, official HTTPS OpenAPI Contract and SHA-256, Anchor selectors, and reusable deployment bindings. Inspection shows deployment choices. The exact ID `docker-compose` selects only that mechanism. Dry-run prints its commands; it does not prove that a service is running.

## Deploy and verify

```
dotx deploy "$DOT_BUNDLE" --deployment docker-compose --name "$DOT_NAME" \
  --env "DOTX_HTTP_PORT=$DOTX_HTTP_PORT"
```

Wait up to 60 attempts (each HTTP attempt has a two-second timeout), then verify the service's build-information API and version:

```
ready=false
for ((attempt=0; attempt<60; attempt++)); do
  if curl --fail --silent --max-time 2 \
    "http://127.0.0.1:$DOTX_HTTP_PORT/-/ready"; then
    ready=true
    break
  fi
  sleep 1
done
test "$ready" = true
curl --fail --silent --show-error --max-time 5 \
  "http://127.0.0.1:$DOTX_HTTP_PORT/api/v1/status/buildinfo" |
  jq -e '.status == "success" and .data.version == "3.15.0"'
```

Expected: `Prometheus Server is Ready.`, followed by `true` from `jq`. Check runtime status:

```
dotx ps
```

Expect a running workload. Open `http://localhost:18909` (or your chosen port) to view Prometheus.

The `build-information` Anchor selects the official OpenAPI operation whose server prefix produces `/api/v1/status/buildinfo`. The `promql-query-api` Anchor selects query operations; both bind to the deployed service's `http` port. Its explicit contractless scrape-target requirement describes metrics ingestion; this example does not configure another DOT as a scrape target.

The pinned image's default configuration starts the service. The descriptor declares a Compose-managed `prometheus-data` volume. Data survives container restarts until the volume is deleted.

# Clean up



For Compose, `dotx destroy` runs `down --volumes`: it deletes this instance's containers, network, and named volume, including collected metrics. Use the same `DOT_NAME` as deployment. Never target another user's instance.

```
dotx destroy "$DOT_BUNDLE" --deployment docker-compose --name "$DOT_NAME" &&
remaining="$(docker ps -aq --filter "$DOT_PROJECT_FILTER")" &&
test -z "$remaining" &&
remaining="$(docker network ls -q --filter "$DOT_PROJECT_FILTER")" &&
test -z "$remaining" &&
remaining="$(docker volume ls -q --filter "$DOT_PROJECT_FILTER")" &&
test -z "$remaining"
```

The final checks succeed silently. If interrupted, restore your original selection variables and repeat cleanup. Downloaded images remain cached.

# Next: connect existing DOTs

For the existing Keycloak/APISIX/OAuth composition, use a full DOT-Design checkout with initialized submodules; see Contributor guide. From the **DOT-Design root**, with `dotx` on `PATH`:

This follow-up uses the development CLI's `compose inspect` command; the original v0.2.5 release calls it `compose plan`. Install the checked-out CLI using the contributor toolchain before continuing.

```
composition=DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml
task cli:install
dotx compose lock "$composition" --check
dotx compose inspect "$composition" --locked
dotx compose apply "$composition" --locked --dry-run
```

The three wires connect Keycloak discovery to APISIX, Keycloak's token endpoint to the OAuth client, and APISIX's protected operation to that client. The plan resolves endpoints and configuration. Live acceptance additionally proves anonymous rejection and authenticated access; the harness prepares Kubernetes, native Secrets, and workloads. Follow Guided demo and Verification and examples for that path.

To author a component, continue with Authoring Anchors. For development toolchains and local/GitHub CI gates, use Contributor guide.

# Troubleshooting

| Symptom | Action |
|---|---|
| Release or clone returns 404/403 | Check `gh auth status` and organization repository access, including SSO authorization if required. Run `gh auth setup-git` for HTTPS cloning. |
| CLI missing or wrong version | Check `command -v dotx`; put `$HOME/.local/bin` first on `PATH`. |
| Docker permission denied | Check `docker info` as your user. Ask the administrator for the intended Docker access, then log in again. Docker group access grants host-level power. |
| Compose unavailable | Install the Compose plugin/package for your Docker distribution and verify `docker compose version`. The legacy `docker-compose` executable is insufficient. |
| Port already allocated | Clean up the failed example, select a free `DOTX_HTTP_PORT`, and deploy again with the same name and explicit `--env` override. |
| Contract download or digest check fails | Check network/proxy access to the declared URL. Never bypass its digest. Private raw GitHub URLs require `MARE_READ_ONLY_TOKEN` or `GITHUB_TOKEN`; `gh` login alone does not supply those to `dotx`. Use `--offline` only with matching verified local Contract assets. |
| Readiness or version fails | Inspect `docker compose -p "$DOT_NAME" logs --tail=100` and `dotx ps --all`, check the selected image/port, then clean up. |

| Symptom | Action |
| --- | --- |
| Interrupted cleanup | Restore the original name and catalogue path, repeat cleanup, and inspect resources carrying `com.docker.compose.project=$DOT_NAME`. |

Download the shared command blocks. Contributors rehearse them from DOT-Design with `task testing:quickstart:dry-run` or the opt-in `task testing:quickstart:live`.

# Guided demo

This guided demo explains how a reusable DOT becomes part of a checked composition. Start with the official ecosystem demo, choose the shorter 10-15-minute standards route, or the five-minute read-only stages later on this page. These paths connect authoritative Contracts, Anchors, deployment choices and explicit wires; they prove different things.

The read-only core path takes about five minutes on an existing checkout. It is read-only apart from temporary files created and removed by the CLI. The live presenter route uses Docker and Kubernetes and performs deployment, protocol probes, inspection and cleanup.



Run the commands from the DOT-Design repository root. Keep evidence levels separate: a valid plan proves deterministic resolution, a dry-run proves that the executor can render supported runtime actions, and only a live probe proves the observed protocol exchange.

## Choose a path

For MARE use cases, start with the PoC composition walkthroughs: real PoC4 monitoring, PoC10 authenticated policy enforcement, then a labelled PoC6 CACAO/OpenC2 rehearsal using an NWDAF double. Its presenter tasks add a bounded post-verification inspection pause and automatic cleanup, with the same live assertions used by acceptance.

For a fully executable local presentation spanning Docker, Kubernetes, and OpenAPI/GraphQL/Protobuf/AsyncAPI wires, follow Docker/Kubernetes four-Contract composition. It includes the rehearsal, real probe outcomes, cleanup, and evidence boundaries.

| Path | Time | Demonstrates | Runtime mutation |
|---|---|---|---|
| Official ecosystem demo | 15-20 minutes of explanation, plus runtime waits | One DOT on two runtimes, secure policy enforcement and stateless recovery; optional four-standard deep dive | Disposable Docker and Kubernetes resources |
| Read-only overview | About 5 minutes | Repository validation, a reusable external DOT, 21 cross-standard wires, and a Kubernetes lifecycle rendering | None |
| Live presenter route | About 10-15 minutes after preparation | Six core concepts, one traced wire, four Contract standards across Docker and Kubernetes, and a real MARE use-case graph | Disposable Docker and Kubernetes resources |
| New developer | About 15 minutes | The read-only overview plus selected plan details, authoritative Contract facts, endpoint materialization, and evidence boundaries | None |

| Path | Time | Demonstrates | Runtime mutation |
|---|---|---|---|
| Live acceptance | Depends on image downloads and host capacity | GraphQL, all four finite gRPC shapes, and MQTT, NATS, Kafka, and AMQP 0-9-1 delivery against standards-native test producers and real brokers | Disposable Docker resources |

# Official ecosystem demo: reuse, compose, enforce, recover

Use this as the recommended presentation route. Budget 15-20 minutes of explanation for the first three chapters; provisioning, readiness and the intentional failed rollout add runtime-dependent waiting. The fourth chapter is an optional standards deep dive. For a short slot, show chapters 1 and 2. The shorter standards-focused route remains available below.

The story is: **one authoritative definition → explicit composition → observed behaviour → controlled recovery**. DOT-Domain and its generated Schemas define the portable models; `dotx` resolves and executes them. Docker/Kubernetes remain the workload engines. A wire connects declared Anchors; it does not magically translate protocols or implement missing producer functionality.

## Prepare and qualify one exact checkout

Run from the DOT-Design checkout root on Linux amd64, with repository access, Git, Go, Task, Python 3, Bash, jq, curl, `ss`, Docker/Compose, kubectl and a Ready local Docker-runtime Minikube node. Select context `minikube`. This advanced developer rehearsal uses the pinned source CLI, not the original `v0.2.5` binary. The simpler Getting started path still needs neither Go, Task nor Kubernetes.

Commit reviewed source changes before using these qualification commands. Choose unused namespaces and ports; keep the same overrides for preflight, rehearsal and presentation:

```
task cli:install
export DOTX_PORTABILITY_NAMESPACE=dotx-demo-portability
export DOTX_PORTABILITY_KUBERNETES_PORT=19083
export DOTX_POC10_NAMESPACE=dotx-demo-secure-api
export DOTX_POC10_FORWARD_PORT=19010
export DOTX_UPGRADE_NAMESPACE=dotx-demo-upgrade
export DOTX_UPGRADE_PORT=19081
task testing:demo:prepare
task testing:demo:preflight
task testing:demo:rehearse
```

`prepare` explicitly pulls only selected digest-pinned official images, builds and tests the standards fixtures from this checkout, and imports their provider image. It may start disposable image-test containers, but never applies a composition or provisions credentials. Fixture preparation shares

the live harness's concurrency guard and label-scoped failure cleanup. Image caches and build cache intentionally persist. Registry access is needed; preparation does not promise offline operation. The command ends with read-only preflight. Use `DEMO=portability`, `upgrade`, `secure-api`, `monitoring` or `four-contracts` with either preparation or preflight for a focused lane; the default is all five.

`preflight` does not pull, build, deploy, adopt or delete resources. It checks the installed CLI provenance, local Contracts/locks, Ready Minikube, selected image caches and relevant namespace/port/runtime conflicts. It reserves nothing. The Compose comparison also needs the descriptor-declared port `18919` free.

`rehearse` runs the quickstart, monitoring, portability, secure API, upgrade/rollback, OAuth gateway and four-Contract paths **twice each**, followed by injected-failure and signal cleanup controls. It preserves every existing live assertion rather than replacing tests with a summary. All three checkouts must be clean; before/after revisions, tracked/untracked fingerprints and the binary SHA-256 must remain identical. Do not edit sources or reinstall `dotx` during it. This is qualification before the talk, not a command to start on stage.

The printed `summary.json` and adjacent `results.txt` consolidate step exits, logs and hash-linked child receipts/results. Receipt-backed live lanes require successful assertions, exact CLI identity and `cleanup=PASSED`; prerequisite, cleanup-control and log-only quickstart evidence stay separately labelled. Preparation has its own report under `DOT-Testing/e2e/target/preparation/` and explicitly claims **no live API acceptance**. Qualification reports live under `DOT-Testing/e2e/target/rehearsal/`. A missing/failed receipt or source change fails the complete rehearsal. These local evidence directories are not published as though they were current online service health.

## Chapter 1: reuse one DOT on two runtimes

```
task testing:demo:portability:present
```

**Say:** both nodes use the same Prometheus UUID, official OpenAPI Contract and immutable image, with two selected deployment mechanisms. There are **zero wires** between these independent servers: this proves reuse, not replication/migration. **Show:** both build-information API versions match the selected image, `GOGC=75`, the declared default configuration and a real self-scrape `up=1`. At the verified pause, compare the runtime observations and the localhost endpoints in the table. Press Enter and require cleanup to pass before chapter 2. See the complete comparison and storage boundaries.

## Chapter 2: compose and enforce a real API boundary

```
task testing:pocs:present:secure-api
```

**Say:** identity, gateway, policy engine, RDF service and client are independently reusable DOTs, connected by reviewed Anchors/wires and deployment-instance inputs. Existing Contracts remain authoritative where present; explicit interfaces are not misrepresented as vendor-official OpenAPI documents. **Show:** anonymous/invalid-bearer rejection, token acquisition and authenticated

SPARQL access to stored RDF. The controlled policy change gives `200 → 403 → 200` and restores the original policy without changing the stored RDF data. Tokens stay out of printed results; native Secret references are inspected by name only. This is the source-adapted portable PoC10 slice, not acceptance of every producer or of the complete MARE security architecture. Enter verifies cleanup. See PoC walkthroughs and their remaining gaps.

## Chapter 3: inspect a change and recover safely

```
task testing:demo:upgrade:present
```

**Say:** reuse the canonical OPA DOT; the change belongs to stateless composition configuration. **Show:** the native upgrade diff, successful stricter policy, deliberately invalid Rego, failed rollout and automatic last-known-good recovery. The recovered version is `1.1.0`, not the original `1.0.0`; a fresh API decision must pass with no pending transaction before the presenter pause appears. This does not demonstrate an OPA software-version upgrade, database migration or universal stateful rollback. Enter verifies cleanup. See the native lifecycle safety boundary and recovery.

## Optional chapter 4: cross-runtime, four-standard interoperability

```
task testing:demo:docker-kubernetes-four-contracts:present
```

**Show:** real OpenAPI/HTTP, GraphQL/HTTP, Protobuf/gRPC and AsyncAPI/MQTT exchanges from the actual Docker consumer's network boundary to Kubernetes. A controlled HTTP-200 value mismatch is rejected, changes the consumer decision, and is then restored and freshly verified. Mosquitto is real; the provider and consumer are labelled standards-native fixtures, **not** production producer acceptance or MARE analytics. See the protocol and network details.

## Close with evidence, not a blanket readiness claim

At every pause, the Mermaid graph is the **locked plan**. The new runtime table is a timestamped observation; a running container or available controller is not API acceptance. Historical probe records keep their own timestamps and age. The final application tables come from the separately completed live assertions. Use the printed cache/Kubernetes context for further inspection. Never substitute a fresh cache when recovering an interrupted run. Enter, Ctrl-C, EOF and timeout take the ownership-safe cleanup path; never delete unrelated resources or force namespace finalizers. Require the cleanup receipt, not just an empty terminal.

Keep matching rehearsal reports as a clearly labelled fallback. Do not replay them as a claimed live result. For the catalogue-wide appendix, run `task testing:contracts:readiness` and consult the verification boundaries. Interface declarations, source-derived Contracts, runtime-only brokers, planning fixtures, stubs and producer doubles do not become production-certified through this demo. DOT-Registry and partner/SFC acceptance remain separate integration work.

The existing **Kubernetes and Helm runtime E2E** workflow accepts the optional `full_demo_readiness=true` manual input to run these same preparation and complete rehearsal Tasks. Ordinary pushes retain their focused runtime matrix; the long full-demo qualification is

explicit. Uploaded reports are redacted evidence, not automatically a public live-service dashboard.

# 10-15-minute live presenter route

Use this route for an audience new to DOTs. The detailed stages below remain the reference walkthrough; this section is the talk track, not another harness. For a ten-minute slot, shorten the wire inspection and show the MARE graph as the closing example rather than starting a second deployment.

**The three messages to leave with the audience:**

- Reuse a product definition; put PoC-specific configuration in the composition.
- Keep existing interface standards authoritative; wires are not protocol translators.
- Distinguish a compatible plan from a verified live exchange and production acceptance.

For technical Q&A, add the five-minute developer extension: inspect a real product's deployment choices and official Contract provenance, demonstrate rejection of an invalid wire, and show where a PoC contribution belongs.

## Before the audience arrives

Follow the mixed-runtime prerequisites and rehearse on the same Linux-amd64 checkout, installed CLI, Docker and local Minikube context that you will present. Install this checkout's CLI for the exact pinned acceptance suite. Published `v0.2.5` includes native topology and presenter support; a release binary does not include later source fixes, and the version string alone does not identify the tested binary. Record the three checkout revisions as shown in that walkthrough.

For the Keycloak/APISIX/OPA/SPARQL follow-up, use `task testing:demo:preflight DEMO=secure-api` before deployment. This read-only gate checks the exact installed CLI revision, local Contracts, cluster readiness, cached images and namespace/port conflicts. Follow the secure-API preflight instructions to keep the same overrides in the presentation terminal. Select `DEMO=four-contracts` for the mixed-runtime fixture, or `DEMO=monitoring` for node-exporter/Prometheus. Select `DEMO=portability` or `DEMO=upgrade` for the additional comparison/recovery lanes. Without a selector, all five are checked. These are point-in-time prerequisite checks, not live acceptance or resource reservations. In particular, four-Contract preflight checks cached images, Docker resource conflicts and selected NodePorts, not end-to-end gateway routing.

Pre-download/build the images by completing the existing live acceptance task before the talk, and require successful cleanup:

```
task testing:demo:docker-kubernetes-four-contracts:live
```

The timing below is a speaking budget, not a deployment-time guarantee. Cold downloads, image builds and a slow cluster can exceed it. Keep the previously recorded, revision-matched rehearsal evidence available and label it as such if the live run fails; never describe a plan or saved result as a new live pass. Use an isolated demo network: the example's plaintext NodePorts and anonymous MQTT are not safe public production defaults. Do not run competing acceptance or cleanup tasks

during the presentation.

## Minutes 0-2: the problem and the six concepts

**Say:** "We want to describe a component once and reuse it across deployments and MARE use cases. DOT-Design connects existing interface standards to explicit deployment and composition choices; it does not replace those standards."

| Concept | Audience meaning | In this demonstration |
|---|---|---|
| DOT | Reusable software component description | Provider, consumer and Mosquitto use canonical catalogue definitions. |
| Contract | Authoritative description of an interface | OpenAPI, GraphQL SDL, Protobuf and AsyncAPI describe different interactions. |
| Anchor | Named provided or required interaction | The provider and consumer select the REST alerts operation from its Contract. |
| Wire | Connection between compatible Anchors | `rest-alerts` connects the provider's Anchor to the consumer's Anchor. |
| Deployment | How a DOT runs on a selected platform | Provider and broker run on Kubernetes; consumer runs on Docker. |
| Composition | Participants, connections and deployment-instance inputs | One locked graph selects those DOTs, deployments, wires and configuration. |

**Point out the ownership:** `DOT-Repository` stores reusable DOT bundles and compositions; `dotx` in `DOT-CLI` validates, plans and executes them; `DOT-Domain` defines the shared model. `DOT-Design` brings together documentation and executable acceptance harnesses. The synthetic demo graph stays in `DOT-Testing`; it is not a new production product.

## Minutes 2-4: read the topology before deploying

```
dotx compose topology DOT-Testing/composition/docker-kubernetes-four-
contracts/scenario.yaml --locked --color always
```

**Show:** three DOTs and four logical wires. Match the diagram's `N` and `W` identifiers to the aligned DOT and connection tables. `CONTRACT` names the interface standard; the separate `PROTOCOL` column names the transport protocol. Explain provider versus consumer and the selected runtime, rather than reading UUIDs aloud. The separate `T4` connection identifies MQTT's broker transport endpoint; it is not a fifth Anchor wire. Arrows mean logical PROVIDES-to-REQUIRES, not packet direction: an HTTP consumer may initiate the request.

**Expected observation:** the graph is available without deploying anything. Parallel wires may share one arrow, but every wire remains in the table. This is a locked plan view, not a live health display.

## Minutes 4-6: trace one wire from Contract to configuration

```
dotx compose inspect DOT-Testing/composition/docker-kubernetes-four-
contracts/scenario.yaml --locked --json |
  jq '.wires[] | select(.name == "rest-alerts") |
      {name, from, to, contractStandard, selector, operationPath,
       compatibility, materializer}'
```

**Say:** "The OpenAPI Contract owns `GET /alerts`. The Anchors select that operation; the wire names the two participants. The materializer puts the resolved operation URL into the consumer's declared `OPENAPI_YAML_URL` environment variable. We are not copying the API definition into the wire."

**Expected observation:** `OpenAPI`, the selected operation, provider/consumer Anchor references, `COMPATIBLE` and the environment-variable materializer. The template's `192.0.2.1` address is deliberately not a live target. The existing harness substitutes the actual Minikube address and locks that run-specific graph before applying it. Do not manually apply the template.

**Limit:** compatibility is checked against declared interfaces. It is not proof of reachability, delivery or application correctness. There is no automatic OpenAPI-to-gRPC-to-MQTT protocol translation.

## Minutes 6-11: deploy, verify and inspect the live composition

```
task testing:demo:docker-kubernetes-four-contracts:present
```

This is the existing acceptance path with an inspection pause, not a parallel demo implementation. It deploys through `dotx`, checks readiness and executes probes from the actual Docker consumer's network namespace to Kubernetes. Presenter mode keeps build and deployment chatter in the printed, owner-only `details.log`. The terminal shows the numbered plan, deployment, probe, inspection and cleanup stages; a failure still reports its stage, exit code and diagnostic-log location. Ordinary noninteractive acceptance remains verbose.

| Standard | Explain the expected live result |
|---|---|
| OpenAPI | HTTP 200 plus the expected JSON alert; the running consumer derives `REVIEW_REQUIRED` from its wired endpoint. |
| GraphQL | `Query.health` returns `true`. |
| Protobuf / gRPC | The selected health RPC returns `serving: true`. |
| AsyncAPI / MQTT | The broker delivers the exact uniquely correlated, schema-valid message after subscription acknowledgement. |

**Show:** the verified pause appears only after all four live assertions and recorded status checks pass. Keep the first terminal at this pause. In a second terminal, copy the exact recovery/status command printed by the harness; it selects this run's private cache and Kubernetes configuration. Inspect the per-wire outcomes and redacted `result.json`, not access tokens or Secrets. The final live-results table separates `CONTRACT`, `PROTOCOL`, `OBSERVED` and `OUTCOME`. It is generated only after each successful probe is also present in recorded lifecycle status; `COMPATIBLE` and running workloads do not populate it. The observations summarize checked assertions, not raw API responses. The same four

rows are saved in `result.json` and the plain-text `results.txt`. Before the pause, point out the controlled negative step: the owned fixture temporarily returns HTTP 200 with severity `low`, the native assertion records `RESPONSE_MISMATCH`, and the running consumer changes its decision. The harness then restores the critical alert and requires a newer `REVIEW_REQUIRED` receipt and successful native assertion. This distinguishes endpoint health from correct application data; it is not a claim of complete schema or policy validation. The graph remains the locked plan, not an inferred network map. Presenter lanes using the shared lifecycle helper, including the four-Contract lane, append timestamped runtime observations with `compose topology --locked --status`. Their results tables still provide the separate application assertions.

For an applied composition, run the following from the same private cache and Kubernetes configuration printed by its harness (do not use another run's state):

```
COMPOSITION=DOT-Repository/Compositions/keycloak-apisix-opa-sparql/composition.yaml
dotx compose topology "$COMPOSITION" --locked --status
```

Explain the layers separately: `APPLIED` is a recorded lifecycle phase; `RUNNING` describes a process, container or VM; `CONTROLLERS_AVAILABLE` describes selected Kubernetes controller status, not rollout or API acceptance. An empty selection is `NO_CONTROLLERS`; query failures are `UNKNOWN` and return nonzero. Historical wire probes show their original timestamps. `RECENT_RECORD` means recently recorded, never a fresh API call; `STALE_RECORD` requires a fresh probe before claiming current acceptance. The default record age boundary is five minutes, adjustable with `--probe-max-age 2m` (positive, at most 24 hours). This observation mode does not deploy, mutate configuration or run probes; it uses the lifecycle coordination lock. Mermaid export remains plan-only.

**State the boundary:** Mosquitto is a real external broker, but provider and consumer are test-only DOTs with standards-native fixture Contracts, not vendor-official or MARE producer APIs. The consumer now executes the REST exchange and derives a synthetic review decision; explicit probes exercise GraphQL, gRPC and MQTT from its network namespace. The MQTT probe subscribes and publishes to prove broker delivery, not a provider application's business-event publication. This verifies interoperability mechanics across runtimes, not a production application or the whole MARE stack.

## Minutes 11-13: relate the mechanism to a real MARE use case

Keep the live demo paused and show this separate, read-only catalogue graph:

```
dotx compose topology \
  DOT-Repository/Compositions/keycloak-apisix-opa-sparql/composition.yaml \
  --locked --color always
```

**Say:** "The same model expresses a PoC10-derived protected data service: Keycloak supplies identity, APISIX gates access, OPA decides policy, Oxigraph serves SPARQL and a client obtains a token and requests the protected capability." Point to the five wires, then explain that this graph has not been deployed by the four-Contract demo.

For a longer session, use the existing `task testing:pocs:present:secure-api` in a separate, rehearsed

run. Its live checks require HTTP `401` for missing and malformed bearer tokens, authenticated POST `403`, and an authorized GET reading the stored `mare-poc10` value. See MARE PoC composition walkthroughs for prerequisites and commands. This is source-adapted evidence: Oxigraph substitutes for GraphDB, the client is synthetic, and SPARQL/OPA use explicit protocol interfaces rather than fabricated official OpenAPI Contracts. It does not accept the complete producer deployment, LDAP, telemetry or automated response.

### Minutes 13-15: clean up and leave a reproducible next step

For deployment reuse, the Prometheus portability lane runs one canonical DOT and official Contract on Docker Compose and Kubernetes, with the same pinned image and fresh API/self-scrape checks. It compares two independent servers; it does not claim a wire, replication or data migration.

For a separate technical follow-up, the stateless upgrade lane reuses the canonical OPA DOT, changes a policy configuration through native `compose upgrade`, and verifies automatic last-known-good recovery after a deliberately broken revision. Rehearse it separately; do not fit its intentional rollout timeout into this 10–15-minute route or imply application-data rollback.

Return to the paused demo terminal and press Enter. Expect a receipt with `cleanup=PASSED` and `exit=0`; the harness checks its owned resources for leftovers. Ctrl-C also invokes cleanup and exits `130`. EOF, timeout and SIGTERM are non-success exits with cleanup, not successful acceptance. The inspection pause defaults to ten minutes; see the mixed-runtime walkthrough for its bound. If cleanup fails, retain the receipt and use that walkthrough's recovery instructions. Never delete unrelated namespaces, containers or volumes.

Close with: "A valid definition is not a running workload; a compatible plan is not a successful exchange. We have shown the live results and their limits."

Direct attendees to Getting started to deploy real Prometheus, the reusable PoC walkthroughs to explore compositions, and Anchors and Composition to author their own integration. Keep the verification matrix available for questions; do not equate catalogue validity with production readiness or claim DOT-Registry REST API or joint SFC acceptance.

## Audience FAQ

### Why not just use Docker Compose, Kubernetes or Helm?

Those tools remain the native deployment mechanisms. DOT adds a reusable component description, authoritative interface references, named interactions, explicit cross-component wiring and a checked composition plan. `dotx` delegates supported deployment actions rather than replacing those platforms. If an existing Compose file is all a use case needs, DOT's additional value is the shared catalogue and interoperability description, not a different container scheduler.

### What is reused, and what changes between PoCs?

The product's canonical `dot.json`, Contracts and Anchor identities are reused. A composition selects participants and deployments and supplies use-case-specific configuration: routes, policies, scrape

targets and credential references. There is no need to create another Keycloak DOT for every PoC. Runtime platform means *where* it runs; deployment mechanism means *how* it is deployed.

## Does connecting two Anchors make incompatible APIs work together?

No. Wires connect declared compatible interactions; they do not translate OpenAPI, GraphQL, gRPC or MQTT automatically. An application-specific adapter, if required, must be implemented and described explicitly. The four-Contract demo verifies four separate standards-native exchanges, not one message automatically converted through four protocols.

## Is a Contract the same as a protocol? Are all Contracts official?

No. OpenAPI is an interface-description standard; HTTP is a protocol used by the selected operation. Protobuf describes messages/services used by the gRPC profile; AsyncAPI describes messaging interactions and transport bindings. The topology table separates `CONTRACT` from `PROTOCOL` for that reason. Use vendor/project-official documents when available. Source-derived Contracts and test-fixture Contracts must be labelled, not advertised as official. An explicit protocol interface is not an invented official Contract; the topology labels that case `explicit interface`. A `-` on a transport-only `T` link does not create another Contract-backed Anchor wire.

## Does validation mean a DOT is production-ready?

No. Base validation checks definitions and declared references; stricter interface checks assess declaration completeness. A locked compatible plan checks the selected graph. A successful deployment establishes a runtime state. A live probe verifies a particular observed exchange. None alone proves full business functionality, resilience, security hardening, scale or acceptance of an unmodified PoC producer. Read the declared scope with each result.

## What does the lock file guarantee?

It pins the composition's portable definition inputs and resolved plan so that unreviewed definition changes are detected. It does not freeze runtime health, network addresses, credentials or upstream service behavior. Contract payload digests verify the referenced bytes; a downloadable URL is not itself evidence that the live implementation conforms to the whole Contract.

## Where do credentials and configuration belong?

DOTs declare supported service configuration. Compositions provide deployment inputs and secret references; secret values and concrete runtime state are not part of a reusable public interface description. The PoC10 rehearsal uses native Kubernetes Secret references. Do not display Secret contents, tokens, raw deployment state or private diagnostic logs during the talk. The four-Contract fixture uses plaintext/anonymous demo settings and is not an example of production security hardening.

## What did this presentation actually demonstrate?

The official route verifies one Prometheus DOT on two runtimes, a source-adapted secure API with policy denial/restoration and unchanged stored RDF, and native stateless OPA upgrade/recovery.

The optional standards chapter uses test-only provider/consumer components and real Mosquitto to verify four exchanges from the consumer's network namespace. The shorter standards-focused route shows the PoC10 graph only unless its separate live task is run; a topology alone never accepts it. Producer doubles, stubs, planning fixtures and source adaptations remain labelled. DOT-Registry REST API, complete producer/PoC acceptance and joint SFC/partner acceptance are not claimed by these demonstrations.

### How can an attendee try this afterwards?

Start with the real Prometheus quickstart, which needs neither a Java/Go development toolchain nor Kubernetes when using the supported released binary. Then explore reusable compositions and the authoring guide. The new topology and presenter features require the installed development checkout described above; do not assume an older release contains them.

## The story in one diagram

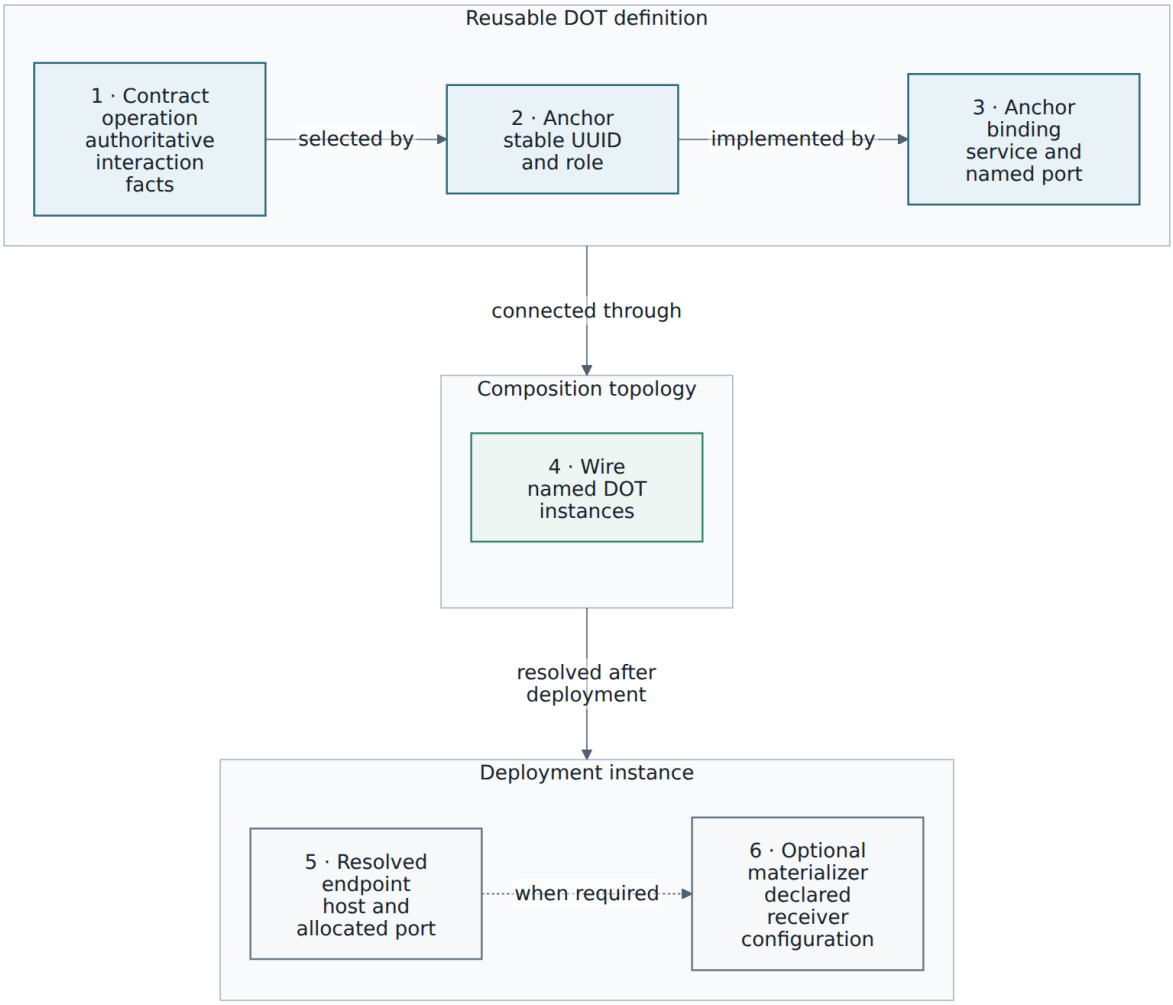



*Figure 2. From an authoritative Contract operation to a deployed connection*

The Contract owns protocol semantics. An Anchor gives one selected Contract operation a stable logical identity. A deployment-service binding identifies the service port that implements the Anchor. A composition wire connects two named DOT instances and may materialize the resolved endpoint into configuration already declared by the consumer.

# Prepare the checkout

For a first deployment with the released CLI, follow Getting started. This presenter path uses a full checkout from Contributor guide with initialized `DOT-CLI` and `DOT-Repository` submodules. The demo needs `dotx`, `jq`, and network access when URL-backed Contracts are not already cached.

```
git submodule update --init --recursive
task cli:install
dotx version
```

Source builds use `DOT-CLI/VERSION`, which may advance independently of the published release. The published release reports:

```
dotx 0.2.5
```

The independently released Linux-amd64 binary, its SBOM, release manifest, and checksums are also available from the DOT-CLI v0.2.5 release. Use `sha256sum --check SHA256SUMS` before executing a downloaded release asset.

# Stage 1: validate the authoritative catalogue

```
dotx validate DOT-Repository
```

This recursively discovers every `dot.json`; there is no separate inventory. The command validates schema shape, identities, Skills, Contract payload integrity, selectors, service ports, Anchor bindings, repository placement, and catalogue-wide uniqueness. Contract-backed external and internal bundles also receive the strict reusable profile.

**What this proves**

The checked-out catalogue is internally consistent and every declared reference resolves. It does not deploy a workload or claim that a third-party service is reachable.

# Stage 2: inspect a reusable external DOT

Keycloak is a useful example because its canonical DOT uses official, digest-pinned Contracts while offering multiple deployment mechanisms.

```
dotx inspect DOT-Repository/DOTs/external/Keycloak-DOT --json |
  jq '{name, version,
       skills: (.skills | length),
       contracts: (.contracts | length),
       anchors: (.anchors | length),
```

```
      deployments: (.deployments | length),
      supportedDeploymentSystems}'
```

The `0.2.4` catalogue produces this compact projection:

```
{
  "name": "Keycloak-DOT",
  "version": "0.2.0",
  "skills": 2,
  "contracts": 3,
  "anchors": 4,
  "deployments": 4,
  "supportedDeploymentSystems": [
    "DOCKER_COMPOSE",
    "DOCKER_IMAGE",
    "HELM_CHART"
  ]
}
```

**What this proves**

One reusable DOT root can describe several supported deployment choices without forking its identity for each PoC. Deployment-instance realms, routes, credentials, and endpoints remain composition inputs rather than copied DOTs.

# Stage 3: resolve the cross-standard composition

The canonical Contract matrix connects six DOT instances with 21 compatible wires. Its committed lock pins every DOT, Contract, configuration input, and portable resolved plan.

```
dotx compose lock \
  DOT-Testing/composition/contract-standards/scenario.yaml --check

dotx compose validate \
  DOT-Testing/composition/contract-standards/scenario.yaml --locked

dotx compose inspect \
  DOT-Testing/composition/contract-standards/scenario.yaml --locked --json |
  jq '{apiVersion, name, version,
       dotInstances: (.dots | length),
       wires: (.wires | length),
       compatibleWires: ([.wires[] |
         select(.compatibility == "COMPATIBLE")] | length),
       standards: ([.wires[].contractStandard] | unique),
       interactions: ([.wires[].providerInteraction] | unique)}'
```

Expected summary:

```
{
  "apiVersion": "dot.mare6g.eu/v1",
  "name": "contract-standards-matrix",
  "version": "1.0.0",
  "dotInstances": 6,
  "wires": 21,
  "compatibleWires": 21,
  "standards": ["AsyncAPI", "GraphQL", "OpenAPI", "Protobuf"],
  "interactions": [
    "BIDIRECTIONAL_STREAM", "CLIENT_STREAM", "MUTATION", "QUERY",
    "REQUEST_RESPONSE", "SEND", "SERVER_STREAM", "SUBSCRIPTION"
  ]
}
```

The project gate asserts the complete result rather than only printing a summary:

```
task testing:compose:contract-standards
```

**What this proves**

Every provider and consumer selector resolves against its authoritative Contract; all 21 pairs are compatible; endpoints, interactions, probes, and materializers can be derived without copying standard-specific semantics into Anchors.

## Follow one AsyncAPI wire

Inspect the Kafka wire to see how standard-derived facts and deployment facts remain separate:

```
dotx compose inspect \
  DOT-Testing/composition/contract-standards/scenario.yaml --locked --json |
  jq '.wires[] | select(.name == "asyncapi-kafka") |
      {name, compatibility, contractStandard, contractFormat,
       providerInteraction, consumerInteraction, channelAddress,
       connectionEndpoint, materializers, probe}'
```

The AsyncAPI Contract supplies the complementary send/receive operations, channel, message schema, Kafka binding, and correlation field. The composition selects the broker instance and materializes its resolved address and topic for the consumer. A plan still does not claim that a message was delivered.

# Stage 4: render a lifecycle without mutating it

The APISIX, Keycloak, and OAuth-client scenario demonstrates dependency order, configuration materialization, native Kubernetes Secret references, rollout readiness, and an exposed Anchor:

```
dotx compose lock \
  DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml --check

dotx compose apply \
  DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml \
  --locked \
  --namespace dot-guided-demo \
  --dry-run \
  --forward protected-metadata-through-gateway=19080
```

The rendering ends with:

```
Dry run complete; composition state was not written.
```

The command may create bounded temporary rendering files and removes them when it exits. It does not create Kubernetes resources, start a port forward, or write composition state. The rendered order is Keycloak, APISIX, then the OAuth client, matching the declared readiness dependencies.

### What this proves

The reference executor accepts the locked inputs and can derive the exact platform-native actions. It does not prove rollout success, credentials, or HTTP behavior; those require the live acceptance layer.

# Stage 5: connect claims to evidence

| Claim | Executable evidence | Boundary |
|---|---|---|
| Catalogue correctness | `dotx validate DOT-Repository` | Static and Contract resolution evidence; no deployment |
| Cross-standard interoperability | `task testing:compose:contract-standards` | 21 deterministic compatible plans; no live traffic |
| Live Contract probes | `task testing:compose:contract-probes:live` | Standards-native test producer and real brokers on disposable Docker resources |
| Runtime configuration portability | `task testing:compose:runtime-configuration` | Dry-run rendering across six adapters |

| Claim | Executable evidence | Boundary |
| --- | --- | --- |
| Workload isolation | `task testing:compose:service-workloads:live` | Live repeated Compose, Kubernetes, and Helm instances |
| Portable-runtime isolation | `task testing:compose:runtime-isolation:live` | Live Vagrant/libvirt and WASI CLI instances |
| Released-product acceptance | 0.2.4 downloaded-binary acceptance | Linux amd64, single-host supported profile |

See Verification and examples for the complete test matrix and the distinction between model, plan, dry-run, source-adapted, and live production evidence.

# Optional: run the live Contract probes



This path builds images, starts disposable Docker workloads, and downloads pinned images when absent. Use the presenter path when time, network, or host capacity is uncertain.

```
task testing:compose:contract-probes:live
```

The harness builds a standards-native GraphQL/gRPC producer and an independent consumer, then deploys the canonical Mosquitto, NATS, Kafka, and RabbitMQ DOTs. It proves GraphQL query, mutation verification/cleanup, and GraphQL-over-SSE; all four finite Proto3 `NO_SIDE_EFFECTS` gRPC shapes through reflection; and correlated MQTT, NATS, Kafka, and AMQP 0-9-1 delivery. It records redacted evidence and destroys composition-owned resources on success, failure, or interruption.

# Five-minute presenter script

1. Show the Contract-to-Anchor-to-wire diagram and explain the four ownership boundaries.
2. Run `dotx validate DOT-Repository` to establish that the catalogue is the checked source of truth.
3. Inspect Keycloak to demonstrate one reusable product identity with several deployment choices.
4. Print the 21-wire plan summary and highlight the four Contract standards and eight interaction forms.
5. Inspect the Kafka wire to distinguish Contract semantics from its resolved broker endpoint.
6. Render the APISIX/Keycloak/OAuth lifecycle with `--dry-run`.
7. Finish on the evidence table and state the supported boundary explicitly.

Keep the published release notes and verification matrix open as fallback evidence if a local network or runtime is unavailable.

# Supported boundary

The stable profile supports Linux amd64 and private single-host composition state. Multi-host orchestration, automatic cross-platform gateway provisioning, general stateful-data migration, AMQPS, AMQP 1.0, native arm64 and non-Linux executables, the DOT-Registry REST API, and joint SFC or partner acceptance are separate activities. Unsupported combinations fail closed rather than being silently approximated.

All eleven MARE PoCs have planning or source-adapted evidence. That evidence is not presented as acceptance of every unmodified partner workload. Continue with Architecture for system boundaries or DOT Anchors and Composition for authoring and lifecycle details.

# Developer demo: reuse and extend

Use this optional five-minute extension after the 10-15-minute presenter route, or during technical Q&A. It adds no new live deployment. Run from the DOT-Design checkout root with its installed `dotx`, initialized catalogue, Bash and `jq`.

The developer takeaway is: **reuse the product definition, describe the real interface, configure a composition, then prove the intended exchange.**

## 1. One real product, several deployment choices

**Ask the audience:** "If two PoCs use Keycloak with different realms and deployment platforms, which parts should they share?"

```
dotx validate DOT-Repository/DOTs/external/Keycloak-DOT \
  --schema DOT-Repository/Schema/schema.json --profile reusable --offline

dotx inspect DOT-Repository/DOTs/external/Keycloak-DOT --json |
  jq -e '{name, version,
          deployments: [.deployments[] |
            {id, platformEngine, type, supported}],
          anchor: (.anchors[] | select(.name == "admin-realm-read") |
            {name, role, contract})} |
         select(([.deployments[].platformEngine] | unique) ==
           ["DOCKER", "KUBERNETES"])'

sed -n '1,12p' \
  DOT-Repository/DOTs/external/Keycloak-DOT/contracts/keycloak-admin-rest.yaml
```

**Show:** one `Keycloak-DOT` identity with Docker image, Docker Compose and Kubernetes/Helm deployment choices. A PoC-specific image is a named deployment variant, not a new product identity. The same selected `admin-realm-read` Anchor refers to the Admin REST Contract and its `GET` selector. The adjacent Contract metadata identifies the official release archive, extracted payload, revision and SHA-256; no API definition is copied into the Anchor.

This existing bundle uses ID-only root references resolved from its local `contracts/` assets. Those

are migration-compatible catalogue references, not permission to publish an unresolved UUID. New complete Contract declarations use the model's structured `payload.url` pointing directly to a raw Contract document, plus its SHA-256; bundle-local verified assets remain supported. The archive URL shown here is provenance, not a raw-document payload URL. Prefer official documents and label derived ones.

**Explain the boundary:** an inspector reporting a supported deployment is not live acceptance of every deployment variant. The PoC10 rehearsal selects pinned native Kubernetes deployments of this same `Keycloak-DOT` and the shared `APISIX-DOT`, with composition-owned realms, routes and policy. It does not certify every variant or every Admin REST operation. Compare each plan's `dotId`, `dotVersion` and `deploymentId` when explaining provenance.

## 2. Show the guardrail, not just the happy path

```
task testing:demo:developer-tour
```

The task repeats the short inspection, validates the existing two-DOT PoC7 planning graph, then attempts its existing deliberately rejected wire. Expect:

```
EXPECTED REJECTION (exit=1): scoring-api-to-collector mixes Contract-backed and
contractless Anchors.
The valid two-DOT graph has zero wires; no runtime was deployed by either validation
command.
```

The task itself exits zero only when the positive graph resolves and the negative graph returns that specific compatibility diagnostic. Missing credentials, a broken checkout, a missing CLI or an unrelated validation error are not counted as a successful negative test.

**Say:** "Putting two boxes on a diagram is not an implemented integration. We cannot make an undeclared collector interface compatible by drawing a wire." At the pinned source boundary, the collector prints flow data rather than calling the scorer. This control demonstrates interface admission, not a live trust-scoring result or a runtime security test.

The first current rejection is Contract-backed/contractless incompatibility; it does not prove that a later undeclared-configuration check ran. After implementing a real consumer, declare its supported endpoint configuration, Anchor and binding, then validate the resulting wire and test the actual call.

## 3. What a PoC developer changes, and what stays shared

| Concern | Owning artifact | Developer action |
|---|---|---|
| Product capability and interfaces | The existing DOT bundle, root Skills/Contracts, Anchors and service bindings | Reuse it. Extend the shared definition only when the software genuinely supports an additional capability or interface. |

| Concern | Owning artifact | Developer action |
| --- | --- | --- |
| API or messaging semantics | Official Contract, or an explicitly labelled source-derived Contract | Select an existing operation. Do not redefine request/response schemas or invent a consumer implementation in a wire. |
| Deployment choice | A selected runtime/deployment already declared by the DOT | Choose a supported deployment ID. Add a reusable variant to the product root if a new mechanism is genuinely needed; test that variant. |
| Realm, route, policy, scrape target or configuration file | Composition inputs and endpoint materializers | Configure the use case without copying or hard-coding the product definition. |
| Credentials and concrete runtime state | Secret references and the deployment instance | Keep secret values outside reusable public descriptors and presentation output. |
| Reproducibility and proof | Reviewed composition lock and executable acceptance harness | Inspect the plan, review changed lock inputs, run dry-run checks, then verify the real workload and its cleanup. Commit definitions, not local runtime state. |

The practical contribution sequence is:

1. Find the product under `DOT-Repository/DOTs/external/` or `DOTs/internal/`; check its README, source revision and declared gaps.
2. Use official interface documents where available. For MARE-owned code, export or derive the Contract from the implementation and review its limits.
3. Declare the provided/required Anchor and the selected deployment's service binding. Ensure the application really reads the declared configuration.
4. Add or update a reusable graph under `DOT-Repository/Compositions/`, with its README, non-secret inputs and reviewed lock. Put synthetic fixtures and live test drivers under `DOT-Testing`, not beside a new duplicate product.
5. Run `dotx validate DOT-Repository`, inspect the locked plan and deployment dry-run, then add a far-end behavioral assertion and success/failure cleanup.

`Skill` is an optional high-level capability classification; it is not an API operation or a command executor. Anchors and authoritative Contracts describe the concrete interoperability boundary.

### 4. Show the broader potential without extending the live claim

| Capability | Existing executable evidence | What to say |
|---|---|---|
| Four Contract standards, 21 selected wires | `task testing:compose:contract-standards` | Broader deterministic interface coverage than the four-wire live demo; a compatible plan is not traffic. |
| GraphQL query/mutation/SSE; finite unary and streaming gRPC; MQTT, NATS, Kafka and AMQP 0-9-1 | `task testing:compose:contract-probes:live` | A separate, longer test-producer/broker acceptance suite. Do not start it unexpectedly during a short presentation or claim partner-producer acceptance. |
| Configuration defaults and overrides across six adapters | `task testing:compose:runtime-configuration` | Docker image/Compose, Kubernetes/Helm, Vagrant and WASI rendering; not proof that every possible DOT runs on every adapter. |
| Repeated multi-service instances and selective cleanup | `task testing:compose:service-workloads:live` | A separate Compose/Kubernetes/Helm isolation test, not multi-host orchestration. |

Choose one row in response to a technical question rather than reading the entire test matrix aloud. The examples demonstrate extension points and tested profiles, not automatic protocol conversion, arbitrary networking, complete PoC production readiness, a DOT-Registry REST API or joint SFC acceptance.

Continue with Anchors and Composition for authoring, Verification and examples for the evidence matrix, and the audience FAQ for short answers.

# MARE PoC composition walkthroughs

Start here when presenting DOT composition against MARE use cases. Run commands from DOT-Design with its pinned submodules. These are integration slices, not complete official PoC outcomes. For generic cross-runtime/four-Contract mechanics, use the Docker/Kubernetes demo.

## Presenter route: inspect, run, prove, clean up

For a 15-20 minute presentation, show PoC4 first, PoC10 second, and PoC6 only with its adapter/double boundary visible. Avoid new feature changes on the day. Each presenter command uses the same locked composition and live assertions as acceptance; there is no second set of DOT definitions or demo-only success path.

```
# Run from a real interactive terminal, one example at a time.
task testing:pocs:present:monitoring
task testing:pocs:present:secure-api
```

```
task testing:pocs:present:remediation
```

Each follows five stages: **inspect topology → deploy → verify → inspect the running result → clean up**. The terminal prints resolved DOT versions, deployments, wires and evidence boundaries. It pauses only **after** every API assertion passes. While paused, open the printed Prometheus targets page or inspect the redacted `result.json`; the secure API must still deny anonymous requests. Do not paste access tokens or Secret values into the presentation.

The topology is generated from the locked plan by `dotx compose topology`, using the MIT-licensed `mermaid-ascii` 1.6.1 Go library embedded in the binary. There is no Python renderer or separate executable to install. DOTs appear once, with connected Unicode boxes. Parallel wires share a connection labelled `W1,W2,⋯`; the connection legend lists every wire, Contract standard and protocol separately. `T` links show broker transport endpoints, not extra Anchor wires or invented application publishers. Aligned tables show full instance names, platforms, deployments and versions, plus each connection's provider, consumer, Contract standard, protocol and name. Long values wrap inside their columns without truncation, using the same width as the graph. Arrows mean logical PROVIDES-to-REQUIRES relationships, not packet direction. The graph reappears after successful live verification.

Interactive diagrams use cyan Kubernetes, magenta Docker and amber connection IDs. Set `NO_COLOR=1` or `DOTX_TOPOLOGY_COLOR=never` for plain output; `always` forces colour when capturing a demonstration (`NO_COLOR` and `TERM=dumb` still disable it). Logs are plain by default. `DOTX_TOPOLOGY_WIDTH=80` controls width (40-240 columns). The renderer tries vertical/compact layouts for narrow screens and fails clearly if it cannot show every connection; it never silently clips.

For all four Contract standards across Docker and Kubernetes, use `task testing:demo:docker-kubernetes-four-contracts:present`; see the mixed-runtime presenter walkthrough.

Press Enter to destroy owned resources and verify cleanup. Ctrl-C also cleans up and returns 130; SIGTERM returns 143. The default inspection limit is ten minutes (`DOTX_POC_PRESENT_TIMEOUT`, integer 1-1800 seconds). Timeout returns 124 and EOF returns 1, both after cleanup: neither is reported as a successful presentation. Headless launches are rejected before deployment, and `DOTX_E2E_KEEP=1` is forbidden in presenter mode.

```
# Safe headless previews: print the real graph and render without deployment.
task testing:pocs:present:monitoring -- --dry-run
task testing:pocs:present:secure-api -- --dry-run
task testing:pocs:present:remediation -- --dry-run
```

The noninteractive `testing:pocs:*:live` tasks remain unpaused. Use the presenter tasks above, not the full failure-injection suite, in front of an audience. For preparation, `task testing:pocs:present:controls` exercises Enter, EOF, timeout and real signals with live Prometheus on overridden port 19041. `task testing:pocs:present:rehearse` requires clean commits, drives all three presenter flows through a real terminal, inspects live status during each pause, then verifies cleanup. It preserves checkout/CLI fingerprints and redacted results under ignored `DOT-Testing/e2e/target/presenter/` and `poc-rehearsals/`. Keep those receipts and the exported PDF as an

explicitly labelled fallback, not evidence that a new live exchange occurred during the talk.

## Prerequisites and fixed inputs

Use Linux amd64, Docker, curl, jq and Git. Acceptance harnesses also require Bash, flock, Python 3 and Task; ShellCheck checks their regressions. PoC10 needs kubectl and a Ready local Docker-driver Minikube node with context `minikube`. PoC6 additionally needs Docker BuildKit and read access to the pinned private producer through Git/`gh`, or `POC6_SOURCE_DIR` at the exact revision below. Automatically acquired PoC6 source uses the private run directory's filesystem, not shared `/tmp`, and is removed after the run. A caller-supplied source checkout remains untouched.

```
git submodule update --init --recursive
task cli:install
dotx version
git rev-parse HEAD
git -C DOT-CLI rev-parse HEAD
git -C DOT-Repository rev-parse HEAD
task testing:pocs:dry-run
```

To display just the graph without running the acceptance harness:

```
dotx compose topology \
  DOT-Repository/Compositions/keycloak-apisix-opa-sparql/composition.yaml \
  --locked --color always
```

Install from the pinned CLI checkout: released 0.2.4 predates repository-wide composition validation. Dry-runs must pass before deployment. The three paths contain respectively 1, 5 and 2 compatible wires. Inspect their exact Anchor IDs, selected deployments and materializers; `COMPATIBLE` is not a live result.

## 1. PoC4: Node Exporter to Prometheus

```
Node Exporter -- metrics scrape wire --> Prometheus -- query API --> presenter
```

Graph: `DOT-Repository/Compositions/node-exporter-prometheus-monitoring/composition.yaml`. Producer context: PoC4 `c03eca71a9d87358603a1d89fa0eca58a9dabe03`. Node Exporter provides metrics; Prometheus requires the scrape endpoint. A `CONFIG_FILE` materializer inserts `ENDPOINT_HOST_PORT` at `/scrape_configs/0/static_configs/0/targets/0`. Prometheus's query Anchor uses its DOT-local Contract. OpenMetrics is not a fabricated OpenAPI operation.

```
dotx compose inspect DOT-Repository/Compositions/node-exporter-prometheus-
monitoring/composition.yaml --locked --json
task testing:pocs:monitoring:live
```

The task waits for readiness, verifies the exact planned scrape target is healthy, checks

`up{job="node-exporter"}` equals `1`, and verifies the exporter version through `node_exporter_build_info`. Expect `PASS`, then a receipt with `exit=0, cleanup=PASSED`. Override `DOTX_POC4_QUERY_PORT` if 19040 is busy.

This grants no host namespaces or host root filesystem mounts. It observes the exporter's container, not the testbed host. Full log/dashboard flow, MITM detection and mitigation remain outside this slice.

## 2. PoC10: authenticated, policy-controlled SPARQL

**Presentation preflight**

After preparing the local cluster and rehearsing once to cache images, run this read-only gate before presenting. It is scoped to this secure-API composition, not the four-Contract fixture or all catalogue deployments:

```
# Keep these overrides in the same terminal for preflight and presentation.
export DOTX_POC10_NAMESPACE=dotx-poc10-presentation
export DOTX_POC10_FORWARD_PORT=19010
task testing:demo:preflight DEMO=secure-api
task testing:pocs:present:secure-api
```

The preflight requires local Go only to inspect the installed binary's embedded build information. It requires clean CLI/catalogue checkouts and a binary built from the CLI's exact revision, rather than trusting the `0.2.5` version label. If mismatched, review/commit CLI changes and run `task cli:install`. It prints the CLI SHA-256, Design/catalogue revisions, verifies local Contract assets through offline catalogue validation, and resolves the reviewed composition lock.

It then checks Docker/Compose access, the `minikube` context, one Ready Linux-amd64 node without resource pressure, an unused namespace and TCP port, and every selected manifest image in the Docker-driver Minikube cache. All five selected images are digest-pinned at their rehearsed versions; tags remain in the reference for readability. Preflight rejects tag-only references and cached images whose reported platform is not Linux amd64. It prints the resolved image IDs without dumping image configuration. This is cache availability, not a remote registry/Contract URL check, image vulnerability scan, or API acceptance. The check is bounded and does not start tools, pull images, create Secrets, deploy, adopt resources or clean anything up. Read-only CLI planning can use temporary files/cache; no composition lifecycle state is written.

If a prerequisite fails, correct only that prerequisite and rerun. Prepare the cluster explicitly with `dotx tool check kubernetes --start` if needed. To warm images, run the existing `task testing:pocs:secure-api:live` acceptance task before the talk; it may download images and must finish with verified cleanup. For an occupied port choose another `DOTX_POC10_FORWARD_PORT`. Never delete a foreign namespace to make preflight pass: choose a new name, or recover only a previous run whose ownership is recorded in its receipt. Preflight is a point in time check, not a reservation; the live harness repeats ownership checks.

**What the live result proves**

The harness now seeds a known RDF triple into the disposable backend, then reads that stored value through the authenticated gateway. Acceptance requires exactly one expected literal, rather than a constant `SELECT` expression that would work with an empty database. An authenticated INSERT attempt must be rejected with 403. Seeding is explicit out-of-band test setup in the owned namespace; it is not a claim that the public read-only interface authorizes writes.

The same live task also requires a reversible authorization transition:

1. Authenticated query of the seeded data: HTTP `200` and the expected literal.
2. Temporarily deny reads in this run's OPA module: the same authenticated query must complete with HTTP `403`, not a redirect or transport failure.
3. Restore the original module, read it back, then issue a fresh authenticated query: HTTP `200` with the same stored value.

The console prints these three stages. The redacted `result.json` contains `policyTransition.httpStatuses: [200, 403, 200]`, `restored: true` and `storedDataUnchanged: true` only after every assertion succeeds. The inspection pause happens after restoration, so the displayed composition remains usable.

Policy administration is an explicit operator-side demonstration action using OPA's Policy API, not a sixth Anchor wire, automatic remediation, or a production policy-management design. The harness requires ownership of the applied namespace and an exact match to the reviewed policy before mutation. A failed or ambiguous update triggers a restoration attempt; any failure still invokes the parent harness's owned-resource cleanup. It never changes another namespace's policy. In production, OPA recommends bundles for policy distribution; this local demonstration uses the API to make one bounded, observable transition.

These arrows show logical PROVIDES-to-REQUIRES wires, not HTTP packet direction:

```
Keycloak -- discovery --> APISIX <-- SPARQL -- Oxigraph
OPA ----- decisions ---> APISIX -- protected capability --> Client
Keycloak ---------------- token ------------------------> Client
```

Graph: `DOT-Repository/Compositions/keycloak-apisix-opa-sparql/composition.yaml`. Producer context: PoC10 `e6a755145bbb71551916163209c0be6e9e996f9c`. Five instances exchange through five wires:

- Keycloak discovery to APISIX;
- Oxigraph SPARQL to APISIX;
- OPA policy decisions to APISIX;
- OAuth token endpoint to the client (`TOKEN_URL`);
- protected SPARQL capability to the client (`CAPABILITY_URL`).

Native Secret references supply credentials. Routes/policy are explicit, composition-owned inputs, not automatic translation. OIDC/OAuth Contracts come from canonical bundles. SPARQL and the

legacy OPA workflow remain explicit protocol interfaces, not invented official vendor OpenAPI documents.

```
kubectl config current-context
kubectl get nodes
dotx compose inspect \
  DOT-Repository/Compositions/keycloak-apisix-opa-sparql/composition.yaml \
  --locked --json
task testing:pocs:secure-api:live
```

Expect a fresh namespace and native test Secrets, HTTP `401` for missing and malformed bearer tokens, authenticated POST `403`, and an authenticated GET returning `mare-poc10` from the real SPARQL engine. Every required client input is checked. The harness destroys its namespace and forward; the receipt must say `cleanup=PASSED`. Override `DOTX_POC10_FORWARD_PORT` if 19010 is busy.

| Observation | Expected result |
|---|---|
| Anonymous request to the gateway | HTTP `401` |
| Deliberately malformed public test token | HTTP `401`; redirects/transport errors are not acceptance |
| OAuth client-credentials token | Acquired inside the client pod; never printed |
| Authenticated POST/write attempt | HTTP `403` from the protected route |
| Authenticated stored-data GET | Exactly one literal `mare-poc10` |
| Lifecycle | Recorded status, destroyed owned namespace/forward, `cleanup=PASSED` |

During the presenter pause, inspect the printed private-cache `compose status` command and `result.json`. The five compatible wires are plan/configuration evidence; the four HTTP assertions above are the live end-to-end evidence. They are not five independent native Contract probes or full response-schema validation. Press Enter to clean up and retain the receipt. Re-run from the cleaned state to check repeatability; never treat a previous receipt as a new live exchange.

The invalid token is fixed public test data; real access tokens are never printed. `invalidBearerHttpStatus` in `result.json` records the checked 401. The bounded check ignores local curl configuration and does not follow redirects. It proves rejection of this malformed token, not expiry/revocation handling, issuer/audience isolation, signing-key rotation, or a security certification.

Oxigraph substitutes for GraphDB; a synthetic machine client uses client credentials instead of a password grant. This does not accept the full producer GraphDB/LDAP/Data Security Service stack, telemetry/events or automated response. HTTP, bootstrap defaults and the read-only policy are demo settings, not production security hardening.

## 3. PoC6: explicitly adapted remediation rehearsal

```
CACAO interpreter [adapter] -- MQTT/OpenC2 via Mosquitto --> actuator
```

```
actuator ------------------- remediation call ---------> NWDAF [DOUBLE]
```

Graph: `DOT-Testing/e2e/poc6-cacao-openc2/scenario.yaml`, not a catalogue example. Producer revision: `cd359b25df1b010324869d7e1c755067070e42dd`.

```
task testing:e2e:poc6-cacao-openc2:dry-run
task testing:pocs:remediation:live
```

A guarded interpreter adapter resolves the CACAO external broker URI. The unchanged pinned actuator receives OpenC2 over real Mosquitto/MQTT through the AsyncAPI wire, then calls the OpenAPI remediation endpoint. The exact job UUID must reach the far end. That endpoint is an **NWDAF double**, not real NWDAF remediation. Telemetry, AI and playbook selection remain unaccepted. Override `DOTX_POC6_INTERPRETER_PORT`/`DOTX_POC6_NWDAF_PORT` if 18261/18262 are busy.

## Repeatability, cleanup and recovery

```
task testing:pocs:rehearse
```

This runs each slice twice, followed by injected failure and real SIGINT/SIGTERM cleanup controls. It records the CLI hash, three checkout revisions, tracked/untracked fingerprints, runtime image identities and per-run receipts under `DOT-Testing/e2e/target/`. Changed inputs invalidate the aggregate receipt. Dirty-worktree evidence does not claim those changes were already published.

The manual **Anchor live E2E** workflow's `first-wave` selection runs the same checks and rehearsal driver. Remote infrastructure and queue failures remain separate from local acceptance; no workflow success is claimed before it runs.

Harnesses refuse existing resources, lock fixed identities, use private caches and check owned leftovers. Anonymous Docker volumes are removed; unrelated resources are untouched. PoC6 test images remain cached. `DOTX_E2E_KEEP=1` records `SKIPPED` cleanup and cannot pass acceptance.

If cleanup fails, preserve the printed run directory. Set `XDG_CACHE_HOME` to its `cache` directory and, for Kubernetes runs, set `KUBECONFIG` to that run's private `kubeconfig`. Do not follow a later global context change or share the saved kubeconfig. Stop if the original run files are missing; a new run's state does not authorize recovery. Inspect `dotx compose status <composition>`, then retry `destroy` on the same graph. Inspect only recorded namespaces and planned identities. Never strip finalizers or delete unrelated workloads. Select an unused port instead of killing another service. Review missing Contracts, inputs or stale locks before intentionally regenerating a lock.

## Full-catalogue appendix

`DOT-Testing/interoperability/FIRST-WAVE.md` and `pocs.yaml` retain all eleven PoCs and remaining gaps. `DOT-Repository/CONTRACT-AUDIT.md` separates official and source-derived Contracts, protocol-only interfaces and runtime-only brokers. Planning fixtures, stubs, adapters and doubles remain labelled. Declaration completeness is not runtime acceptance or production certification.

This page publishes through `next`; stable-tag documentation is unchanged. Rehearsal evidence is a

presentation fallback, not a substitute for a claimed current live result or a green remote workflow.

# Docker/Kubernetes four-Contract demo

This `next` development walkthrough reuses three canonical catalogue definitions: the standards-native provider, consumer, and official Mosquitto broker. Named deployment variants select Kubernetes for provider/broker and Docker for consumer. No product descriptor is duplicated.

The composition is named `docker-kubernetes-four-contracts`. Its template lives at `DOT-Testing/composition/docker-kubernetes-four-contracts/scenario.yaml`: this is a synthetic test graph, unlike the reusable catalogue examples `keycloak-apisix-oauth` and `keycloak-prometheus-openapi-planning`.



Provider and consumer are explicitly test-only components, not MARE analytics or vendor products. Real HTTP, GraphQL.js, gRPC-js and Mosquitto exchanges demonstrate interoperability mechanics, not producer acceptance, production security, or automatic protocol conversion. Probe clients use the actual consumer's Docker network namespace; its application process consumes REST data and derives a synthetic review decision. MQTT traffic is generated by the correlated delivery probe, not the GraphQL application. The MQTT runner uses the pinned official `mosquitto_rr` client in MQTT 3.1.1 mode, subscribing and publishing on the same Contract-selected channel. It waits for SUBACK before publishing, ignores retained messages, uses a clean session, and requires the exact uniquely correlated payload. This proves broker delivery, not an application-level request/response implementation.

## What runs where

| Wire | Authoritative Contract | Kubernetes endpoint | Docker-side assertion |
|---|---|---|---|
| `rest-alerts` | OpenAPI 3.1, `GET /alerts` | Provider, NodePort `31080` | HTTP 200 plus the expected fixture alert UUID and `critical` severity |
| `graphql-health` | GraphQL SDL, `Query.health` | Provider, NodePort `31400` | `/data/health` equals `true` |
| `grpc-health` | Protobuf, side-effect-free `AlertService.Check` | Provider, NodePort `31551` | Reflected response has `serving: true` |
| `mqtt-alert-delivery` | AsyncAPI 3.0, MQTT-bound alert channel | Mosquitto, NodePort `31883` | Schema-valid, uniquely correlated QoS 1 delivery |

Anchors select Contract operations. Wires select participants and materialize resolved endpoints into the consumer's declared environment. Each wire declares `EXTERNAL_GATEWAY`: the selected DOT declares the NodePort Service, and Minikube provides the reachable node address. Docker normally isolates its bridge networks: the harness explicitly attaches only its disposable consumer to the existing `minikube` Docker network. Cross-runtime probes use `--network container:<consumer>` so they exercise that actual consumer route, not a different runner's network. The attachment is external to `dotx` state, and consumer destruction removes it; the harness verifies its removal. No

host listener, firewall change, tunnel, or Minikube-container modification is made.

The REST provider returns one deterministic, read-only synthetic alert: `a4a70000-0000-4000-8000-000000000099`, severity `critical`. The native `probe.openapi.expect` assertion compares the complete JSON array, requiring a declared successful status, JSON media type, and the expected identity/severity. Use the installed development CLI; the released `v0.2.5` binary predates this feature. A supplementary independent application assertion reads `OPENAPI_YAML_URL` from the deployed consumer and fetches it from that consumer's actual network namespace. It requires HTTP 200, JSON content type, exactly one alert, and the expected identity/severity. Redirects, wrong data, oversized bodies and a stalled response fail acceptance. The 10-second, 1-MiB-bounded Node.js client uses the already-built provider image; it is a test client, not a new DOT or an implicit GraphQL-to-REST translator. Only its fixed redacted summary is saved as `rest-body.json`; arbitrary response fields are not printed. The independent check also verifies the actual materialized consumer configuration; it is not needed to enable the native CLI response assertion.

## Show application behaviour and a controlled failure

The Docker consumer now runs a small synthetic application, not `sleep`. It repeatedly fetches the wire-materialized `OPENAPI_YAML_URL`, validates the bounded response, and derives `REVIEW_REQUIRED` when any alert is critical; otherwise it derives `NO_CRITICAL_ALERTS`. It does not hard-code the demo UUID or emit a canned decision. Its owner-only, atomically replaced receipt lives inside the disposable consumer at `/tmp/dotx-consumer-result.json`. No new public endpoint or duplicate DOT definition is introduced.

The normal live and presenter commands automatically rehearse this sequence:

1. Observe a fresh consumer receipt: one critical alert, `REVIEW_REQUIRED`.
2. Change only the owned synthetic provider's test-control file to serve severity `low`. Its API still returns HTTP 200 and schema-valid JSON, so a readiness check alone would not expose the changed application behaviour.
3. Require `dotx compose probe` to exit 1 with recorded HTTP 200 and `RESPONSE_MISMATCH`, not a timeout, transport error, or arbitrary failure. Independently observe the running consumer change to `NO_CRITICAL_ALERTS`.
4. Restore the critical fixture; require a successful native assertion and a newer consumer receipt with `REVIEW_REQUIRED` before presenting results.

These steps run automatically through the existing live and presenter commands below; no separate control command or second demonstration is required.

The file `/tmp/dotx-rest-severity` is an explicitly test-only fault switch in the disposable provider container. The harness controls it only after owning the run's namespace, restores it before accepting the negative control, and destroys the owned composition on any unexpected failure. It never changes a real vendor DOT, Kubernetes context, or unrelated workload. `low` is valid for the Contract's string field; the rejection proves the reviewed composition's value assertion, not full response-schema validation or production policy.

Evidence includes `consumer-before.json`, `consumer-negative.json`, `consumer.json`, `rest-negative-status.txt` and `response-recovery.json`. Increasing receipt sequence numbers prove observation

after each transition. `result.json` includes the recovered consumer decision and negative-control summary. Interrupted or failed checks cannot publish a successful result. The application and fixtures are synthetic: `REVIEW_REQUIRED` is a demonstration decision, not real security analytics, mitigation, or producer acceptance.

## Prepare before presenting

Use the full checkout in Contributor guide, and run commands from its root. Requirements: Linux amd64, Go and Task to install this checkout's CLI, Git, Bash, jq, curl, GNU coreutils, `flock` (util-linux), Docker with Compose, kubectl, and a running local Minikube cluster using its Docker driver and `minikube` Docker network. The Kubernetes context must be `minikube`. The deterministic demo checks also require ShellCheck and Python 3. On Ubuntu, install the `shellcheck` package before running `task testing:demo:checks`. The introductory Prometheus quickstart still needs no Go or Kubernetes.

```
git submodule update --init --recursive
task ci:prepare
git rev-parse HEAD
git -C DOT-CLI rev-parse HEAD
git -C DOT-Repository rev-parse HEAD
go version -m "$(command -v dotx)"
docker info --format '{{.ServerVersion}}'
docker compose version
kubectl config current-context
kubectl get nodes --request-timeout=10s
minikube ip
```

`dotx 0.2.4` alone is insufficient: this example uses post-release development changes. Install the pinned submodule and keep the checkout fixed. Evidence records revisions, CLI binary SHA-256, root/CLI/catalogue working-tree patch hashes, and provider/consumer image IDs. Non-empty patches mean the commit alone does not describe the tested checkout. Broker/probe images use digest references; the test images are locally built from pinned Dockerfiles/dependency locks, not claimed published immutable images. The provider uses `imagePullPolicy: Never` after explicit loading; the harness verifies actual running image IDs. Rehearse with network access for metadata/dependency/image downloads; caching does not promise fully offline execution.

After a successful live rehearsal has built/loaded the fixture images and cleaned up, run the read-only check for this route:

```
task testing:demo:preflight DEMO=four-contracts
```

It checks the exact installed CLI revision, clean CLI/catalogue checkouts, offline Contract validation and locked plan, the Ready Linux-amd64 Minikube node, selected cached images, Docker identity/network conflicts, and existing Services occupying the four selected NodePorts. It does not build/pull images, reserve ports, attach networks, deploy anything or prove gateway reachability. The live acceptance below remains necessary. Use `DEMO=monitoring` for the node-exporter/Prometheus route, `DEMO=portability` or `DEMO=upgrade` for those lanes, or omit `DEMO` to check all five presenter routes.



Use an isolated demo machine/network. NodePorts are plaintext; MQTT permits anonymous access and disables persistence. Do not expose them to untrusted networks. These are demo defaults, not production authentication, TLS, retention, or availability settings. Never delete an unrelated namespace or Service.

## Inspect without deploying

```
exec bash DOT-Testing/e2e/multi-runtime-demo/test.sh --dry-run
```

This resolves three DOTs/four wires, writes and verifies a lock, checks both platforms and all four standards, and renders deployment actions in a private directory under `DOT-Testing/e2e/target/multi-runtime-demo/`. The template uses documentation-only address `192.0.2.1`; never manually apply that template.

## Run the working example

### Presenter mode: inspect before cleanup

```
task testing:demo:docker-kubernetes-four-contracts:present
```

Run in a real interactive terminal. The plan-derived Mermaid graph is laid out by the Go `mermaid-ascii` library embedded in `dotx`: Unicode boxes, cyan Kubernetes labels, magenta Docker labels and amber connection IDs. Parallel wires share a labelled arrow and each retains its own legend entry. The node inventory shows selected runtimes/deployments/versions; a `T4` link identifies the MQTT transport endpoint, not another Anchor wire or an application publisher. Arrows mean logical PROVIDES-to-REQUIRES relationships, not packet direction. The graph is printed again after all four real probes and recorded status checks pass.

Only then does the harness pause. Use the printed, shell-quoted recovery/status command in a second terminal to inspect the exact live composition. `plan.json`, `topology.txt`, `topology.mmd`, probe logs, `status.txt`, `results.txt` and `result.json` remain in the printed private run directory. The result summarizes checked fixture assertions, not real MARE analytics. `rest-body.json` holds the separate REST data assertion. Presenter mode shows concise numbered progress and an aligned results table with separate Contract, protocol, observed assertion and outcome columns. Each row requires both completed probe and recorded status evidence. Detailed build/deployment/cleanup output remains in owner-only `details.log`; a failure prints the original failing stage, exit code and its log location. Do not share raw private logs. Normal live acceptance keeps its verbose output. Both normal and presenter runs save `result.json` and `results.txt`; summaries are published only after all probes, the REST data assertion and recorded status checks pass. A failed recheck removes earlier result summaries rather than leaving stale green output. Cleanup is reported separately in `evidence.json`; an assertion summary alone never proves successful cleanup. See the audience FAQ for the model's value, reuse, standards and evidence boundaries. The graph deliberately omits endpoint hosts, configuration and credential values. `topology.mmd` can also be rendered in Antora. `NO_COLOR=1` disables colours; `DOTX_TOPOLOGY_COLOR=auto|always|never` controls them (`NO_COLOR` or `TERM=dumb` takes precedence). `DOTX_TOPOLOGY_WIDTH=80` sets a 40-240-column target; narrower screens use

vertical/compact layouts. No connection may disappear to fit. The saved `topology.txt` is always plain text. Rendering needs only the `dotx` binary: no Python, Node, browser or separate `mermaid-ascii` executable. The Go module pins the upstream 1.6.1 source revision and verifies dependencies through `go.sum`; no renderer is downloaded at runtime.

```
# Read-only topology preview; Docker/Kubernetes need not be running.
dotx compose topology DOT-Testing/composition/docker-kubernetes-four-
contracts/scenario.yaml --locked
dotx compose topology DOT-Testing/composition/docker-kubernetes-four-
contracts/scenario.yaml --locked --color always --width 80
dotx compose topology DOT-Testing/composition/docker-kubernetes-four-
contracts/scenario.yaml --locked --format mermaid
```

Press Enter to clean up normally. Ctrl-C exits 130, SIGTERM 143, EOF 1 and timeout 124, all through the existing ownership-checked cleanup path. Inspection defaults to 600 seconds; set `DOTX_POC_PRESENT_TIMEOUT` to an integer from 1 to 1800 to change it. Headless presenter launches and `DOTX_E2E_KEEP=1` are rejected before runtime mutation. CI and the normal live task never pause, even when presenter environment variables are inherited.

`task testing:demo:present:rehearse` drives all five terminal outcomes against the real composition, verifies status while resources are still running, and requires verified cleanup. It is an acceptance preparation task, not the talk.

#### Noninteractive acceptance

```
exec bash DOT-Testing/e2e/multi-runtime-demo/test.sh --live
```

Equivalent Task: `task testing:demo:docker-kubernetes-four-contracts:live`. The harness locks the real Minikube address, builds/loads images, creates an isolated namespace and state cache, deploys through `dotx compose apply --locked`, checks readiness, and probes each wire with a 45-second timeout. It verifies recorded status, destroys the composition, removes its namespace, and checks for owned Docker leftovers.

Expect `PASSED` lines for `rest-alerts`, `graphql-health`, `grpc-health`, and `mqtt-alert-delivery`, plus exit zero and `cleanup=PASSED`. Inspect the printed run directory's `plan.json`, `render.txt`, `status.txt`, and `evidence.json` to explain the locked plan, endpoint materialization and observations. Evidence JSON contains no passwords/tokens or response bodies. Each run retains its evidence; the top-level `evidence.json` is only the latest result, possibly an injected failure.

### Rehearse the complete presentation

Present Prometheus first, authenticated Keycloak/APISIX/OAuth second, then the cross-runtime standards example:

```
task testing:demo:dry-run
task testing:demo:mqtt-suback:live
```

```
task testing:demo:rehearse
```

The full rehearsal now repeats seven live paths twice: quickstart, monitoring, Prometheus portability, secure API, OPA upgrade/rollback, OAuth and this fixture. Use the integrated ecosystem route for explicit preparation, preflight and evidence interpretation. The quickstart covers default/overridden ports and failure cleanup. OAuth verifies HTTP 401 without a token, token acquisition, protected access, native Secret references, status, and cleanup. The separate MQTT regression uses the actual pinned client and a loopback-only fault-injection peer: delayed, denied, or absent SUBACK, stale retained messages, and incorrect correlated payloads must behave correctly. This peer is not a replacement for the full composition's real Mosquitto acceptance. Its ephemeral credentials and Keycloak development mode are not production defaults. The final mixed-runtime run deliberately fails after deployment: exit 42 and successful cleanup are required. Two further live runs receive actual SIGINT and SIGTERM after deployment and network attachment. They must exit 130/143, record cleanup success, and pass independent namespace/container/network checks. Run those checks separately with `task testing:demo:interruption-cleanup`.

The rehearsal prints one per-run `summary.json` under `DOT-Testing/e2e/target/rehearsal/run-*/`, with a separate log for every step, exit codes, durations, and before/after checkout and CLI fingerprints. Tracked patches and untracked-file content fingerprints are recorded for all three checkouts; no source or credential contents are copied into the receipt. Changed inputs during rehearsal fail acceptance. The convenience `latest.json` points to that per-run receipt, including failures; it is not a green badge. A dirty checkout remains explicitly labelled by its recorded status.

This is pre-presentation acceptance, not the short live talk itself. A per-user live-run lock rejects overlapping instances before deployment, including across checkouts using the same runtime/cache environment. `DOTX_DEMO_LOCK_DIR` is an override for isolated lock tests only; do not override it to run concurrent copies of this composition.

## Recover safely

Normal failure, Ctrl-C and termination attempt cleanup. Do not use `kill -9` or delete state while cleanup runs. After interruption, use the exact printed path:

```
RUN_DIR=/absolute/path/printed/by/the/failed/run
(
  test -f "$RUN_DIR/kubeconfig" && test -f "$RUN_DIR/scenario.yaml" || exit 1
  XDG_CACHE_HOME="$RUN_DIR/cache" KUBECONFIG="$RUN_DIR/kubeconfig" \
    dotx compose status "$RUN_DIR/scenario.yaml"
)
```

Keep the saved kubeconfig private; it contains credentials. These commands use the run's frozen Kubernetes context, not whichever context is currently selected in another terminal. If either required file is missing, stop: do not substitute the default cache/context or guess a namespace. Execute the commands one at a time and stop on an unexpected error.

Only after the original harness has stopped and ownership is confirmed, request cleanup

separately (using the same RUN_DIR):

```
(
  test -f "$RUN_DIR/kubeconfig" && test -f "$RUN_DIR/scenario.yaml" || exit 1
  XDG_CACHE_HOME="$RUN_DIR/cache" KUBECONFIG="$RUN_DIR/kubeconfig" \
    dotx compose destroy "$RUN_DIR/scenario.yaml"
)
```

Inspect the namespace in that run's render.txt; delete only that test-owned namespace if it remains. An apply failure may already have rolled back with no applied state; independently verify resource absence. Never prune Docker, remove Minikube, or delete unrelated volumes. Find the owner of an occupied NodePort before proceeding. Fix Docker user permissions before rerunning. Gateway timeouts require checking node IP, routing/firewall, readiness and Services; planning does not prove reachability. Contract download failures require repository access or verified bundle-local assets, never bypassing SHA-256 checks.

#### After a manually cancelled GitHub job

The live workflow queues subsequent pushes rather than cancelling a rehearsal mid-cleanup. Manual cancellation or loss of the runner can still interrupt teardown. An occupied-name/network preflight failure is not a request to delete whatever was found, and rerunning the job alone cannot repair those leftovers.

First confirm that no job or local demo still owns the resources. Correlate the cancelled attempt's private run path and timestamps with the container's exact name, image and dotx.runtime.identity label, the dedicated network's dotx.composition.id label and attached containers, and the namespace's workloads. Inspect mounts before considering removal. If the runner checkout no longer contains the run's state, do not use a new run's cache for recovery: an operator must verify each exact leftover independently. Remove only confirmed test-owned resources, verify their absence, and rerun the failed job on the fixed revision. Leave the shared Minikube network/container, cached images, unrelated volumes and namespace finalizers untouched. Successful recovery is not successful live acceptance; the subsequent workflow must still finish its probes and cleanup.

## Catalogue readiness appendix

```
task testing:contracts:readiness
jq '.dots | group_by(.status) | map({status: .[0].status, count: length})' \
  DOT-Testing/e2e/target/catalogue-contract-readiness.json
```

Audited snapshot after product-root consolidation: 67 valid definitions, 27 strict interface-profile passes and 40 gaps. Removing duplicate demo products reduces the definition/pass counts; it does not resolve the remaining gaps. Use the generated report for the actual checkout. Broker-only and negative fixtures do not necessarily need a business API. Official/source-derived Contracts, source-adapted PoCs, planning fixtures, stubs and producer doubles remain distinct evidence levels. See catalogue CONTRACT-AUDIT.md and Verification and examples. This demonstration does not promote the entire catalogue to production-ready.

# Stateless upgrades and recovery

This optional technical-demo lane demonstrates native `dotx compose upgrade` with the existing external OPA DOT's Kubernetes deployment. The OPA image and DOT UUID/version stay unchanged; the **composition version and policy input** change. This is a configuration revision, not an OPA software-version upgrade. There are no copied product descriptors, persistent volumes or application-data migrations in this example.

The focused test graph lives under `DOT-Testing/composition/opa-policy-upgrade/`, not the reusable production catalogue: one revision deliberately contains invalid Rego. Its three composition documents and locks resolve the same canonical `DOT-Repository/DOTs/external/OPA-DOT` bundle. This lane has one DOT and an exposed Anchor, not multiple wired DOTs; use the four-Contract and secure-API lanes for those claims.

## Prerequisites and preview

Use the DOT-Design checkout root on Linux amd64, with its exact pinned CLI installed (`task cli:install`), Task, Bash, jq, curl, GNU coreutils, `ss`, Docker, kubectl and a Ready local Linux-amd64 Minikube node. Select context `minikube`. Building that pinned CLI requires Go; follow the contributor installation instructions. Go is not needed by the installed binary during this live lane. The canonical OPA image is digest-pinned; Kubernetes must have it cached or be able to obtain it. Use an isolated demo cluster: this OPA API intentionally has no production authentication policy and must not be exposed publicly.

```
task testing:demo:upgrade:dry-run
```

This checks all three locks and renders their native deployment commands without deploying anything. It does not parse Rego or prove upgrade/rollback acceptance: the invalid policy is intentionally a runtime-negative fixture. The native upgrade diff requires an applied baseline, so the live lane runs it after apply.

## Run, inspect, and clean up

```
# Optional overrides; keep them in the same terminal.
export DOTX_UPGRADE_PORT=19081
export DOTX_UPGRADE_NAMESPACE=dotx-opa-upgrade-presentation
task testing:demo:upgrade:present
```

Choose a fresh namespace and a free port; the harness refuses existing resources and does not adopt them. `DOTX_UPGRADE_ROLLOUT_TIMEOUT` defaults to `90s` per Kubernetes rollout. Allow a few minutes: the broken revision intentionally waits for a rollout failure before native recovery.

The task automatically executes this sequence:

| Stage | Native lifecycle operation | Required observation |
|---|---|---|
| Baseline `1.0.0` | Locked `compose apply` | A viewer's GET authorization decision is HTTP `200`, `result: true`. |
| Review | `compose upgrade --dry-run --json` | Exactly one `UPDATE` for the existing OPA instance, with different input digests. |
| Target `1.1.0` | Locked `compose upgrade` from `1.0.0` and its exact lock | Viewer decision becomes `false`; operator decision remains `true`. |
| Broken target `1.2.0` | Locked `compose upgrade` from `1.1.0` | The exact negative fixture fails parsing with the running OPA binary; its deployment fails rollout and the native command reports automatic rollback. |
| Recovered `1.1.0` | Native status plus fresh API requests | Last-known-good version remains `1.1.0`, no upgrade transaction is pending, viewer remains denied and operator remains allowed. |

OPA's decision API returns HTTP `200` for both boolean decisions; `false` is an authorization decision, not HTTP `403`. In the separate secure-API composition, APISIX enforces that decision and returns HTTP `403` to its caller.

**Say:** "A valid DOT and lock do not guarantee valid application configuration. The runtime rejected this policy, and the CLI restored the last verified configuration before reporting failure. We then tested the recovered service."

The normal lane calls native CLI commands directly. It does not implement a second upgrade or rollback engine. Recovery is to the last successful `1.1.0` revision, not an undo of the previously successful promotion from `1.0.0`. The baseline and target locks are required for both changes.

After successful recovery, presenter mode holds the running OPA instance for inspection. It prints the private run directory and a status command using that run's cache and frozen kubeconfig. Inspect `upgrade-diff.json`, redacted `result.json`, API decision receipts and status snapshots. The saved `failed-upgrade.log` contains the expected failed rollout and native recovery; that failure is not silently treated as a successful upgrade. Routine native apply/upgrade output stays in private `apply.log`/`upgrade.log`; the terminal shows ordered progress stages and a final observed-decision table.

Press Enter to clean up. EOF, Ctrl-C and the bounded presenter timeout also invoke owned-resource cleanup. Successful acceptance requires a final `cleanup=PASSED` receipt. For noninteractive rehearsal use:

```
task testing:demo:upgrade:live
```

With Python 3 installed, `task testing:demo:upgrade:rehearse` uses the existing terminal-driven acceptance driver to verify the pause, inspect native status, send Enter and require verified cleanup. It records a summary and before/after checkout fingerprints; it fails if inputs change during rehearsal.

### Boundaries and recovery

This proves one stateless native-Kubernetes configuration update and automatic recovery, not database rollback, zero downtime, all deployment adapters, policy distribution security or production certification. The exposed API Contract is the bundle's ecosystem-derived PoC1 normalization, not an official comprehensive OPA OpenAPI document. The live checks verify the actual boolean decisions.

Current upgrade admission rejects changed stateful deployments and changed Helm/Vagrant deployments whose storage safety cannot be established. A pending transaction after interruption requires the native `--resume` or `--rollback` route documented in Composition lifecycle. Do not rewrite its state file, delete unrelated resources or bypass these checks.

If cleanup is interrupted, inspect the printed run's evidence and use **both** its `XDG_CACHE_HOME` and `KUBECONFIG` for status/destroy. Never substitute a new cache or a changed global cluster context. Do not force namespace finalizers.

The existing **Anchor live E2E** workflow offers an explicit `opa-policy-upgrade` selection with the same harness and private evidence upload. The default PoC `all` matrix remains the established twelve scenarios; this optional technical lane is selected separately. Dry-run CI checks the fixtures through the existing presentation-validation job.

## One DOT, two runtimes

This optional presenter lane answers: "Can we reuse the same DOT without copying its product definition when the deployment runtime changes?"

The catalogue composition `prometheus-docker-kubernetes-portability` selects the existing `configuration-docker-compose` and `configuration-kubernetes-manifest` profiles of the **same** Prometheus DOT: UUID `7af6bbe8-fb09-52a7-a0fd-2ee35d6966f9`, version `0.2.0`. Both resolve its official OpenAPI Contract, use the same digest-pinned upstream image, and declare the same default self-scrape configuration and `GOGC=75`. No product descriptors, Contracts or default configurations are copied for this demonstration.



These are two independent servers with **zero wires**. This demonstrates deployment and configuration portability, not replication, data migration, or cross-runtime communication. Use the four-Contract demo for actual Docker-to-Kubernetes exchanges, and the secure-API composition for authentication and authorization between several products.

### Prepare and preview

Run from the DOT-Design checkout root on Linux amd64. Install its exact pinned CLI using `task cli:install`; Go is needed for installation, not by the running binary. Follow the contributor checkout/tool guidance. The released `v0.2.5` binary predates this lane's cross-runtime repetition and restrictive-umask config-mode fixes; the version string alone is insufficient. Requirements: Task, Git, Bash, jq, curl, `ss`, GNU coreutils, Docker with Compose, kubectl and a Ready Linux-amd64 Minikube node with Docker runtime. Select context `minikube`. No Java, Kubernetes cluster

provisioning or credentials are needed once those local prerequisites are installed and ready.

```
task testing:demo:portability:dry-run
task testing:demo:portability:prepare
```

The first command validates the reviewed lock and renders the two selected deployments without deploying anything. Temporary default-config files are materialized in an owned preview cache and removed on exit. The second command is **explicit preparation**, not read-only preflight: it pulls only the selected digest-pinned image into Docker and the Minikube node cache. It does not deploy the composition. Preparation requires registry access and does not imply live API acceptance or universally offline operation.

## Present the comparison

```
# Optional: choose a fresh namespace and unused Kubernetes forwarding port.
export DOTX_PORTABILITY_NAMESPACE=dotx-prometheus-portability-presentation
export DOTX_PORTABILITY_KUBERNETES_PORT=19083
task testing:demo:portability:present
```

The Compose profile's declared localhost port is `18919`. Both host ports must be unused. Never delete an unknown workload to free them. The task refuses existing namespaces and Compose project resources; it does not adopt them.

The task then:

1. Resolves the same canonical DOT/Contract for both instances and requires the same immutable image pin. It reads the image's own `--version` output in a disposable, network-disabled container to establish the expected API version.
2. Applies both native profiles through `dotx compose apply` and records status.
3. Verifies the actual container/pod image identities, bounded readiness, official build-information version, runtime `GOGC`, default configuration and a real healthy self-scrape on **both** servers.
4. Displays the locked topology, a live comparison table and inspection URLs, then holds the verified services for bounded inspection.
5. Destroys only this run's resources and verifies no Compose containers, networks, volumes, namespace, forwards or composition state remain.

The default selected image currently reports Prometheus `3.15.0`; the assertion derives the version from its pinned bytes rather than assuming that a version label or two matching server responses prove provenance. The runtime receipts record actual image IDs and observation timestamps. The terminal table reports `GOGC=75` and `up=1` for each independently verified server.

**Say:** "The DOT identity and API Contract stay the same. The selected deployment mechanism, resource identities and endpoint change. That is reuse of one product definition, not two products that happen to have similar names."

While paused, open the displayed localhost URLs at `/targets`, or inspect the redacted `result.json`

and each runtime's receipt in the printed private run directory. The printed native status command includes the run's cache and frozen kubeconfig; keep both when inspecting or recovering that run.

Press Enter to clean up. EOF, Ctrl-C and presenter timeout also invoke cleanup, but return a nonzero status. A successful demonstration requires the final `cleanup=PASSED` receipt. Native deployment details stay in `apply.log`; early Compose process failures retain bounded startup diagnostics before cleanup.

For noninteractive acceptance and an automated inspection/Enter rehearsal:

```
task testing:demo:portability:live
# Requires Python 3; also exercises a Kubernetes port override.
task testing:demo:portability:rehearse
```

## Storage and acceptance boundaries

Data is disposable. Native Compose destroy removes this test project's named TSDB volume. The selected Kubernetes profile has no persistent volume. This comparison does not move history between servers, prove persistence/backup semantics or certify every Prometheus API and deployment variant.

The CLI admits one native Kubernetes instance of this DOT alongside isolated Docker instances. It still rejects two native Kubernetes instances with fixed object names, including when a Docker instance separates them in the document. Do not bypass that guard or infer arbitrary same-namespace object isolation.

If interrupted, inspect the receipt and use the recorded cache/kubeconfig for native status/destroy on the same composition. Never substitute a new run's cache, delete unrelated resources or strip namespace finalizers. The **Anchor live E2E** workflow offers this lane explicitly as `prometheus-portability`; dry-run CI exercises the same script. Runtime and fixture acceptance remain distinct from production certification.

# Contributor guide

This page covers building DOT-Design and running its shared local/GitHub gates. To use an existing DOT without installing development toolchains, start with Getting started.

## Prerequisites

| Tool | Required for | Project baseline |
|---|---|---|
| Java and Maven | `DOT-Domain`, schema generation, and Java tests | Java 27 (`27.0.0+35-open` in `.sdkmanrc`); Maven 3.10.0 from `PATH` |
| Go | Building and testing `dotx` | Go 1.27.2 |
| Task | Repository command aliases | Optional but recommended |
| Docker, Kubernetes tools, Helm, or Vagrant with libvirt/KVM | Only the matching deployment adapters and live scenarios | Install only what you intend to execute |
| Antora, the Antora PDF extension, Asciidoctor PDF, and Mermaid CLI | Building the HTML/PDF documentation site and Mermaid diagrams | CI uses Node.js 26 and Ruby 4.0; `task docs:check` reports the locally installed tool versions; see Build the documentation |

## Check out the repository

```
git clone https://github.com/MARE-EU-Project/DOT-Design.git
cd DOT-Design
git submodule update --init --recursive
```

The source repository and its two submodules may require access granted by the MARE-EU-Project organization. `DOT-CLI` and `DOT-Repository` remain separately cloneable; the generated Schema and high-level Skill classifications are tracked inside `DOT-Repository` and require no additional checkout.

## Verify the domain model

The domain module uses plain JUnit tests and does not start a service:

```
mvn -pl DOT-Domain test
```

Run the complete Java build with:

```
mvn clean install
```

```
# equivalent Task alias
task mci
```

The build generates CycloneDX JSON and XML software bills of materials during the Maven `verify` phase.

# Install and test dotx

`DOT-CLI` is an independent Go module and is not part of the Maven reactor:

DOT-Design `VERSION`, its Maven artifacts and the pinned `DOT-CLI/VERSION` share the same version number. Run `task testing:version:alignment` to verify the checkout. Independent repositories and publication jobs do not imply independent version numbers; source revisions still identify unreleased changes. Antora keeps current development content under `Next`, preserving stable tags.

Install the standalone release with Go 1.27 or later:

```
go install github.com/MARE-EU-Project/DOT-CLI/cmd/dotx@v0.2.5
dotx version
```

The GitHub release also provides a checksummed Linux-amd64 binary, build metadata, and a manifest that pins the exact DOT-CLI source plus the DOT-Design and DOT-Repository acceptance revisions. For source development or the complete cross-repository suite, build the checked-out submodule instead:

```
task cli:build
task cli:test
task cli:install
dotx --version
```

`task cli:install` installs the pinned checkout into `GOBIN` or `GOPATH/bin` and installs Bash completion. Keep that directory on `PATH`; project tasks, test harnesses, examples, and CI all invoke the same `dotx` command rather than a repository-local binary.

# Run GitHub workflow gates locally

GitHub Actions and local pre-push validation share the root Taskfile test bodies. Actions still provisions clean toolchains and publishes artifacts, but the build, validation, conformance, coverage, runtime, and documentation commands are not CI-only shell fragments.

```
task ci:prepare
task ci:deterministic
task ci:compatibility-matrix
```

`ci:prepare` installs the pinned DOT-CLI checkout and verifies that `dotx version` matches the root

`VERSION`. `ci:deterministic` runs every deterministic non-live release workflow gate, including Java and Go tests, schema drift, official Contract validation, repository and Anchor conformance, Composition v1, lifecycle recovery, upgrade, all PoC dry-runs, external-DOT rendering, coverage, Antora, and PDF generation. Individual equivalents are available as `ci:build`, `ci:contract-conformance`, `ci:repository-conformance`, `ci:coverage`, and `ci:docs`.

On a Linux host with all runtime prerequisites, run the mutable workflows too:

```
task ci:live
task ci:all
```

The live gate requires Docker, Minikube, Kubernetes, Helm, Wasmtime, Vagrant/libvirt, writable KVM access, Rust 1.98.1 with the `wasm32-wasip2` target, and network access for the first official WASI HTTP sample build. The shared task fetches the exact pinned `sample-wasi-http-rust` revision into an ignored test cache, builds it with Cargo, and passes it to the runtime test, so local and GitHub executions cannot silently omit the WASI HTTP assertions. Set `WASI_HTTP_COMPONENT` only to reuse an already built component. The live gate performs runtime mutation and cleanup; `ci:deterministic` is the normal pre-push default.

For a Vagrant runtime on Ubuntu, install or reconcile the complete provider boundary and then open a new login session so the `libvirt` and `kvm` groups are active:

```
dotx tool install vagrant
dotx tool check vagrant
```

The installer selects `qemu-kvm` when the distribution provides it and falls back to `qemu-system-x86` on releases such as Ubuntu 26.04. The readiness check also verifies the pinned `vagrant-libvirt` plugin, the `qemu:///system` connection, the default libvirt network, and `/dev/kvm` access.

# Validate a DOT

`dotx validate` reads `dot.json` at the project root, falling back to `dot/dot.json`. The selected file is the unique DOT deployment definition; unreferenced Dockerfiles, Compose files, Kubernetes manifests, Helm charts, and scripts are not inferred.

```
dotx validate DOT-Repository/DOTs/testing/contract-standards/provider \
  --schema DOT-Repository/Schema/schema.json
dotx validate DOT-Repository/DOTs/external/Keycloak-DOT \
  --schema DOT-Repository/Schema/schema.json --profile reusable
dotx inspect DOT-Repository/DOTs/testing/contract-standards/provider --json
```

Base validation checks the JSON Schema and DOT-level Anchor, Contract, service-port, CDI, and sensitive-default relationships. Every declared Contract must either be complete in `dot.json` or resolve to an asset below that DOT's `contracts/` directory. The validator never searches repository-level Contract directories or sibling DOTs. A bundle can therefore be copied and validated independently with an explicit schema path. A URL-backed definition declares one structured

payload:

```
"payload": {
  "url": "https://raw.githubusercontent.com/vendor/project/v1.2.3/openapi.yaml",
  "sha256": "0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef"
}
```

Raw Contract text and the former flat `payloadUrl`/`payloadSha256` shape are rejected. Validation downloads the exact bytes, rejects non-HTTPS or mutable raw-GitHub branch URLs, verifies the digest, parses the declared standard/format, and resolves Anchor selectors. Missing IDs, incomplete declarations, digest mismatches, and unresolved selectors always fail validation. Use `--profile reusable` before publishing a stable reusable DOT. That profile additionally requires at least one pinned Contract, rejects unused Contracts, and requires every deployment to bind each Contract-backed provided Anchor exactly once through a service that declares the same Contract. Explicit contractless Anchors retain base-validation evidence only; they are never silently promoted to Contract-backed interoperability.

For an external product, prefer a complete root Contract declaration pointing to an official versioned vendor raw file. A full Git commit or release-version tag is accepted for `raw.githubusercontent.com`; `main`, `master`, and other mutable branch names are rejected. When no official raw machine-readable Contract exists, do not fabricate vendor provenance: retain a clearly attributed derived/adapter Contract until it can be migrated. Adjacent wrappers keep internal, testing, sample, and composition fixtures self-contained. Shared Contract UUIDs must resolve to identical metadata and payload bytes in every bundle; repository validation rejects conflicts. Use `dotx validate --offline` only when a matching local wrapper is present; offline validation verifies that its identity, metadata, and payload SHA-256 match the URL-backed declaration without contacting the vendor. For private raw GitHub URLs, `dotx` reads `MARE_READ_ONLY_TOKEN`, falling back to `GITHUB_TOKEN`, and sends it only to `raw.githubusercontent.com`; credentials must never appear in `payload.url`. Inspection provides a normalized, machine-readable view without deploying anything.

# Validate and plan a composition

The canonical example connects Keycloak, APISIX, and an OAuth client:

```
dotx compose lock \
  DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml --check

dotx compose validate \
  DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml

dotx compose inspect \
  DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml

dotx compose inspect \
  DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml --json

dotx compose apply \
```

```
  DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml --dry-run
```

All five commands above are non-mutating. `lock --check` verifies the committed content-addressed input graph; `validate` answers whether references and compatibility can be resolved; `plan` shows the exact instances, endpoints, security status, and materializers; `apply --dry-run` renders supported runtime actions without writing composition state.

Read Validation, planning, and lifecycle before a live apply.

# Verify the authoritative repository

Recursively discover and validate every definition. External/internal paths record ownership; Contract-backed external/internal bundles receive the strict reusable profile:

```
dotx validate DOT-Repository
dotx repository inspect \
  d4a70000-0000-4000-8000-000000000001@1.3.0 \
  --repository DOT-Repository --json
```

The same command is the required DOT-Repository GitHub Actions gate for every change below `DOTs/` or `.gitmodules`. The filesystem is authoritative; no separate inventory or validation wrapper is maintained.

Run `task testing:contracts:portability` to copy every bundle into an isolated temporary directory and validate it offline against the explicit catalogue schema. This is also part of `ci:repository-conformance`; it prevents a valid repository checkout from concealing a DOT's missing local Contract asset.

# Run the composition conformance gate

```
task testing:conformance
```

This gate combines Java domain tests, Go tests, repository-wide schema validation, Contract checks, composition validation/planning, and dry-run scenario checks. It does not claim that external partner systems or production networks were exercised.

Run only the versioned composition production profile with:

```
task testing:compose:acceptance
```

This gate writes revision-pinned, secret-free JSON evidence below `DOT-Testing/e2e/target/`.

# Build the documentation

Install the documentation toolchain once:

```
task docs:setup
```

Then generate the Antora HTML site and PDF export:

```
task docs:check
task docs:build
```

Generated artifacts are written under `DOT-Docs/build/site` and are intentionally ignored by Git. To build and serve the site locally:

```
task docs:run
```

Open `http://localhost:8000`. Do not edit files under `DOT-Docs/build`; update the AsciiDoc sources instead.

Mermaid sources are stored as Antora attachments and rendered locally by `task docs:diagrams`. The renderer produces SVG for the HTML site and high-resolution PNG for PDF export; both outputs are committed so a clean checkout can resolve every Antora resource. Each source includes an accessible title and description, and each AsciiDoc image macro supplies alternative text.

`task docs:build` regenerates the diagrams before building either output. The toolchain is version-pinned and does not send architecture content to an external rendering service. Edit the `.mmd` source, never the generated SVG or PNG, then run the full documentation build to verify both renderers.

# Where to continue

- Guided demo for a presentation-ready walkthrough from catalogue validation to cross-standard planning and optional live probes.
- Architecture for system and ownership boundaries.
- DOT Anchors and Composition for the model overview and task-oriented composition guides.
- Verification and examples for evidence levels and MARE PoC scenarios.
- DOT-Domain for Java model details.
- Integration handoff for catalogue and composer implementers.

# Architecture

DOT-Design separates reusable component description from deployment-instance topology. That separation allows a DOT to be authored and validated once, then instantiated in multiple compositions without embedding addresses, credentials, or orchestrator-specific connection objects in the DOT aggregate.

The diagrams use a consistent visual language: blue identifies authored or persistent artifacts, teal identifies validation and resolution activities, amber identifies decisions or incomplete evidence, green identifies successful outcomes, and red identifies rejection. Labels and line styles carry the same meaning when the documentation is printed without color; dashed lines indicate optional, indirect, or evidence-only relationships.

## System context

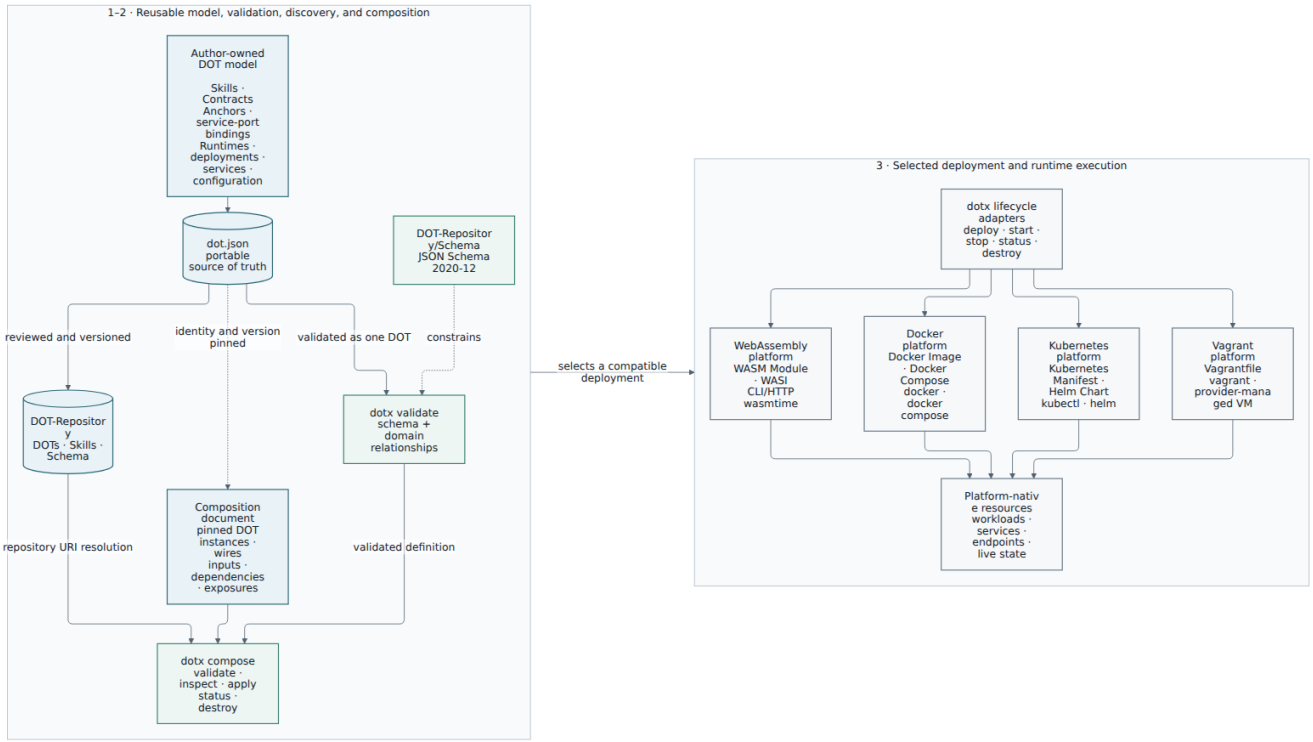



*Figure 3. DOT-Design system context and runtime support*

`DOT-Repository` is the authoritative catalogue of reusable definitions. `dotx` is the reference validation, resolution, deployment, and composition client. Platform-native systems remain responsible for workload execution and live state.

The runtime portion deliberately distinguishes **where** a workload runs from **how** it is deployed. The domain model supports four `PlatformEngine` values and six compatible manifest mappings:

- Docker accepts `DOCKER_IMAGE` and `DOCKER_COMPOSE`;
- Kubernetes accepts `KUBERNETES_MANIFEST` and `HELM_CHART`;
- Vagrant accepts `VAGRANTFILE`; and
- WebAssembly accepts `WASM_MODULE`.

The current dotx lifecycle implements Docker Image, Docker Compose, Kubernetes manifest, Helm chart, Vagrantfile, and WebAssembly execution. The WebAssembly adapter uses Wasmtime with an explicit WASI_CLI or WASI_HTTP world, verifies the module's SHA-256 before execution, and grants only declared guest environment and directory capabilities.

## Source-of-truth boundaries

All portable model definitions originate in DOT-Domain, including the separate Composition aggregate. The existing generation pipeline is Java/Jackson/Jakarta constraints → Victools JSON Schemas → quicktype Go input bindings. DOT-Repository/Schema/ stores the generated artifacts; the CLI embeds the same Composition schema. There is no independent CLI input model or hand-authored production schema to keep in sync. Plans, locks and runtime state are executor representations, not alternative domain definitions.

| Concern | Authoritative source | Explicitly excluded |
|---|---|---|
| DOT identity and supported capability | dot.json: id, name, version, Skills, and Contracts | Composition-local instance names and live state |
| Contract semantics | OpenAPI, AsyncAPI, Protobuf, or GraphQL Contract payload | Repeated protocol, media, schema, or security fields on a contract-backed Anchor |
| Logical composition point | Dot.anchors[] | Hosts, URLs, credentials, placement, health, and policy decisions |
| Deployable endpoint declaration | DeploymentService.ports[] and anchorBindings[] | Allocated addresses, forwarded ports, and certificates |
| Runtime configuration surface | DeploymentService.config | Operator-supplied values already applied to one deployment |
| Multi-DOT topology and readiness | Composition document wires and dependsOn | Changes to either participating dot.json |
| Runtime state | Platform-native resources plus private dotx composition state | Contract or Anchor metadata |

## Repository architecture

```
DOT-Design
 ├──── DOT-Domain       Java model and validation
 ├──── DOT-CLI          independent Go CLI and reference executor
 ├──── DOT-Repository   DOTs, high-level Skills, and generated Schema (submodule)
 ├──── DOT-Testing      conformance, composition, interoperability, and E2E evidence
 └──── DOT-Docs         Antora sources, UI, playbooks, and generated diagrams
```

The Maven build contains the DOT-Domain Java module. DOT-CLI has its own Go module and lifecycle. DOT-Testing is deliberately outside both module hierarchies. DOT-Repository is the only admitted source for valid deployable DOT definitions used by shared tests. It also owns the generated JSON

Schema and the optional high-level Skill classification catalogue, avoiding separate repositories for assets that evolve with the DOT catalogue.

## DOT aggregate

`Dot` is the aggregate root. Its primary relationships are:

```
Dot
 ├──── Skills[]                human-readable capabilities
 ├──── Contracts[]             authoritative interaction definitions
 ├──── Anchors[]               stable logical composition points
 └──── Runtimes[]
     ├──── Platform            where execution occurs
     └──── Deployments[]
         ├──── DeploymentManifest   how deployment occurs
         └──── DeploymentService[]
             ├──── Contract references
             ├──── ServicePort[]
             ├──── AnchorBinding[]
             └──── DeploymentConfig
```

`Skill` and `Contract` are independently identified domain entities. Root `Dot.contracts` carries authoritative Contract declarations (with legacy UUID-only catalogue references accepted during migration), while deployment services continue to reference those Contracts by ID only. Anchors, Runtimes, Deployments, services, bindings, and configuration declarations are embedded values owned by the DOT aggregate.

The class-level `@ValidAnchor` constraint delegates aggregate traversal to `AnchorValidator`. It enforces Anchor identity, Contract membership, selector syntax, and service-port binding consistency without placing cross-reference logic in `Dot`.

## Contract, Anchor, binding, and wire

These four concepts form a resolution chain but have different owners:

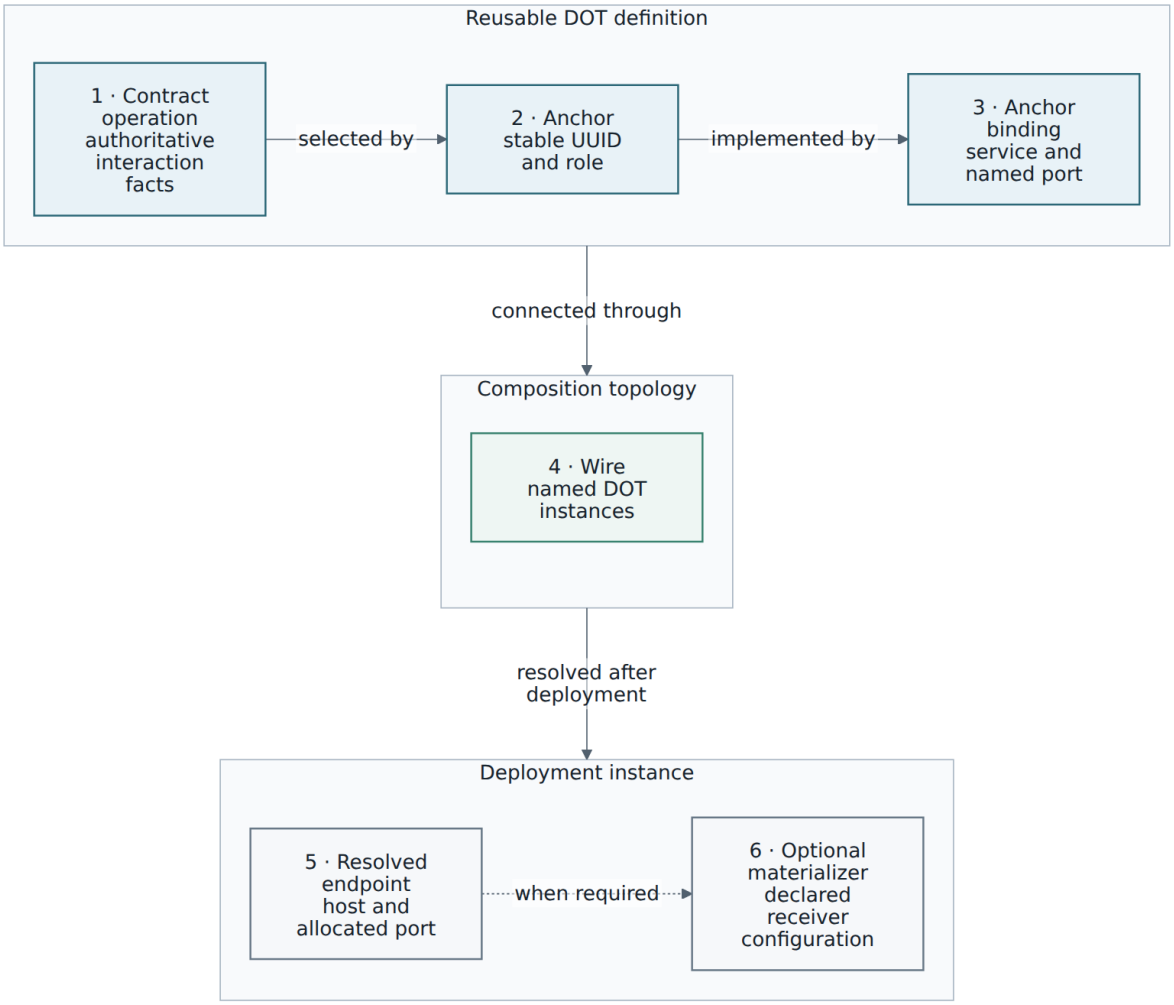



*Figure 4. From an authoritative Contract operation to deployed configuration*

- The **Contract** defines the interaction.
- The **Anchor** identifies one logical participation point.
- The **binding** says which declared service port implements that Anchor.
- The **wire** connects named deployment instances.
- The optional **materializer** writes a resolved value into configuration that the receiving service has already declared.

See DOT Anchors and Composition for the field model, compatibility rules, and examples.

# Runtime and deployment separation

The model separates two orthogonal questions:

| Concept | Field | Answers |
|---|---|---|
| `PlatformEngine` | `Runtime.platform.engine` | Where does the workload run? For example Docker, Kubernetes, Vagrant, or WebAssembly. |

| Concept | Field | Answers |
| --- | --- | --- |
| `DeploymentManifestType` | `Deployment.manifest.type` | How is it deployed? For example Docker image, Kubernetes manifest, Helm chart, Vagrantfile, or WASM module. |

Every manifest type declares its target platform. Runtime validation rejects a deployment whose manifest mechanism does not align with its platform.

`DeploymentService` is the unit that binds a deployed service to Contracts, named ports, Anchors, and supported configuration. Configuration is service-scoped; there is no deployment-level shared configuration surface.

# Contract resolution

The reference resolver maps an Anchor selector to an authoritative Contract element:

```
AnchorContract(id, selector)
        |
        v
load Contract by UUID
        |
        v
standard-specific resolver
        |
        v
identifier + interaction + protocols + media types
          + input/output schema references + security scheme names
```

Resolution is read-only. Derived facts are returned to consumers and never written back into the Anchor. Contractless Anchors already carry their explicit direction, protocols, and media types and therefore have no selected Contract element.

# Composition resolution

The stable Composition v1 executor uses a compile-like pipeline:

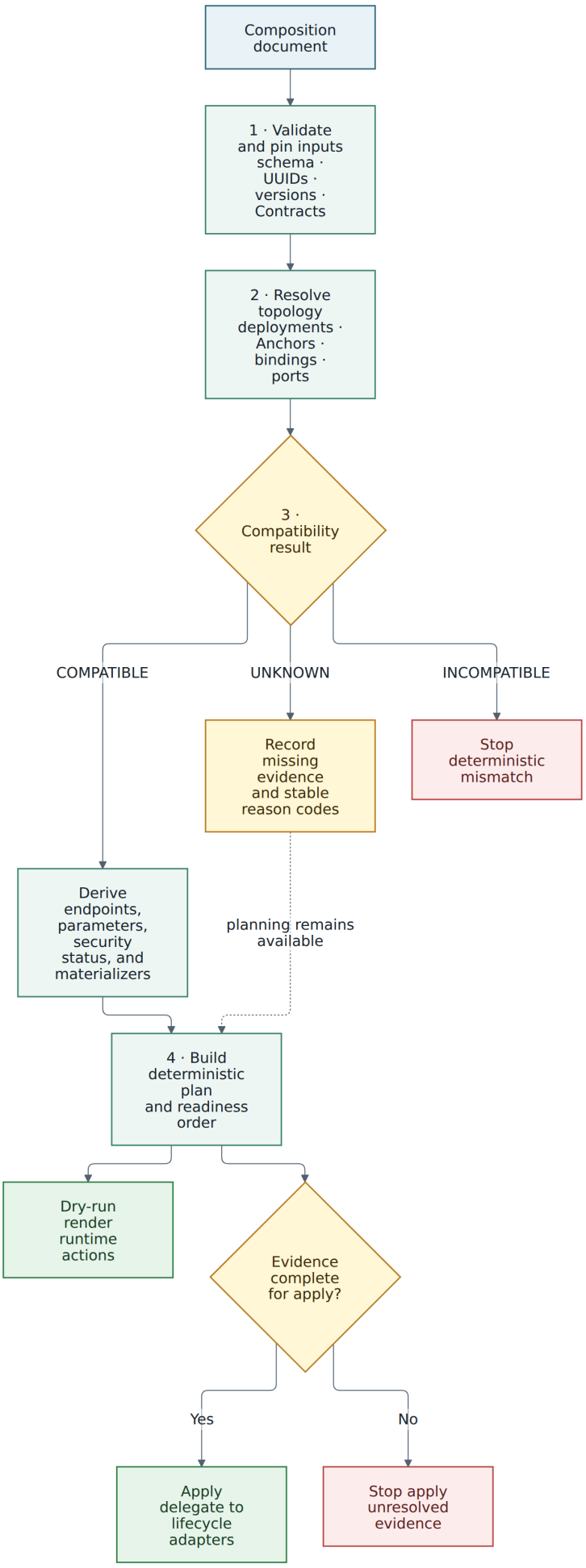



*Figure 5. Fail-closed composition resolution pipeline*

1. Validate the strict Draft 2020-12 composition shape.
2. Load each local DOT source and verify the pinned UUID and version.
3. Load each Contract input and verify its UUID.
4. Select one deployment for every named DOT instance.
5. Resolve Anchor references and service-port bindings.
6. Evaluate three-state compatibility: `COMPATIBLE`, `INCOMPATIBLE`, or `UNKNOWN`.
7. Derive connection endpoints, explicit cross-platform network ownership, parameter values, security status, and materializers.
8. Build a deterministic plan and readiness order.
9. On apply, prepare configuration and delegate workload operations to the existing single-DOT lifecycle adapters.

Planning may retain `UNKNOWN` with reason codes so missing evidence is visible. Apply rejects `UNKNOWN` rather than treating absence of evidence as compatibility.

# Security boundaries

DOT-Design models declarations and explicit grants; it does not infer trust.

- Contract security requirements remain in the Contract.
- Sensitive deployment values are supplied by reference, not embedded in a composition or persisted as resolved values.
- OCI capabilities, host namespaces, volumes, and CDI devices require exact DOT declarations and exact operator grants.
- A configured wire does not prove credential validity, authorization, remote attestation, policy compliance, or network reachability.
- An `EXTERNAL_GATEWAY` binding records operator ownership and `NONE` cleanup; it does not assert that `dotx` provisioned or verified the gateway.
- Cross-domain admission decisions belong to deployment or policy systems.

Active traffic and trust evidence must be collected by scenario-specific probes after deployment.

# Catalogue and API boundaries

`DOT-Repository` versions reusable definitions and its catalogue metadata in Git. `dotx repository` validates and resolves those entries; `dotx compose` provides the executable compatibility reference. A future remote catalogue or API is free to use another language, framework, storage engine, and transport provided it preserves the domain semantics and passes the consumer acceptance profile.

`dotx compose` stores private, local lifecycle state under the user's cache directory. Per-composition locking prevents concurrent local lifecycle commands from racing that state. The current model is therefore explicitly single-host; shared orchestration state remains future work.

# Design principles

- Reuse established Contract standards instead of inventing parallel protocol models.
- Keep stable logical identity separate from runtime location.
- Make adapters, transformations, privileges, and secret access explicit.
- Fail closed on ambiguity and missing security evidence.
- Preserve independently versioned DOT definitions; keep topology in composition.
- Treat generated plans and test evidence according to what they actually prove.

# DOT Anchors and Composition

This guide explains how independently authored Dynamic Open Trust (DOT) components participate in composition by referring to authoritative Contracts rather than duplicating contract-defined facts. It distinguishes the stable DOT Anchor model from the stable, versioned `dotx compose` format and follows the complete path from a logical capability to a deployed connection.

For a technology-neutral consumer boundary, see integration handoff.

## Read this first

Keep these three boundaries intact:

1. A **Contract** describes what can be exchanged.
2. An **Anchor** gives one Contract operation a stable DOT-local identity and role.
3. A **wire** connects deployed DOT instances and remains outside both DOTs.



A structurally valid, deployable DOT is not automatically interoperability-ready. `dotx validate <bundle> --offline --profile interoperable` additionally requires a Contract for every Anchor and a Contract-backed provided Anchor for each exposed service port. The older `reusable` profile preserves mixed evidence during migration. Neither profile discovers undeclared implementation endpoints or replaces live producer acceptance. The catalogue Contract audit records official sources, source-derived internal descriptions, licensing boundaries, and remaining gaps. From DOT-Design, `task testing:contracts:readiness` produces a per-bundle strict-profile report; known gaps remain failures, not an interoperability certification.

If a value changes between installations—for example a host, URL, port, credential, realm, broker, namespace, or device—it does not belong in an Anchor.

| If you want to… | Start with |
|---|---|
| Publish a composable DOT | Authoring two composable DOTs, then Contract-backed Anchors and deployment binding. |
| Connect existing DOTs | Composition documents and DOT instances, direct request/response wires, and materializers. |
| Model messaging through a broker | Brokered wires and path and channel parameters. |
| Handle authentication or secrets | Security and secrets. |
| Operate a composition | Validation, planning, and lifecycle and troubleshooting. |
| Design a new 6G integration type | Anchor boundaries, extension points, and 6G security composition patterns. |

# Release status and supported profile

The DOT-owned Anchor model, contract selectors, service-port bindings, validation, generated schema, repository resolution, and conformance fixtures are release candidates. The separate composition document and its local lifecycle executor implement the stable `dot.mare6g.eu/v1` profile.

The current executable profile supports:

- local DOT sources pinned by UUID and version, and local Contract payloads pinned by UUID, in the composition;
- OpenAPI, AsyncAPI, Protobuf, and GraphQL contract selectors;
- contractless protocol/media declarations when no supported contract exists;
- named deployment selection and service-port bindings;
- explicit environment-variable and config-file materializers;
- explicit externally owned gateway bindings for cross-platform wires, with ownership and no-cleanup semantics recorded in plan, lock, and state;
- Docker Image composition networking and per-instance identity;
- Kubernetes and Helm deployment/readiness through the existing single-DOT lifecycle;
- reference-only process-environment and Kubernetes Secret inputs;
- rollback, private local state, resumable partial apply, status, and reverse-order destroy;
- contract-authorized active probes for safe OpenAPI operations, GraphQL query/mutation/SSE lifecycles, finite side-effect-free unary/client/server/ bidirectional gRPC calls, and correlated AsyncAPI MQTT/NATS/Kafka/AMQP delivery within one Docker network or Kubernetes namespace; and
- fail-closed handling of ambiguity, unknown compatibility, unsupported privilege grants, and unreachable cross-platform materialization.

The authoritative machine-readable boundary is `DOT-Testing/acceptance/composition/v1/profile.json`. Run `task testing:compose:acceptance` to compare the installed CLI help and emitted schema with that profile, verify every committed lock, exercise all twenty-one cross-standard wires and six runtime dry-run adapters, and emit revision-pinned, secret-free JSON evidence.

It is not a general multi-host production orchestrator. Shared state, composition-owned cross-platform gateways or tunnels, external secret-store plugins, multi-scheme authentication, unsafe or unbounded gRPC, AsyncAPI event protocols other than MQTT, NATS, Kafka, and AMQP 0-9-1, AMQPS/AMQP 1.0, and cross-platform probe adapters remain outside this profile. Safe anonymous OpenAPI probes, single-scheme OAuth 2.0 client-credentials probes, GraphQL query/mutation/SSE probes, all four finite side-effect-free gRPC shapes, and AsyncAPI MQTT/NATS/Kafka/AMQP delivery probes within a shared Docker network or Kubernetes namespace are supported.

Single-DOT live lifecycle evidence for a runtime adapter is not automatically multi-DOT composition acceptance. The profile records validation, planning, dry-run, single-DOT live lifecycle,

and composition live lifecycle separately for each adapter.

# The five distinct objects

| Object | Owner | Responsibility |
|---|---|---|
| Contract | DOT author | Authoritative operation, message, payload, protocol, and security description in OpenAPI, AsyncAPI, Protobuf, or GraphQL. |
| Anchor | DOT author | Stable UUID and composition role selecting one contract element, or declaring explicit communication facts when no contract exists. |
| Anchor binding | DOT author | Maps one Anchor UUID to a named port of a deployment service. It does not contain a live address. |
| Wire | Composer/operator | Connects one named DOT instance and Anchor UUID to another. It may select a provider, consumer, or broker service-port endpoint. |
| Materializer | Composer/operator | Explicitly writes a resolved connection value into configuration declared by a participating service. |

## Where each fact lives

| Fact | Source of truth | Referenced by |
|---|---|---|
| API operation, message, schemas, protocol, media type, security requirement | Contract payload | `Anchor.contract.selector` |
| Logical composition point and provider/consumer role | `Dot.anchors[]` | Wire `from` or `to` |
| Service and named port implementing an Anchor | `DeploymentService.anchorBindings[]` | Resolved plan endpoint |
| Configuration the service permits a deployer to supply | `DeploymentService.config` | Deployment input or materializer target |
| Selected DOT version, topology, readiness, parameter values, and secret references | Composition document | `dotx compose` inspection and lifecycle |
| Concrete runtime address and port | Explicit composition network binding or resolved deployment plan | Materializer or exposed-Anchor forwarding |

| Fact | Source of truth | Referenced by |
| --- | --- | --- |
| Live reachability, authentication result, and policy decision | Runtime probe or external policy system | Operational evidence; never persisted as an Anchor fact |

The ownership chain is:

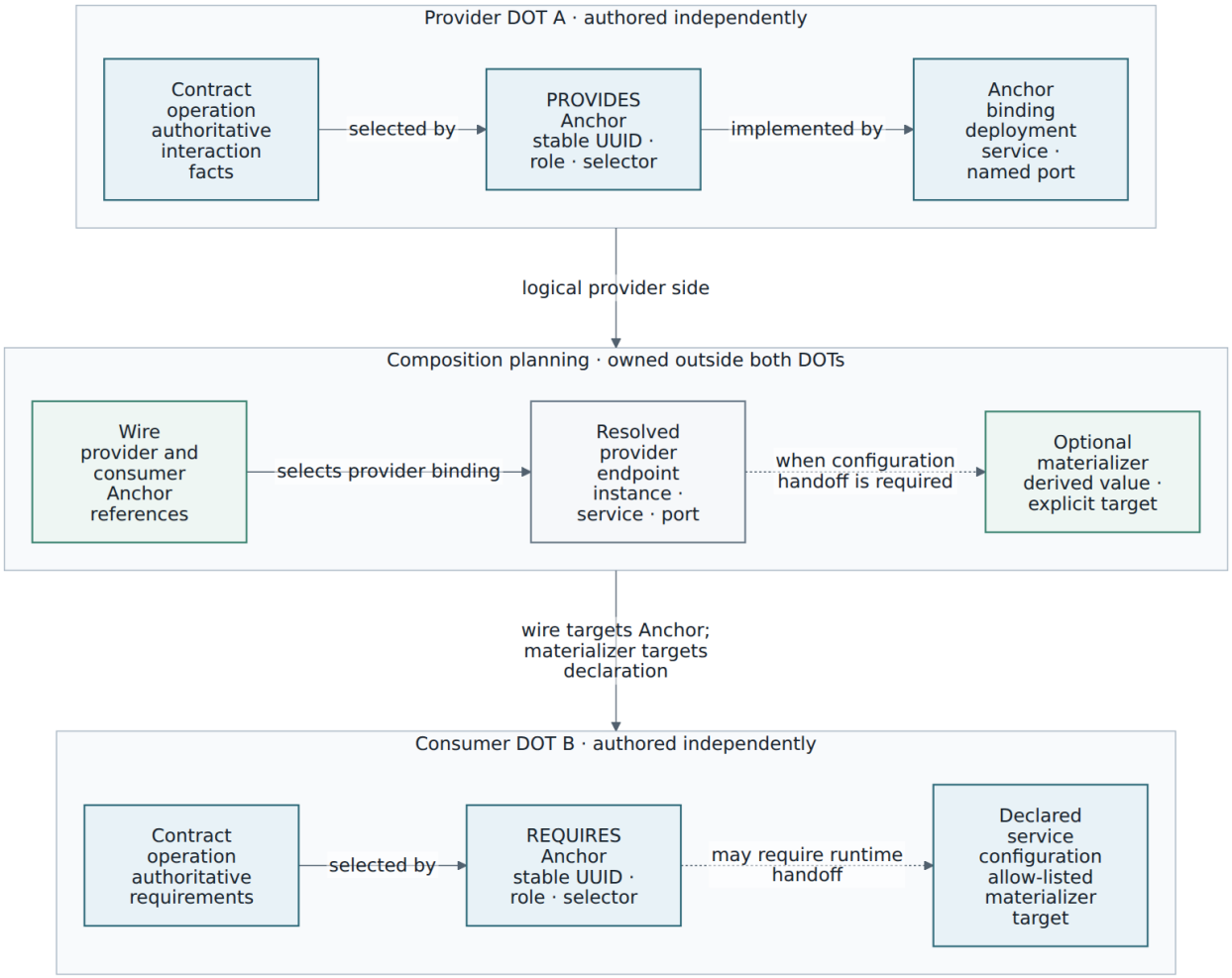



*Figure 6. Provider-to-consumer ownership chain*

These objects must not collapse into one schema. In particular, a wire is not part of either DOT, and a live URL is not part of an Anchor.

## Resolution pipeline

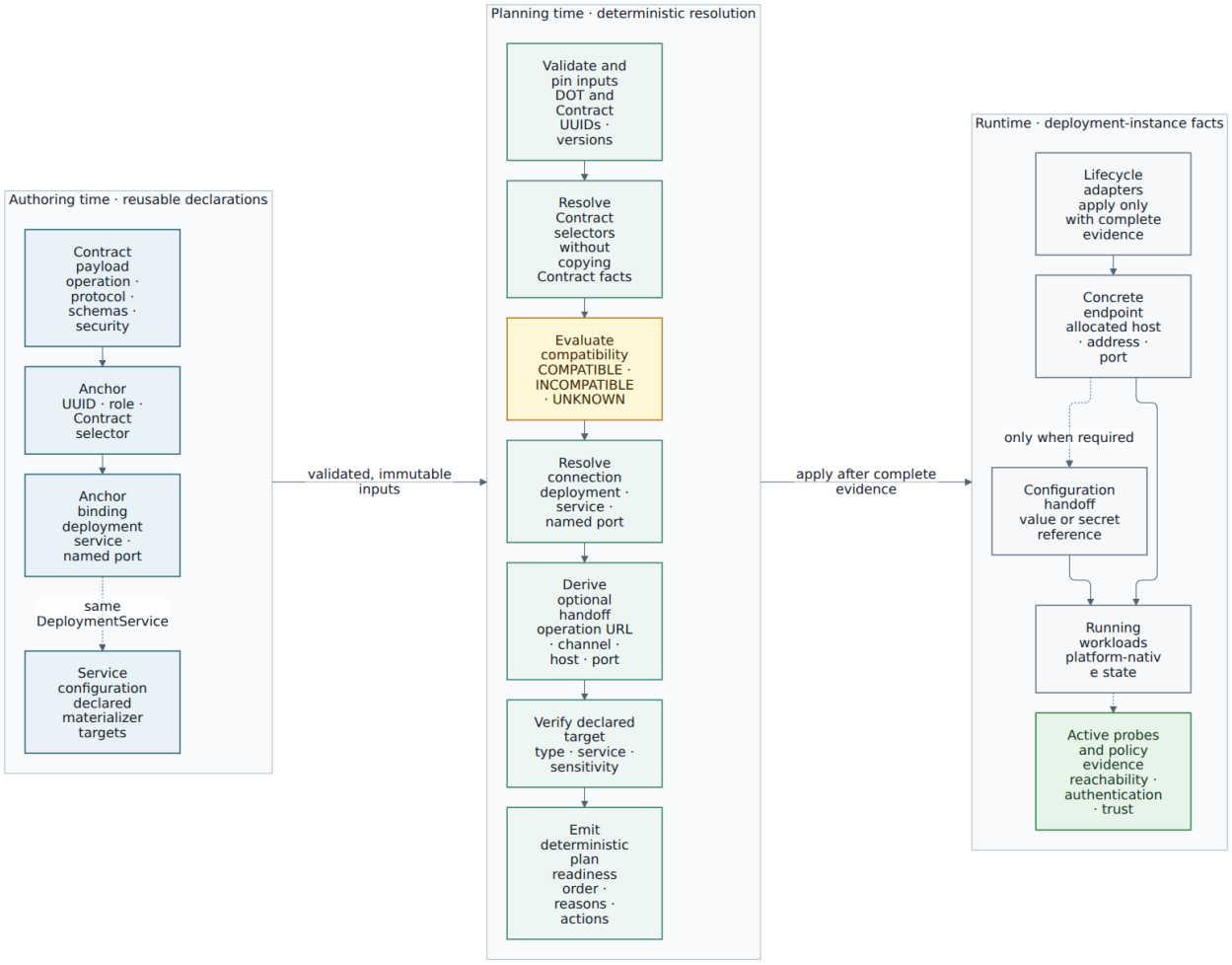



*Figure 7. Authoring, planning, and runtime resolution*

This resembles compilation: validate the independently authored inputs, link stable references, produce a deterministic deployment plan, and only then mutate runtime state. A failure at one phase is not evidence that a later phase was attempted.

# Continue through the workflow

The detailed guide is organized around the work being performed:

| Task | Guide |
|---|---|
| Publish stable composition points in a DOT | Authoring Anchors |
| Define pinned DOT instances and composition topology | Composition documents and instances |
| Connect endpoints, provide security inputs, and materialize configuration | Wiring, materialization, and security |
| Validate, plan, apply, inspect, and evolve a composition | Composition lifecycle and design guidance |

Readers implementing a separate composer should also consult the integration handoff.

# Authoring Anchors

This page covers the DOT publisher's side of composition. Start with the model overview if the distinction between Contracts, Anchors, bindings, and wires is not yet familiar.

## Authoring two composable DOTs

Use this sequence when introducing a new connection:

1. Put the operation or message boundary in an OpenAPI, AsyncAPI, Protobuf, or GraphQL Contract. Use the contractless form only when none of those standards can describe the boundary.
2. Add a stable `PROVIDES` Anchor to the provider DOT and a stable `REQUIRES` Anchor to the consumer DOT. Select the authoritative Contract element from each Anchor; do not copy its protocol or payload fields.
3. On every deployable implementation, bind each Anchor UUID to the named `ServicePort` that implements it.
4. Declare every consumer configuration input on its `DeploymentService`, including sensitivity. This declaration is the allow-list for composition.
5. Validate each DOT independently with `dotx validate` before writing a composition.
6. Add named, UUID/version-pinned DOT instances and ID-pinned Contract inputs to a separate composition document.
7. Connect the Anchor UUIDs with a wire. Add an explicit broker endpoint, parameter values, security binding, or materializer only when the connection requires it.
8. Run `dotx compose lock` and commit the generated lock beside the composition. Review digest changes as changes to the resolved DOT-to-DOT graph.
9. Run `dotx compose validate` to check the document and `dotx compose inspect` to inspect the resolved topology. Treat `UNKNOWN` compatibility as missing evidence, not as success.
10. Use `apply --dry-run` before execution. After apply, use `compose probe` for safe Contract-derived traffic and `status` for native runtime status, configuration evidence, and the latest redacted probe evidence. Dedicated acceptance tests remain necessary for authentication, authorization, mutable operations, and application-specific outcomes.

The reusable three-DOT worked example is under `DOT-Repository/Compositions/keycloak-apisix-oauth/`: `composition.yaml` owns the wiring and references canonical DOT definitions and their local Contracts; `runtime/` holds non-secret instance inputs. The executable acceptance harness remains in `DOT-Testing/e2e/api-gateway-oidc/`.



Concrete identifiers in this guide use the canonical lowercase UUID text form with UUIDv4 version and IETF variant bits defined by RFC 9562. Deterministic values are retained where they identify executable repository fixtures, making the documentation traceable to the tested examples. Generate new UUIDs for production DOTs; do not reuse documentation or fixture identifiers.

## Contract-backed Anchors

A contract-backed Anchor carries stable identity and selects one authoritative contract element. It deliberately omits protocol, direction, media type, payload schema, and security fields that the Contract already defines.

```
{
  "id": "a0000000-0000-4000-8000-000000000001",
  "name": "submit-alert-http",
  "role": "PROVIDES",
  "semanticTypes": ["urn:mare:security:alert"],
  "contract": {
    "id": "11111111-1111-4111-8111-111111111111",
    "selector": "#/paths/~1alerts/post"
  }
}
```

The opposite DOT declares its own `REQUIRES` Anchor. It may select the same Contract ID or an independently authored Contract. Compatibility is deterministic when authoritative facts are comparable; otherwise the result is `UNKNOWN`, never an invented match.

### Reading compatibility results

The plan compares the provider and consumer across independently explainable dimensions:

| Dimension | Meaning |
|---|---|
| Role and direction | The connection is from `PROVIDES` to `REQUIRES`, and the selected interaction directions can communicate. |
| Semantics | Declared semantic-type URIs intersect when both sides provide them. |
| Protocol and media | Resolved protocol and media-type sets have a usable intersection. |
| Interaction | Request/response, publish/subscribe, streaming, query, mutation, or subscription behavior agrees. |
| Schema | Provider output and consumer input schema identities are compared in the correct direction when both select the same authoritative Contract; cross-Contract structural comparison remains `UNKNOWN`. |
| Security | Contract requirements are understood and, for execution, a complete supported binding has all required declared inputs. |

`INCOMPATIBLE` means comparable evidence conflicts. `UNKNOWN` means evidence is missing or the current resolver cannot prove the dimension. Validate and plan may explain `UNKNOWN`; apply rejects it. An explicit mediator DOT is preferable to weakening compatibility when protocols or payloads genuinely differ.

### Selector syntax

| Standard | Selector | Example |
|---|---|---|
| OpenAPI | RFC 6901 URI-fragment JSON Pointer to an Operation Object | `#/paths/~1alerts/post` |
| AsyncAPI | RFC 6901 URI-fragment JSON Pointer to an Operation Object | `#/operations/publishAlert` |
| Protobuf | Fully qualified service method | `mare.security.AlertService.PublishAlert` |
| GraphQL | Root-field schema coordinate | `Mutation.publishAlert` |

For OpenAPI and AsyncAPI, `/` in a JSON Pointer token becomes `~1` and `~` becomes `~0`. Concrete path or channel template values belong to a composition wire's `parameters`, not to the Anchor.

#### Supported Contract representation matrix

| Standard | Contract `format` | Authoritative interaction | Composition evidence |
|---|---|---|---|
| OpenAPI | `YAML` or `JSON` | Request/response operation, including path and security requirements | One shared selector; `REQUEST_RESPONSE` for both participants |
| AsyncAPI | `YAML` or `JSON` | Message operation and Channel Object | Distinct complementary selectors; `SEND` and `RECEIVE`, plus the resolved channel address |
| Protocol Buffers (Proto3; including gRPC service declarations) | `PROTOBUF` | Fully qualified RPC and unary or streaming shape | One shared method selector; `REQUEST_RESPONSE`, `CLIENT_STREAM`, `SERVER_STREAM`, or `BIDIRECTIONAL_STREAM` |
| GraphQL | `GRAPHQL` | SDL root field | One shared schema coordinate; `QUERY`, `MUTATION`, or `SUBSCRIPTION` |

`dotx compose inspect --json` exposes the resolved `contractStandard`, `contractFormat`, `providerInteraction`, and `consumerInteraction` on each contract-backed wire. These fields are derived from the supplied Contract and therefore do not duplicate authoring metadata in either Anchor or composition. The deterministic `DOT-Testing/composition/contract-standards/` matrix connects one provider DOT to one consumer DOT through every row above.

#### Semantic types

`semanticTypes` is a list of URI identifiers describing the security or domain meaning of the interaction independently of its transport. Prefer stable, governed ontology or vocabulary URIs when one exists; a project URN is suitable for concepts that have no external vocabulary yet.

Current matching is deliberately conservative: both Anchors must declare at least one semantic type and their sets must intersect exactly. Missing values produce `UNKNOWN`; different values produce `INCOMPATIBLE`. The evaluator does not infer subclass, equivalence, or broader/narrower ontology

relations. Such reasoning belongs in a separately reviewed semantic-policy layer.

## Contractless Anchors

Use a contractless Anchor only when no supported authoritative contract exists, for example a raw packet stream or a legacy metrics boundary. It must state the facts that cannot otherwise be derived:

```
{
  "id": "a0000000-0000-4000-8000-000000000006",
  "name": "packet-stream",
  "role": "PROVIDES",
  "semanticTypes": ["urn:mare:security:packet-stream"],
  "direction": "EGRESS",
  "protocols": ["TCP"],
  "mediaTypes": ["application/octet-stream"]
}
```

Contract-backed and contractless forms are mutually exclusive. A contract-backed Anchor must not repeat `direction`, `protocols`, or `mediaTypes`. A contractless Anchor must declare all three.

#### Role is not traffic direction

`role` answers who offers the selected capability:

- `PROVIDES` means this DOT makes the capability available;
- `REQUIRES` means this DOT depends on that capability.

It does not mean that bytes move in only one direction. An OpenAPI request/response operation has a providing server and requiring client while both send data. An AsyncAPI provider may select a `send` operation and its consumer a complementary `receive` operation. For a contract-backed Anchor, the resolver derives that interaction from the selected operation. Only a contractless Anchor states `INGRESS`, `EGRESS`, or `BIDIRECTIONAL` explicitly.

When two DOTs implement the same published Contract, referencing the same Contract ID gives the strongest deterministic evidence. Independently authored Contracts are allowed, but identical-looking schema names do not prove structural compatibility across Contract identities; that comparison remains `UNKNOWN` until a schema-aware adapter or policy evaluates it.

## Stable identity and version evolution

- The UUID, not `name`, identifies an Anchor in a wire.
- A rename or documentation correction preserves the UUID.
- A compatible DOT version preserves the UUID for the same logical Anchor.
- An incompatible semantic replacement receives a new UUID.
- Removal and any future `deprecated`/`replacedBy` metadata are publishing governance decisions; they are not inferred from contract changes.

- Names remain useful in logs and displays but are never foreign keys.

## Binding an Anchor to a deployment

An Anchor is logical until a selected deployment maps it to a named service port:

```
{
  "id": "security-function",
  "name": "Security Function",
  "host": "security-function",
  "ports": [
    {"name": "http", "protocol": "HTTP", "port": 8080}
  ],
  "anchorBindings": [
    {
      "anchorId": "a0000000-0000-4000-8000-000000000001",
      "port": "http"
    }
  ],
  "config": {
    "environment": [],
    "configFiles": [],
    "volumes": []
  }
}
```

This says which logical service and named port implement the Anchor. The deployer still decides the concrete container DNS name, cluster Service, allocated address, forwarded port, certificate, or network path.

`@ValidAnchor` and `AnchorValidator` reject duplicate Anchor UUIDs, undeclared Contract references, invalid selector syntax, undeclared ports, undeclared Anchor bindings, and duplicate service-local bindings.

### Runtime configuration portability

Every `DeploymentService` declares its supported configuration surface through `environment`, `configFiles`, and logical `volumes`. A composition may supply only values whose names or target paths the selected service declares. Required environment variables and files without defaults fail before runtime mutation; sensitive environment values require a `secretRef` instead of a literal. When an input is omitted, `dotx` reuses the non-sensitive `defaultValue` or `defaultContent` declared by that service. Explicit composition values and wire materializers override defaults; conflicting explicit inputs fail closed. Declarations marked `sensitive` are forbidden from embedding either kind of default and must use a deployment-time secret reference or runtime-native secret facility. Dry runs print resolved environment values in stable name order for every adapter, but replace every value declared sensitive with `<redacted>`.

Each composition input carries the stable `service` id it targets. Authors may omit `service` only when exactly one service in the selected deployment declares that environment name, file path, or

volume name; planning resolves and records the service explicitly. If two services declare the same item, omission is an error. Duplicate detection is service-local, so two services can intentionally receive different values for the same environment-variable name.

Environment variables and configuration files are rendered by every currently executable `dotx` deployment adapter:

| Manifest | Environment | Configuration files |
|---|---|---|
| `DOCKER_IMAGE` | Container environment | Read-only bind mounts |
| `DOCKER_COMPOSE` | Per-service override using process-environment staging | Generated per-service Compose override mounts |
| `KUBERNETES_MANIFEST` | Exact workload/container patch; existing Secret keys remain `secretKeyRef` | Service-qualified ConfigMap or Secret plus exact workload/mount patch |
| `HELM_CHART` | Exact post-install workload/container patch; existing Secret keys remain native | Service-qualified ConfigMap or Secret plus exact workload/mount patch |
| `VAGRANTFILE` | Vagrant process/provisioner environment | Files provisioned into the managed VM and exposed to the Vagrantfile |
| `WASM_MODULE` | Allow-listed WASI guest environment; sensitive values inherited by name | Staged configuration root plus explicit Wasmtime directory preopens |

Configuration remains service-scoped. A Docker image or Vagrant deployment is one workload or VM, so several logical services may safely share its declared configuration surface. Docker Compose, Kubernetes, and Helm may materialize several independent workloads. Their composition therefore supplies `deploymentInputs.workloads`: a Compose binding names one native service; a Kubernetes/Helm binding supplies exact native labels and optionally a container. Dotx intersects those labels with its composition-owned instance label and requires exactly one workload. Missing, duplicate, cross-runtime, ambiguous workload, and ambiguous container bindings fail closed. This prevents environment values, secrets, or mounted files from reaching the wrong security function without adding platform-specific selectors to the reusable DOT model.

```
deploymentInputs:
  workloads:
    - service: api
      kubernetes:
        matchLabels:
          app.kubernetes.io/component: api
        container: api
```

Storage and privilege configuration is deliberately runtime-native. Docker image deployments

support composition-scoped named volumes, explicit OCI bind sources, exact capability/host-namespace grants, and CNCF CDI devices. Compose payloads own their volume graph; Kubernetes and Helm own claims, storage classes, and volume topology; Vagrantfiles own disks and synced folders. A composer must not translate a host bind path into a cluster volume or silently grant privileged mode.

`WASM_MODULE` uses a reviewed Wasmtime/WASI profile. Each module is SHA-256 pinned and loaded from a relative/file or HTTPS URI. `WASI_CLI` executes command modules/components; `WASI_HTTP` serves a component on one declared TCP/HTTP service port bound to loopback. Dotx exposes only declared environment values, staged configuration files, and explicitly bound host directories. It does not translate OCI capabilities, host namespaces, or CDI devices into broader WASI permissions. Sensitive values are never printed or persisted. Because the Wasmtime CLI cannot enforce read-only permissions on a directory preopen, that request fails closed unless the host filesystem already supplies the boundary. The deterministic `DOT-Testing/composition/runtime-configuration/` matrix exercises every executable adapter twice: once using only declared non-sensitive defaults and once using explicit composition overrides. It checks the materialized default configuration document, the resolved service-scoped plan, and the runtime-specific dry-run renderer. The five native cases reuse the external Prometheus DOT; WASI retains a separate conformance fixture. The real Prometheus acceptance runner verifies loaded configuration and environment through its official status APIs, PromQL results and invalid-query responses, stop/start, Git acquisition, and failure cleanup:

```
task testing:compose:runtime-configuration
task testing:compose:runtime-configuration:live RUNTIME=docker-image
```

Select `docker-compose`, `kubernetes-manifest`, `helm-chart`, or `vagrantfile` for the other native mechanisms. These are ephemeral single-server examples, not production monitoring configurations. The `DOT-Testing/composition/service-workloads/` matrix verifies isolation between two services across Compose, Kubernetes, and Helm dry-run adapters.

## Discovering and resolving Anchors

The authoritative test and reference definitions are stored in `DOT-Repository` and addressed as `repository://<dot-id>@<version>`. Consumers can inspect one immutable definition or validate the complete repository:

```
dotx repository inspect <dot-id>@<version> --repository DOT-Repository --json
dotx validate DOT-Repository
```

`dotx compose validate` and `dotx compose inspect --json` resolve Anchor UUIDs, service-port bindings, and selected Contract elements conservatively. The plan reports `COMPATIBLE`, `INCOMPATIBLE`, or `UNKNOWN`; missing evidence is never treated as a match. Contract payloads remain authoritative and are not copied into Anchors or composition wires.

This command boundary is the reference behavior, not a requirement to embed dotx. Another catalogue or composer may use any language, framework, storage, or API shape if it preserves the same identities, fail-closed resolution, and ownership boundaries. See integration handoff.

Next, define the concrete participants and topology in a composition document.

# Composition Documents and DOT Instances

A composition document is owned by the composer or operator, not by any member DOT. It selects immutable DOT identities and creates deployment instances whose topology may vary between environments.

## Reusable composition catalogue

`DOT-Repository/Compositions/` distributes reusable multi-DOT graphs, reviewed locks, READMEs, and non-secret deployment inputs. Start with `keycloak-apisix-oauth/` (Keycloak, APISIX, OAuth client) or `keycloak-prometheus-openapi-planning/` (official Keycloak and Prometheus OpenAPI operations). Each directory has `composition.yaml` and `composition.lock.json`.

From the standalone DOT-Repository checkout root:

```
dotx validate . --offline
composition=Compositions/keycloak-apisix-oauth/composition.yaml
dotx compose inspect "$composition" --locked --json
dotx compose apply "$composition" --locked --dry-run
```

Repository-wide composition validation requires the current source-built CLI; the released 0.2.4 binary predates this gate. It recursively checks declarations, semantic plans, adjacent locks, and READMEs without deploying or resolving Secrets. `task compositions:render` additionally renders every catalogue composition. There is no separate composition inventory or Scenario domain type.

A scenario is the broader use-case narrative and acceptance procedure described by the README. Synthetic matrices, negative controls, source-adapted PoC fixtures, executable harnesses, and evidence remain in DOT-Design's `DOT-Testing/`. See Getting started for the gateway's three Wires and Guided demo for live acceptance. Catalogue inclusion or a valid plan never implies production or live producer acceptance.

## Composition documents and DOT instances

A composition is a separate versioned document. It names concrete instances of DOT definitions and pins their identities through the authoritative `DOT-Repository` catalogue. The `repository://<dot-id>@<version>` form is stable across checkout locations and prevents a scenario-local copy from drifting away from the reviewed definition:

This structural excerpt omits the required instance inputs and wires. Use `DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml` for the complete runnable document; Contract paths below are relative to that directory.

```
apiVersion: dot.mare6g.eu/v1
kind: Composition
id: e3000000-0000-4000-8000-000000000001
```

```
name: keycloak-apisix-oauth
version: 1.0.0
dots:
  - name: identity
    source: repository://49b0c883-e77a-535e-8589-d8a5c01b0a59@0.2.0
    dotId: 49b0c883-e77a-535e-8589-d8a5c01b0a59
    dotVersion: 0.2.0
    deployment: kubernetes
  - name: gateway
    source: repository://da100000-0000-4000-8000-000000000002@1.0.0
    dotId: da100000-0000-4000-8000-000000000002
    dotVersion: 1.0.0
    deployment: kubernetes
    dependsOn: [identity]
contracts:
  - id: c3000000-0000-4000-8000-000000000001
    source: ../../DOTs/external/Keycloak-DOT/contracts/c3000000-0000-4000-8000-
000000000001.yaml
```

`name` identifies an instance inside this composition. `source` selects the repository definition by UUID and exact version. `dotx compose` locates the nearest `DOT-Repository` containing a `DOTs/` directory, discovers its definitions recursively, resolves the UUID/version to its reviewed bundle, and then verifies that `dotId` and `dotVersion` match the loaded `dot.json`. Neither a repository definition nor a local path can silently substitute a different DOT. `deployment` selects a deployment by its stable deployment ID or unambiguous manifest alias.

Local discovery is the convenience default. `--repository <path-or-Git-URL>` selects another authoritative catalogue and `--git-ref <revision>` selects a Git revision. Locked and runtime-mutating commands require a full 40-character commit SHA for remote repositories; branch and tag names are allowed only for unlocked, read-only exploration. Exact commits are cached in a private user cache by normalized repository identity and commit, so the same plan and lock can be reproduced offline. Plans, locks, and lifecycle state record only the credential-free identity and resolved commit. Credential-bearing URLs are rejected; authentication remains external through Git credential helpers, SSH agents, or environment-backed CI credentials.

Relative paths remain supported for authoring experiments and external Contract documents. Reusable stable, sample, and test-only DOT definitions belong in `DOT-Repository`, under their corresponding catalogue scope. Contracts remain authoritative protocol documents and are referenced independently; placing a DOT in the repository does not copy OpenAPI, AsyncAPI, Protobuf, or GraphQL facts into its Anchors.

`dependsOn` is a readiness graph, not a connection graph. A wire does not imply startup order. This distinction permits reciprocal peer wires and brokered topologies without creating false dependency cycles.

#### Composition field reference

The emitted Draft 2020-12 JSON Schema is normative for the serialized `v1` shape. This table is the study guide for its top-level ownership:

| Field | Required | Meaning |
| --- | --- | --- |
| `apiVersion`, `kind` | Yes | Fixed discriminators: `dot.mare6g.eu/v1` and `Composition`. |
| `id` | Yes | Stable UUID of this composition, also used to derive local state and runtime identities. |
| `name`, `version` | Yes | Human-readable identity and version of the composition document. |
| `dots[]` | Yes, at least one | Named deployment instances. Every entry pins `source`, `dotId`, and `dotVersion`; published scenarios use an exact `repository://<uuid>@<version>` source. `deployment`, `dependsOn`, and `deploymentInputs` refine execution. |
| `contracts[]` | When a referenced Contract is needed | Maps an exact Contract ID to a local Contract document. The document's embedded `id` must match. |
| `wires[]` | No | Connections between one `PROVIDES` and one `REQUIRES` Anchor. A composition with no wires can still expose a standalone capability. |
| `exposedAnchors[]` | No | Composition-local names for member Anchors that an operator may expose or forward. |

Each wire requires `name`, `from`, and `to`. Both ends contain an instance name and Anchor UUID. Optional `endpoint` selects a third-party or otherwise non-default service port; `parameters` supplies only Contract-declared template values; `materializer` or `materializers` writes resolved values; and `security` binds Contract security requirements to declared participant configuration. Unknown fields are rejected.

#### Deployment inputs and native workload bindings

`deploymentInputs.environment`, `configFiles`, and `volumes` identify the logical `DeploymentService` that owns each input. `ociRuntime` and `cdiDevices` authorize only requirements already declared by that selected DOT deployment. These fields supply instance values and grants; they do not expand the DOT's declared configuration surface.

When a Docker Compose, Kubernetes, or Helm deployment contains several logical services with configuration, `deploymentInputs.workloads` is required for every configured service. The binding is deliberately instance-owned because native service names, labels, and containers can differ between deployment variants or environments:

```
deploymentInputs:
  workloads:
    - service: api
      compose:
        service: backend
```

The Kubernetes/Helm form uses exact-match standard labels and may select one container:

```
deploymentInputs:
  workloads:
    - service: api
      kubernetes:
        matchLabels:
          app.kubernetes.io/component: api
        container: api
```

Labels are combined with dotx's instance ownership selector. A live apply must resolve exactly one Deployment, StatefulSet, or DaemonSet, and an omitted container is accepted only when that workload has one container. Two logical services cannot target the same Compose service or the same Kubernetes workload/container pair. Plans expose the binding and secret identifiers, and locks cryptographically pin them; neither contains resolved secret values.

#### DOT definition versus DOT instance

One pinned DOT definition can appear more than once in a composition. Each entry has a distinct composition-local `name`, runtime identity, deployment inputs, and wires while retaining the same `dotId` and `dotVersion`:

```
dots:
  - name: site-a
    source: ./federation-agent
    dotId: d5000000-0000-4000-8000-000000000001
    dotVersion: 1.0.0
    deployment: docker
    deploymentInputs:
      environment:
        - {service: federation-agent, name: FEDERATION_NODE_ID, value: site-a}
  - name: site-b
    source: ./federation-agent
    dotId: d5000000-0000-4000-8000-000000000001
    dotVersion: 1.0.0
    deployment: docker
    deploymentInputs:
      environment:
        - {service: federation-agent, name: FEDERATION_NODE_ID, value: site-b}
```

Wires reference `site-a` or `site-b`, never the source path. Docker Image deployments receive deterministic per-instance container, DNS, network, and named-volume identities. Apply rejects repeated instances for deployment mechanisms whose manifest-owned resource names cannot yet be isolated safely. One native Kubernetes instance may coexist with isolated Docker Image/Compose instances of that same DOT because they occupy separate resource scopes. Two native Kubernetes instances of the same DOT remain rejected, including when a Docker instance separates them in document order. See the Prometheus portability demonstration.

The executable example is `DOT-Testing/composition/federation-peers/scenario.yaml`.

After defining the participants, wire and configure their selected Anchors.

# Wiring, Materialization, and Security

Wires express composition topology outside every participating DOT. They refer to stable Anchor UUIDs, resolve deployable endpoints through service bindings, and materialize only values that the target service has declared as configurable.

## Direct request/response wire

A wire uses instance names and stable Anchor UUIDs:

```
wires:
  - name: oauth-token-to-client
    parameters:
      realm: {value: dotx}
    from:
      instance: identity
      anchorId: a1000000-0000-4000-8000-000000000002
    to:
      instance: client
      anchorId: a5000000-0000-4000-8000-000000000001
    materializer:
      type: ENVIRONMENT_VARIABLE
      valueFrom: OPERATION_URL
      target:
        instance: client
        service: oauth-client
        environmentVariable: TOKEN_URL
```

Planning performs the following sequence:

1. Resolve the two pinned DOT instances.
2. Resolve both Anchor UUIDs and require complementary `PROVIDES`/`REQUIRES` roles.
3. Resolve the selected Contract elements.
4. Evaluate semantic, interaction, protocol, media, schema, and security compatibility.
5. Resolve the connection endpoint through the selected deployment's `anchorBindings`.
6. Verify that the target service declares `TOKEN_URL`.
7. Build `OPERATION_URL` from the runtime endpoint and the authoritative OpenAPI server/operation path.

The resulting URL is deployment data. It is never copied back into `dot.json` or the Anchor.

### Cross-platform external gateway

A wire whose participants use different runtime platforms must declare a reviewed network

adapter. The first supported adapter represents a gateway that the operator provisions outside `dotx`:

```
wires:
  - name: identity-discovery-to-proxy
    from: {instance: identity, anchorId: a1000000-0000-4000-8000-000000000003}
    to: {instance: proxy, anchorId: aa100000-0000-4000-8000-000000000002}
    network:
      adapter: EXTERNAL_GATEWAY
      host: identity-gateway.example.net
      port: 8080
```

The Contract and bound service port remain authoritative for protocol and interaction semantics. The binding replaces only the deployment-specific host and port used by materializers. The plan, lock, and private lifecycle state record `ownership=EXTERNAL` and `cleanup=NONE`; apply, resume, status, and destroy never claim to create, inspect, or remove the gateway. Cross-platform wires without a binding, same-platform wires with a redundant binding, unknown adapters, URL-shaped hosts, and invalid ports fail before runtime mutation.

Composition-owned gateways, ingresses, tunnels, and brokers require dedicated lifecycle adapters and remain unsupported.

A wire and a materializer are intentionally separate. The wire says *these Anchors are connected*; the materializer says *write this derived value into this declared consumer setting*. A wire needs no materializer when the runtime already provides discovery, a service mesh performs routing, both participants use a fixed broker declared elsewhere, or the composition is documenting an Anchor relation without an executable handoff. Apply still requires every actually required service configuration value to have a default or explicit input.

For the example above, the human-readable plan makes every resolution step inspectable before apply:

```
oauth-token-to-client:
  identity/a1000000-0000-4000-8000-000000000002
    -> client/a5000000-0000-4000-8000-000000000001 [COMPATIBLE]
  endpoint: keycloak:8080
    (HTTP, instance=identity service=keycloak port=http)
  contract: c1000000-0000-4000-8000-000000000001
    selector=#/paths/~1realms~1%7Brealm%7D~1protocol~1openid-connect~1token/post
  materializer: OPERATION_URL -> client/oauth-client env:TOKEN_URL
```

The same plan with `--json` is the machine-readable automation output. It includes stable instance and Anchor references, compatibility reasons, the selected Contract, resolved endpoint, derived operation/channel value, materializer targets, security status, and exposed-Anchor endpoints. Consumers should parse JSON rather than scrape the human-readable form.

### Brokered wire

For messaging, the semantic provider, semantic consumer, and connection endpoint may be three different instances. Select the broker explicitly:

```
wires:
  - name: openc2-command-via-mqtt
    from:
      instance: interpreter
      anchorId: a6000000-0000-4000-8000-000000000003
    to:
      instance: actuator
      anchorId: a6000000-0000-4000-8000-000000000006
    endpoint:
      instance: broker
      service: mqtt
      port: mqtt
    materializers:
      - type: ENVIRONMENT_VARIABLE
        valueFrom: ENDPOINT_HOST
        target:
          instance: actuator
          service: openc2-actuator
          environmentVariable: MQTT_BROKER_HOST
      - type: ENVIRONMENT_VARIABLE
        valueFrom: ENDPOINT_PORT
        target:
          instance: actuator
          service: openc2-actuator
          environmentVariable: MQTT_BROKER_PORT
```

The AsyncAPI operations and message schema stay in the Contract. The broker's host and port stay in the resolved plan. If a channel has a templated address, the wire supplies only values for parameters declared by the Contract.

### Materializers

Materializers are explicit adapters between a resolved endpoint and a configuration surface already declared by a service.

| Type | Behavior |
|---|---|
| `ENVIRONMENT_VARIABLE` | Writes one resolved scalar to a declared environment variable. |
| `CONFIG_FILE` | Mounts a declared file and optionally replaces one YAML/JSON value selected by an RFC 6901 JSON Pointer. |

Supported `valueFrom` values are:

| Value | Result |
| --- | --- |
| `ENDPOINT_HOST` | Resolved service host only. |
| `ENDPOINT_PORT` | Resolved numeric service port only. |
| `ENDPOINT_HOST_PORT` | `host:port`. |
| `ENDPOINT_URL` | `scheme://host:port`, using the bound port protocol. |
| `OPERATION_URL` | Endpoint URL plus the authoritative OpenAPI server base and operation path. |
| `CHANNEL_ADDRESS` | Address of the authoritative AsyncAPI Channel Object. |

An optional RFC 6570 `uriTemplate` can derive an application-specific URI from one resolved scalar. It is explicit composition behavior, not an Anchor or Contract fact.

#### Configuration-file materialization

A `CONFIG_FILE` materializer always names an operator-controlled source and a file path already declared by the target service. Without `valueFrom`, the source is supplied unchanged. With `valueFrom`, the source must be YAML or JSON and `jsonPointer` selects the scalar to replace:

```
materializer:
  type: CONFIG_FILE
  source: ./prometheus/prometheus.yml
  valueFrom: ENDPOINT_HOST_PORT
  target:
    instance: prometheus
    service: prometheus
    path: /etc/prometheus/prometheus.yml
    jsonPointer: /scrape_configs/0/static_configs/0/targets/0
```

This produces a deployable Prometheus configuration without putting the provider's runtime address in either DOT. The reusable executable graph is `DOT-Repository/Compositions/node-exporter-prometheus-monitoring/composition.yaml`; the broader PoC4 planning fixture remains under `DOT-Testing/composition/`.

#### URI-template derivation

Use `uriTemplate` only when the application needs an explicit deterministic derivation that is not already the selected Contract operation URL:

```
materializer:
  type: ENVIRONMENT_VARIABLE
  valueFrom: ENDPOINT_URL
  uriTemplate: "{+value}/realms/drax"
  target:
    instance: proxy
    service: proxy-xapp
```

```
    environmentVariable: OIDC_ISSUER
```

The RFC 6570 variable is named `value`. The template belongs to the composition because it adapts one deployment to another; it does not amend the Contract or redefine the Anchor.

Materialization fails when the target configuration is undeclared, the Contract path/channel remains unresolved, alternative server paths are ambiguous, two wires produce conflicting values for one target, or the endpoint and target are on different runtime platforms without an explicit network binding.

## Path and channel parameters

Only deployment-specific scalar values belong in the wire:

```
parameters:
  realm:
    value: drax
```

The Contract remains authoritative for the parameter name, location, requirement, schema, default, and serialization rules. Current composition bindings cover OpenAPI path parameters and AsyncAPI channel parameters. Query/header/cookie parameters, complex serialization, and runtime-derived parameter values require explicit future adapters.

## Security and secrets

Contract security remains authoritative. A wire may demonstrate how one declared scheme is satisfied by deployment configuration:

```
security:
  bindings:
    - scheme: bearerAuth
      mechanism: https://www.rfc-editor.org/rfc/rfc6749#section-4.4
      participant: {instance: client, service: oauth-client}
      parameters:
        tokenUrl: {environmentVariable: TOKEN_URL}
        clientId: {environmentVariable: CLIENT_ID}
        clientSecret: {environmentVariable: CLIENT_SECRET}
```

Sensitive inputs are references, never secret values. They are scoped to the DOT instance whose selected deployment consumes them:

```
dots:
  - name: client
    source: repository://d3a00000-0000-4000-8000-000000000003@1.0.0
    dotId: d3a00000-0000-4000-8000-000000000003
    dotVersion: 1.0.0
    deployment: kubernetes
```

```
    deploymentInputs:
      environment:
        - name: CLIENT_SECRET
          secretRef:
            store: kubernetes
            name: oauth-client
            key: client-secret
```

Plans and composition state keep only the reference. Docker deployments can resolve `store: environment` at final handoff; Kubernetes and Helm preserve `store: kubernetes` as `secretKeyRef` without reading the Secret. A `CONFIGURED` plan proves configuration completeness, not token validity, issuer trust, mTLS, workload identity, or remote attestation.

### Exposed Anchors

A composition may re-export a member Anchor under a composition-local name:

```
exposedAnchors:
  - name: protected-metadata-through-gateway
    target:
      instance: gateway
      anchorId: a4000000-0000-4000-8000-000000000003
```

At apply time, an operator may request a local forward:

```
dotx compose apply scenario.yaml \
  --forward protected-metadata-through-gateway=19080
```

The exposure name belongs to the composition boundary. It does not rename or replace the member Anchor UUID.

Before runtime mutation, follow the validation and lifecycle workflow.

## Composition Lifecycle and Design Guidance

The lifecycle deliberately separates reproducible input locking, model validation, deterministic planning, runtime mutation, recorded state, and active acceptance evidence. This keeps review and CI read-only while making execution recoverable and auditable.

Use `dotx compose inspect composition.yaml --locked` to explain the resolved DOTs, deployments, Contracts, wires, endpoints and configuration destinations before execution. It does not deploy workloads or probe services, although it may fetch and cache referenced definitions and Contracts. Use `topology` for a graph, `apply --dry-run` for deployment commands, and `status` for recorded runtime state. `inspect --json` retains the existing execution-plan JSON format.

## Validation, planning, and lifecycle

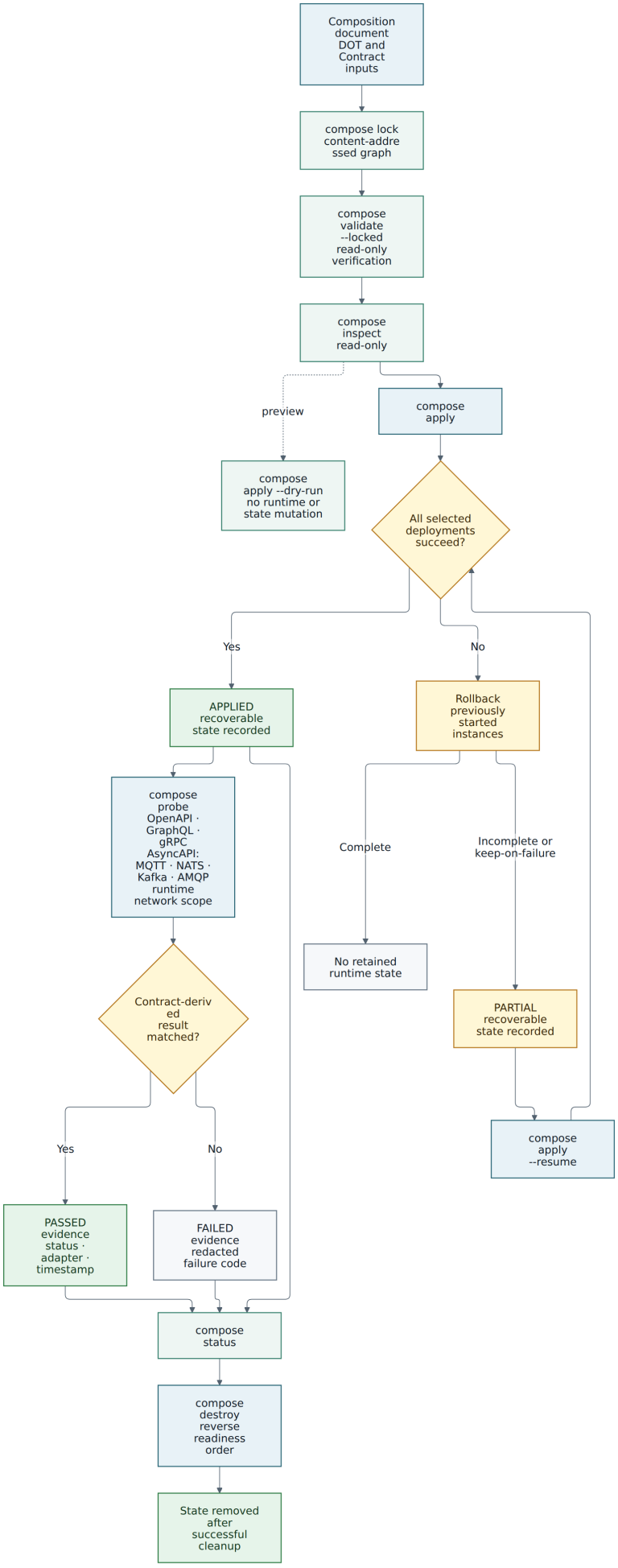



*Figure 8. Composition lifecycle*

### What each command proves

| Command | Proves | Does not prove | Mutates runtime/state |
|---|---|---|---|
| `compose lock` | The exact composition, selected DOT definitions and deployments, Contracts, configuration inputs, and portable resolved plan are content-addressed. | Runtime availability, credentials, or policy acceptance. | Writes only the requested lock file; `--check` is read-only |
| `compose validate` | The strict document shape, references, selected deployments, Anchors, Contracts, compatibility evidence, and materializer declarations resolve. | The exact runtime actions or live availability. | No |
| `compose inspect` | Everything validated above plus the deterministic instance order, runtime identities, endpoints, Contract standard/format and participant interaction, derived values, security status, and materializers. | That the planned hosts are reachable or credentials work. | No |
| `compose apply --dry-run` | The composition reaches lifecycle preparation and can render supported deployment actions without persisting composition state. | Successful deployment or application-level traffic. | No |
| `compose apply` | The supported deployment adapters completed and recoverable state was recorded, subject to rollback behavior. | End-to-end Contract behavior unless separately probed. | Yes |
| `compose upgrade --dry-run` | The exact current and target locks resolve, the target SemVer increases, and a deterministic per-instance `ADD`/`UPDATE`/`REMOVE`/`UNCHANGED` diff passes runtime, storage, ownership, and recovery preflights. | That an application-specific persistent-data migration is safe. | No |

| Command | Proves | Does not prove | Mutates runtime/state |
|---|---|---|---|
| `compose upgrade` | The target revision was applied through a checkpointed transaction and the last-known-good control-plane revision remained available until promotion. | Reversal of application data writes; stateful mutations are rejected. | Yes; failures automatically restore the verified previous revision, while interrupted transactions require `--resume` or `--rollback` |
| `compose probe` | A selected safe OpenAPI operation, GraphQL lifecycle, finite side-effect-free gRPC method, or correlated AsyncAPI MQTT/NATS/Kafka/AMQP message is observable from the composition's shared Docker network or Kubernetes namespace. GraphQL mutations/subscriptions include mandatory verification or trigger and cleanup. | Complete application semantics, unsafe or unbounded gRPC, other AsyncAPI protocols, multi-scheme authentication, authorization-policy quality, or unprobed operations. | Creates and removes an ephemeral probe runner without changing member workloads; writes redacted probe evidence to composition state |
| `compose status` | Native status for recorded instances, wire/exposure configuration evidence, and the latest recorded active-probe outcomes. | Current reachability after the probe timestamp, authentication mechanisms not explicitly identified by the evidence, authorization semantics, response bodies, or policy compliance. | No runtime mutation; reads state while holding the lifecycle lock |
| `compose destroy` | Owned instances were destroyed in reverse order and successful cleanup removed composition state. | Cleanup of resources not owned or recorded by this composition. | Yes |

Current-source upgrade admission inspects the selected deployment in both the previous and target revisions, not unrelated deployment alternatives. Compose YAML/JSON is parsed structurally; Kubernetes persistent volumes, claims, StatefulSets and non-ephemeral volume sources are refused. Missing, ambiguous or malformed inputs fail closed. Changed Helm and Vagrant instances are refused until chart defaults and dynamic Ruby storage can be established safely; unchanged instances and normal apply/status/destroy remain supported. This hardening is newer than the published CLI `v0.2.5`: use the reviewed fixed source for upgrades until a later release includes it. No control-plane rollback promises to undo application writes or preserve data in writable container layers.

`validate` answers *is this composition internally resolvable?* `plan` answers *what exactly would be connected and materialized?* Keeping both commands read-only makes them suitable for review and CI.

`lock` makes those answers reproducible. Its JSON output preserves the original `repository://` and relative references while recording SHA-256 digests for the composition, every resolved `dot.json`, every supplied Contract, configuration inputs, and the portable resolved plan. For an explicit remote repository it also records the normalized repository identity and exact commit. It never records secret values, runtime addresses, or checkout-local absolute paths. Validate, plan, and apply verify an adjacent lock automatically; `--locked` additionally requires it, and `--lock <path>` selects an explicit file. Any mismatch fails before runtime mutation.

`probe` deliberately reuses the interaction selected by the wire instead of introducing health-check fields in an Anchor or DOT. For OpenAPI, it derives the request from the authoritative operation, accepts only `GET`, `HEAD`, and `OPTIONS`, and requires one fully bound path. The OAuth 2.0 adapter reuses one configured RFC 6749 client-credentials binding that satisfies a complete, single-scheme OpenAPI Security Requirement alternative. It resolves the provider-derived token URL, client ID, scopes, and client-secret reference at execution time, then requires a Contract-declared `2xx` response from the protected operation. Kubernetes Secret values remain `secretKeyRef` references; Docker secrets are supplied only through the runner environment.

Development builds newer than the published CLI `v0.2.5` also accept an optional native JSON response assertion on safe OpenAPI `GET` or `OPTIONS` wires:

```
probe:
  openapi:
    expect: {pointer: /data/status, equals: success}
```

The existing Contract remains authoritative for the operation and declared status codes. With this fixture, the response must additionally be `2xx`, have `application/json` or `application/*+json` content type, and contain valid JSON whose RFC 6901 pointer equals the non-sensitive expected value. Array indices are supported; an empty pointer compares the entire document. Redirects are not followed, execution uses the probe timeout, and runner output is limited to 1 MiB. Only redacted outcomes and failure codes are recorded, never response bodies or actual values. This is explicit value checking, not complete OpenAPI response-schema validation. Without the fixture, existing status-only behavior is unchanged. The assertion is a lock-covered composition input, not a new DOT or Anchor property. `HEAD` cannot carry a JSON body assertion.

For GraphQL, the composition supplies the deployment-specific HTTP path and a non-sensitive operation document with optional operation name and variables. Before planning succeeds, `dotx` parses the authoritative SDL, validates every document and variable set, requires exactly one direct top-level field, and requires that field to equal the Anchor-selected coordinate. It never guesses `/graphql`. Queries use HTTP directly. Mutations additionally declare an observable verification query and cleanup mutation. Subscriptions use the GraphQL-over-SSE protocol on a connection established before a separate trigger mutation and finish with a cleanup mutation. Every phase must return HTTP `2xx` without GraphQL errors; the expectation is evaluated against the query response, mutation verification result, or delivered subscription event.

For gRPC, the composition supplies one protobuf-JSON `request` for unary and server-streaming methods, or an ordered `requests` sequence for client and bidirectional streams. The selected Proto3 RPC must declare the standard `NO_SIDE_EFFECTS` idempotency level and be discoverable through explicitly enabled server reflection. Every streaming fixture declares a positive exact `responseCount`; an optional expectation uses `ALL`, `ANY`, `FIRST`, or `LAST` matching. Requests are checked against the authoritative input descriptor before the pinned `grpcurl` runner starts, then every returned document is checked against the authoritative output descriptor. Unsafe, unbounded, wrong- count, or descriptor-incompatible calls fail closed.

For AsyncAPI, the active adapters require a provider `send` operation, an explicit MQTT/MQTT5, NATS, Kafka, or AMQP channel binding, one concrete destination, and a JSON-Schema message payload. The composition declares the message and RFC 6901 correlation pointer; MQTT additionally declares QoS. The Kafka adapter honors the official binding-level `topic` override and restricts topics to the Kafka-supported ASCII alphanumeric, dot, underscore, and hyphen syntax. The digest-pinned official Mosquitto or NATS client subscribes before publishing a fresh UUID-correlated non-retained message. Kafka creates a unique consumer group, waits for its topic assignment before publishing the exact correlated record through the official Apache Kafka JVM image, and removes the group after the bounded check. Kafka records remain subject to broker/topic retention and are never represented as transient. The AMQP 0-9-1 adapter applies the official 0.3.0 channel and operation bindings, declares the exact direct/topic/fanout exchange, binds an exclusive auto-delete queue before publishing, acknowledges the correlated delivery, and closes the connection to remove the queue. It accepts credentials only as a password secret reference. Queue-bound channels, headers exchanges, AMQPS, AMQP 1.0, a binding/protocol mismatch, or MQTT-only QoS on NATS/Kafka/AMQP fail closed.

All adapters execute digest-pinned runners inside the common Docker network or Kubernetes namespace. Runner output is capped at 1 MiB before parsing. Secrets and bearer tokens never appear in the argument vector or state. Contractless, unresolved, unsupported-protocol, and cross-platform wires are reported as unsupported rather than inferred healthy. Diagnostics, documents, variables, requests, messages, and response bodies are not persisted; state contains only normalized identifiers, adapter, result metadata, failure code, and timestamp.

The live contract-probe fixture runs a real GraphQL.js producer with distinct GraphQL-over-SSE connections, a reflected `grpc-js` server implementing all four RPC shapes, Eclipse Mosquitto, NATS, Apache Kafka, and RabbitMQ. It demonstrates the full supported boundary; it does not turn those test services or interaction fixtures into reusable DOT metadata.

```
# Composition-level probe input; the DOT and Anchor still contain only the
# authoritative Contract reference and selector Query.health.
probe:
  graphql:
    path: /graphql
    operationName: Health
    document: |
      query Health {
        health
      }
    expect: {pointer: /data/health, equals: true}
```

```
# Finite bidirectional gRPC probe. The selected Proto method in the Contract
# declares stream input/output and NO_SIDE_EFFECTS; these fields supply only
# instance-specific test messages and acceptance criteria.
probe:
  grpc:
    plaintext: true
    serverReflection: true
    requests:
      - {component: contract-provider, count: 1}
      - {component: contract-consumer, count: 1}
    responseCount: 2
    responseMatch: LAST
    expect: {pointer: /sequence, equals: 2}
```

```
# NATS transport delivery. The selected AsyncAPI channel owns the subject,
# NATS binding, message schema, and send/receive semantics.
probe:
  asyncapi:
    protocol: NATS
    correlationPointer: /id
    message: {id: replaced-at-runtime, severity: critical}
```

```
# Kafka retained-stream delivery. The selected AsyncAPI Kafka 0.5.0 channel
# binding owns the physical topic override and the message schema.
probe:
  asyncapi:
    protocol: KAFKA
    correlationPointer: /id
    message: {id: replaced-at-runtime, severity: critical}
```

```
# Emit the canonical strict Draft 2020-12 schema
dotx compose schema > composition-v1.schema.json

# Generate once, commit, and verify without rewriting
dotx compose lock scenario.yaml
dotx compose lock scenario.yaml --check

# Structure, references, selectors, compatibility, and materializer declarations
dotx compose validate scenario.yaml --locked

# Deterministic resolved topology; no runtime mutation
dotx compose inspect scenario.yaml --locked
dotx compose inspect scenario.yaml --locked --json

# Resolve the catalogue from one immutable remote Git revision
dotx compose inspect scenario.yaml --locked --json \
```

```
  --repository https://github.com/MARE-EU-Project/DOT-Repository.git \
  --git-ref <40-character-commit>

# Render runtime actions without state or runtime mutation
dotx compose apply scenario.yaml --locked --dry-run

# Execute, inspect recorded evidence, and clean up owned resources
dotx compose apply scenario.yaml --locked
dotx compose probe scenario.yaml --locked --wire oidc-provider-to-gateway
dotx compose probe scenario.yaml --locked --wire protected-gateway-to-client
dotx compose status scenario.yaml
dotx compose destroy scenario.yaml

# Review and execute a locked upgrade
dotx compose upgrade scenario-v2.yaml \
  --from scenario-v1.yaml \
  --from-lock scenario-v1.lock.json \
  --locked --dry-run --json
dotx compose upgrade scenario-v2.yaml \
  --from scenario-v1.yaml \
  --from-lock scenario-v1.lock.json \
  --locked
```

Apply records each pending instance before runtime mutation, rolls back earlier instances on failure, and preserves recoverable state when rollback is incomplete or `--keep-on-failure` is selected. `apply --resume` requires an identical composition source, resolved plan, runtime options, inputs, ordering, and previously applied runtime state. Resume also reconciles an interrupted composition-network creation: it reuses the correctly owned network or creates the still-missing network before continuing. Destroy checkpoints every successfully removed instance, so a retry targets only cleanup that did not finish.

Upgrade is a separate lifecycle; `apply` never mutates an already recorded plan. Upgrade requires the exact previous and target locks, the same composition UUID, and a strictly greater target SemVer. Existing instance names must retain their platform and deployment mechanism. Docker image, Docker Compose, Kubernetes manifest, Helm, Vagrantfile, and WASM module instances use their stable managed identities. The private transaction journal stores only paths, identifiers, phases, and digests—not resolved secrets or input payloads. Stateful and host-bound mutations fail before the first runtime action because restoring a deployment specification cannot undo workload data changes.

The deterministic recovery profile injects an interruption immediately before and after network creation, instance deployment, instance destruction, and network removal. It also verifies lock contention, lock release after a panic, partial cleanup, idempotent resume, and fail-closed plan/runtime drift:

```
task testing:compose:lifecycle-recovery
task testing:compose:upgrade
```

The task writes revision-pinned, secret-free JSON evidence under `DOT-Testing/e2e/target/` and is enforced by the Composition lifecycle recovery workflow.

Apply, probe, status, and destroy acquire a private per-composition process lock. A concurrent lifecycle command fails immediately rather than racing local state or runtime mutations.

`compose status` reports:

- the recorded phase and native runtime status of each DOT instance;
- recorded wire compatibility, security configuration, selected endpoint, and materializer count; and
- recorded exposed-Anchor bindings; and
- the latest redacted active Contract-probe result for each probed wire.

Wire and exposure entries remain configuration evidence. Only entries under active Contract-probe evidence demonstrate a runtime request and declared OpenAPI response match, bounded by their timestamp and adapter scope.

## Fail-closed rules

Apply rejects:

- `UNKNOWN` wire compatibility;
- missing, duplicate, or mismatched IDs and versions;
- ambiguous deployments, bindings, services, ports, or OpenAPI server paths;
- unresolved OpenAPI paths or AsyncAPI channel templates;
- undeclared materializer targets and conflicting target values;
- readiness dependency cycles;
- unsupported secret-store/runtime combinations;
- ungranted OCI capabilities, host namespaces, volumes, or CDI devices;
- repeated DOT instances on deployment mechanisms that cannot isolate their resource names; and
- materialized cross-platform wires without an explicit network binding.

Planning may retain `UNKNOWN` to explain incomplete evidence. Execution never silently upgrades `UNKNOWN` to compatible.

## What does not belong in an Anchor

Do not add these as optional Anchor fields:

- hosts, URLs, numeric ports, broker addresses, or Kubernetes names;
- credentials, tokens, certificates, or secret values;
- Camel/Kamelet route syntax;

- transformation logic;
- placement, slice, tenant, trust-zone, latency, or availability policy;
- current health or live authorization state;
- host paths or operator device names; or
- orchestration dependencies.

Use Contracts for interaction semantics, service bindings for deployable ports, composition inputs for operator-controlled configuration, wires for topology, materializers or mediator DOTs for explicit adaptation, and policy engines for deployment/security decisions.

## Choosing an extension point

| Requirement | Preferred representation |
|---|---|
| New API/message operation | Add or update the authoritative Contract and select it from an Anchor. |
| Consumer needs a provider address | Wire materializer targeting declared service configuration. |
| Payload/protocol transformation | Explicit mediator DOT with two Contracts and two Anchors. |
| Startup/readiness ordering | Composition `dependsOn`. |
| Secret or certificate | Reference-only deployment input and platform/external secret provider. |
| Privileged runtime resource | Typed deployment requirement plus exact composition grant. |
| Broker service | Explicit wire `endpoint` selecting the broker service port. |
| Network attachment, eBPF/XDP, WASM import/export, or lifecycle event | Separately reviewed polymorphic binding type when a real use case requires it; do not grow `Anchor` with nullable fields. |

## 6G security composition patterns

The same model supports security-function composition without baking one orchestrator's topology language into every DOT:

- **Zero-trust API chain:** identity-provider and policy-decision DOTs expose token, discovery, and authorization Anchors; an API gateway consumes them and exposes a protected application Anchor. Contracts carry OAuth/OpenID Connect and application semantics, while the composition supplies client-specific secret references and issuer URLs.
- **Closed-loop detection and response:** telemetry, inference, CACAO interpretation, and OpenC2 actuation remain separate DOTs. HTTP and MQTT Contracts define the boundaries; the wire selects the broker endpoint and supplies only deployment-specific channel values.
- **Multi-domain federation:** peer DOT instances can have reciprocal wires while `dependsOn` contains only real readiness dependencies. Trust-domain, roaming, and federation policy

belongs in policy inputs or a dedicated policy DOT, not in the Anchor.

- **Near-real-time RAN control:** E2 or management APIs are Contract-backed Anchors. Host networking, accelerators, eBPF/XDP attachments, and radio-device access are explicit deployment requirements and grants, never inferred from a wire.
- **Observability and digital twins:** metrics, logs, events, and integrity APIs are distinct Anchors. A storage or message broker is selected explicitly as a connection endpoint; a shared host path is not treated as a portable integration boundary.

For cross-domain 6G deployments, a successful plan proves model consistency, not trust. Admission policy must still evaluate publisher identity, artifact provenance, attestation, slice/tenant placement, data-sovereignty constraints, and least-privilege grants before execution. Runtime acceptance should then probe the selected Contract operation over the actual network path.

## Troubleshooting

| Symptom | What to check |
|---|---|
| Anchor UUID is not found | The wire references the pinned DOT version, the UUID is stable, and the Anchor is present in that version's `dot.json`. |
| Contract or selector cannot be resolved | The composition maps the exact Contract ID to a readable source, the source document's `id` matches it, and the selector uses the standard-specific syntax. |
| Anchor has no deployable endpoint | The selected deployment contains exactly one service binding for the Anchor, and its named port exists on that same service. |
| Compatibility is `UNKNOWN` | Inspect `compatibilityReasons` in `compose inspect --json`; add authoritative Contract evidence or use an explicit mediator. Do not override the result. |
| Materializer target is rejected | The target service declares the environment variable or config-file path, the JSON Pointer is valid, and no other wire writes a conflicting value. |
| Security remains unresolved | Bind a Contract-required supported scheme, declare all participant variables, and supply sensitive values by `secretRef` rather than plaintext. |
| Apply reports the composition is busy | Another apply, status, or destroy holds the per-composition lock. Let that command finish; do not delete the lock while a lifecycle process is running. |
| Status shows a configured wire but no active evidence | Run `dotx compose probe scenario.yaml --wire <wire>`. If the wire is skipped, its operation requires a dedicated adapter or application-level acceptance test. Configuration evidence alone is not reachability evidence. |

| Symptom | What to check |
| --- | --- |
| An OAuth probe reports `AUTHENTICATION_ERROR` | Check the bound token URL and client ID, the referenced Secret name/key or environment-store source, required scopes, issuer reachability, and client authorization. Dotx deliberately does not persist the token endpoint's body. |

## Executable examples

- `DOT-Testing/conformance/anchors/` covers all four Contract standards, contractless Anchors, validation failures, bindings, and compatibility.
- `DOT-Testing/e2e/api-gateway-oidc/` is the canonical authenticated request/response lifecycle example.
- `DOT-Testing/composition/poc6-closed-loop/` demonstrates an explicit MQTT broker endpoint and OpenC2/AsyncAPI operations.
- `DOT-Testing/composition/federation-peers/` demonstrates reciprocal wires without false readiness cycles.
- `DOT-Testing/e2e/poc1-ric-aaa/` demonstrates identity, OPA authorization, forwarding, metrics feedback, and deny-list update across seven wires.
- `DOT-Testing/interoperability/pocs.yaml` is the evidence ledger for all eleven MARE PoCs and records every remaining acceptance gap.

Run the repository gate with:

```
task testing:conformance
```

The source-adapted E2E tests prove only their documented boundaries. They do not claim partner-runtime, live-RIC, production-network, trust-policy, or SFC acceptance unless that exact behavior is asserted by the scenario.

# Verification and examples

DOT-Design keeps model correctness, resolved planning, deployment dry-runs, and live behavioral acceptance as separate evidence levels. A passing lower level never implies a higher one.

## Evidence model

| Level | Demonstrates | Does not demonstrate |
|---|---|---|
| Model and schema conformance | `dot.json` shape, Bean Validation constraints, stable references, selectors, service-port bindings, and generated-model compatibility | Contract payload resolution or multi-DOT topology |
| Contract resolution and compatibility | Selected operations exist and normalized semantic, interaction, protocol, media, schema, and security facts can be evaluated conservatively | Runtime reachability or valid credentials |
| Composition validation and plan | Pinned DOTs, wires, endpoints, materializers, parameters, grants, and readiness ordering resolve deterministically | Successful deployment |
| Lifecycle dry-run | The reference executor accepts the supported profile and renders runtime actions without writing composition state | Workload startup or traffic |
| Source-adapted E2E | Pinned upstream logic or artifacts execute behind explicit test adapters and satisfy the assertions documented by that scenario | Unmodified partner runtime, production infrastructure, or full PoC acceptance |
| Live production acceptance | The intended workloads, network, identity, policy, and Contract operations work in the target environment | Nothing beyond the probes and acceptance criteria actually executed |

## Primary verification commands

| Command | Scope | External requirements |
|---|---|---|
| `mvn -pl DOT-Domain test` | Domain value objects, manifests, Anchors, and aggregate validation | Java and Maven only |
| `go -C DOT-CLI test ./…` | CLI parsing, validation, planning, deployment adapters, and lifecycle behavior | Go; individual tests use doubles where appropriate |

| Command | Scope | External requirements |
| --- | --- | --- |
| `task testing:dots:repository` | Every `dot.json` discovered recursively below `DOT-Repository/DOTs/`, with the reusable profile for external/internal bundles that publish adjacent Contract evidence, ownership-path isolation, mandatory per-bundle README files, schema validity, CLI cross-references, and identity uniqueness | Go and the checked-out `DOT-CLI` and `DOT-Repository` revisions; repository validation uses its co-versioned `Schema/`, validates the authoritative `Skills/` catalogue, and resolves every DOT Skill UUID to one categorized capability |
| `task testing:dots:external:dry-run` | Reusable Keycloak/Prometheus DOTs plus runtime-only external Loki/Wireshark definitions: profile-appropriate validation, inspection, deployment dry-run, exact-version Helm rendering, expected service/workload shape, and pinned OCI image metadata | `dotx`, Helm, Python/PyYAML, Docker, and network access to the pinned public artifacts |
| `task testing:quickstart:dry-run` | Execute the command blocks included by Getting started: validate and render Prometheus, then check the locked gateway composition plan and dry-run | `dotx`, `jq`, initialized catalogue, and network access for official Contracts |
| `task testing:quickstart:live` | Run Prometheus at the default and overridden port, assert the build-information API/version, and verify container/network/volume cleanup on success and injected failure | Docker Compose, `dotx`, `curl`, `jq`, network access, and free host ports `18909`, `18919`, and `18929`; all example metrics are deleted |
| `task testing:dots:external:stress` | Repeated Linux deployment, readiness, HTTP/HTTPS access, destroy, and cleanup across two mechanisms per external product: three Helm/Compose pairs plus the Wireshark Docker-image/Compose pair; Keycloak and Prometheus additionally call the exact operations selected from their official Contracts | Docker, Minikube/Kubernetes, Helm, network access, and enough resources for Keycloak, Loki, Prometheus, and Wireshark; emits revision-pinned evidence |
| `task testing:compose:keycloak-prometheus-openapi-planning` | Reusable-profile validation and a locked three-DOT plan proving the stable Keycloak and Prometheus Anchors resolve against exact official OpenAPI payloads, compose compatibly, bind the Keycloak realm parameter, and materialize the expected consumer operation URLs | `dotx`, `jq`, and the checked-out repository/schema revisions; no live runtime |

| Command | Scope | External requirements |
| --- | --- | --- |
| `task testing:compose:contract-standards` | Two canonical test DOT definitions connected through OpenAPI 3.0/3.1/3.2 and AsyncAPI 3.0/3.1 YAML/JSON, Protocol Buffers (including gRPC service declarations), and GraphQL SDL; asserts selectors, participant interactions, endpoints, probe fixtures, and materializers across twenty-one wires | `dotx`, `jq`, and the checked-out repository/schema revisions |
| `task testing:compose:contract-probes:live` | A standards-native test producer and four real brokers prove GraphQL query, mutation verification/cleanup, GraphQL-over-SSE delivery, all four finite side-effect-free gRPC shapes through reflection, correlated non-retained AsyncAPI MQTT/NATS delivery, correlation-scoped Kafka delivery under broker retention, and acknowledged AMQP 0-9-1 delivery through an exclusive probe queue; rejects existing resources and emits redacted success evidence only after verified cleanup | Docker, network access for pinned images, and the checked-out `dotx` |
| `task testing:compose:acceptance` | Authoritative `v1` command/schema parity, every committed lock, twenty-one Contract wires, contractless compatibility, all six runtime dry-runs, service-scoped configuration, recovery/status/destroy, and focused fail-closed controls; emits revision-pinned JSON evidence | A globally installed `dotx` from the checked-out CLI revision, Go, Task, and fixture-specific dry-run CLIs |
| `task testing:contracts:standards` | Extracts the twelve distinct Contract payloads and validates them with dated official OAI schemas, the official AsyncAPI CLI, `protoc`, and GraphQL.js; each family also has an invalid control which must be rejected | Python with PyYAML/jsonschema, Node.js, network access for pinned validators, and `protoc` |
| `task testing:compose:runtime-configuration` | The same catalogued DOT receives declared environment and configuration-file inputs through Docker image, Docker Compose, Kubernetes manifest, Helm chart, Vagrantfile, and Wasmtime/WASI dry-run adapters | `dotx`, `jq`, and runtime CLIs in dry-run rendering mode |

| Command | Scope | External requirements |
| --- | --- | --- |
| `task testing:compose:service-workloads` | A two-service DOT receives isolated defaults, overrides, configuration files, and secret references through exact Compose, Kubernetes, and Helm workload bindings | `dotx`, `jq`, and runtime CLIs in dry-run rendering mode |
| `task testing:compose:service-workloads:live` | Two concurrent copies of the same multi-service workload prove exact environment, secret, config-file, and volume isolation plus readiness, status, resume, selective destroy, surviving-peer health, and cleanup across Docker Compose, Kubernetes, and Helm | Docker Compose, Minikube/Kubernetes, Helm, and network access for runtime images; emits redacted revision-pinned evidence |
| `task testing:compose:runtime-isolation:live` | Two locked instances of one Vagrant DOT and one WASI CLI DOT prove distinct runtime identities, concurrent state, status, selective destroy, peer survival, and cleanup. Repeated fixed-port WASI HTTP fails before mutation. | Vagrant 2.4.9 with libvirt/KVM and Wasmtime 49.0.2; emits redacted, revision-pinned evidence |
| `task testing:cli:coverage` | Complete DOT-CLI test suite plus machine-readable total and high-value package statement-coverage profiles | Go; writes reports below `DOT-Testing/e2e/target/` |
| `cd DOT-CLI && task ci:security` | Scans the Go module dependency graph and reachable code against the current OSV vulnerability database with the exact pinned scanner used by CI | Go and network access; intentionally separate from deterministic `task ci` |
| `task testing:runtime:kubernetes-helm:live` | Real Kubernetes-manifest and Helm-chart deploy, rollout readiness, status, service-scoped environment materialization, and destroy lifecycle | Docker, Minikube, Kubernetes, Helm, and network access for runtime images |
| `bash DOT-Testing/runtime/wasm/live.sh` | Real Wasmtime lifecycle for WASI CLI; CI additionally builds the pinned Bytecode Alliance `wasi:http/proxy` sample and proves an HTTP response | `dotx`, Wasmtime, and the optional prebuilt HTTP component path |
| `task testing:conformance` | Cross-language Anchor and composition gate, including all deterministic PoC fixtures and lifecycle dry-runs | Java, Maven, Go, Task, and fixture-specific local tools |
| `mvn clean install` | Full Java domain build, schema generation, coverage gates, and SBOMs | Java and Maven only |
| `task testing:e2e:<scenario>` | Scenario-specific live behavior | Read that scenario's README first; commonly Docker and network access |

# Test asset layout

```
DOT-Testing/
 ├──── acceptance/        versioned, implementation-neutral consumer profiles
 ├──── conformance/       deterministic valid and invalid DOT fixtures
 ├──── composition/       read-only validate, inspect, and dry-run scenarios
 ├──── coverage/          executable code-coverage ratchets
 ├──── e2e/               opt-in deployable behavioral scenarios
 ├──── interoperability/  revision-pinned evidence and remaining-gap ledger
 └──── runtime/           isolated native-runtime lifecycle harnesses
```

`testing` is repository-level infrastructure, not a Maven module and not a deployable DOT. Authoritative test-only definitions live under `DOT-Repository/DOTs/testing/` and are selected from compositions with `repository://<dot-id>@<version>`. Intentionally invalid conformance inputs stay under `DOT-Testing/conformance/`; they must never be admitted to the repository catalogue. Generated secrets, logs, reports, coverage profiles, and materialized configuration belong under scenario-local `target/` directories and must not be committed.

The focused `task testing:anchors:acceptance` gate verifies the versioned DOT Anchor consumer profile. It resolves pinned test definitions from `DOT-Repository`, verifies the composition lock, and asserts the complete cross-standard plan without a deployed workload.

The separate `task testing:compose:acceptance` gate freezes the complete `dot.mare6g.eu/v1` production profile. Its machine-readable source is `DOT-Testing/acceptance/composition/v1/profile.json`; the gate binds that source to CLI help, the tracked schema, repository locks, positive evidence, and negative fail-closed behavior. Runtime rows distinguish underlying single-DOT live lifecycle proof from multi-DOT composition lifecycle proof.

The focused `task testing:compose:contract-standards` gate complements that API profile with an executable composition matrix. It proves all six supported standard/format representations between the same provider and consumer DOTs; it does not promote planning evidence to live protocol or application acceptance.

The separate `task testing:contracts:standards` gate validates those payloads outside the DOT resolvers with standards-native tooling. The positive fixtures must pass and deliberately malformed OpenAPI, AsyncAPI, Protobuf, and GraphQL controls must fail. This catches both invalid examples and accidentally permissive validator invocations.

The runtime-configuration gate separately proves that selecting another executable deployment does not discard a service's declared environment or configuration-file surface. Each adapter is exercised first with its declared `defaultValue` and `defaultContent`, then with explicit composition overrides; the gate validates the cached default document, resolved service id, override values, and runtime-specific dry-run. Runtime-native storage, secret, privilege, and device semantics retain dedicated tests rather than being flattened into a misleading universal volume abstraction.

The multi-service live profile is the higher evidence layer for the exact workload bindings. Compose projects and their named volumes are keyed by the composition runtime identity; Kubernetes repetitions use separate namespaces; Helm repetitions use identity-derived release names in one

namespace. The profile verifies that destroying one copy leaves the other healthy and that dotx-created ConfigMaps/Secrets are cleaned without deleting referenced operator-owned Secrets. The tracked Helm chart is packaged from the pinned checkout, and its SHA-256 is included in the redacted evidence artifact.

# Coverage claims and gates

Coverage is reported by evidence dimension; the project does not collapse different meanings of "covered" into one percentage:

| Dimension | Gate | Interpretation |
|---|---|---|
| Repository inventory | `dotx validate DOT-Repository`, `dotx repository audit`, and the DOT repository conformance workflow | Must be 100%: every filesystem definition is discovered, parsed, schema-valid, cross-reference-valid, uniquely identified, and stored in a supported category; no deployable copy may remain outside `DOT-Repository`. |
| CLI statements | `task testing:cli:coverage` | The whole CLI and every package containing executable statements must retain 100% statement coverage. The constant-only `feature/sources` package has no testable statements. This is a code-execution metric, not proof that a workload or security control works. |
| Java instructions and branches | `mvn verify` through JaCoCo 0.8.15 | The Java domain module must retain 100% instruction and branch coverage. JaCoCo HTML/XML/CSV reports are uploaded with the Java build evidence. |
| Native runtime lifecycle | WebAssembly and Kubernetes/Helm runtime E2E workflows | Proves real Wasmtime CLI and HTTP behavior plus Minikube manifest/Helm lifecycle. It does not imply that every external DOT workload is accepted. |
| Composition scenarios | `task testing:e2e:all:dry-run` | Every listed deterministic scenario must validate, inspect, and complete its supported lifecycle dry-run. |
| PoC interoperability | `DOT-Testing/interoperability/pocs.yaml` plus scenario-specific live workflows | Each of the eleven PoCs has traceable evidence, while remaining adapters, unavailable partner components, and unproved production behavior stay explicit. |

The repository validation gate fails on duplicate identities, mis-scoped definitions, invalid DOTs, invalid Skill/category assets, or unresolved DOT Skill references. The workspace audit separately classifies uncatalogued, identical, divergent, and malformed definitions and fails on every deployable `dot.json` outside `DOT-Repository`. `DOT-Testing/conformance/` fixtures are reported but exempt because invalid definitions are their purpose. During the staged migration, run:

```
dotx repository audit --repository DOT-Repository --workspace . --json
```

The coverage gates enforce 100% Java instruction/branch coverage and exact 100% Go statement coverage overall and for every package containing executable statements. Those thresholds do not replace negative tests, lifecycle checks, live probes, or partner acceptance.

# Canonical Anchor scenario

`DOT-Testing/e2e/api-gateway-oidc/` is the primary tutorial and lifecycle fixture. Its composition resolves three independently valid, test-scoped definitions from `DOT-Repository`:

- Keycloak provides OpenID Connect discovery and token operations;
- APISIX requires discovery information and provides a protected operation; and
- an OAuth client requires the token and protected operations.

The composition uses three OpenAPI Contracts, three wires, explicit security configuration, Kubernetes Secret references, configuration-file and environment materializers, readiness dependencies, and one exposed Anchor.

```
task testing:compose:keycloak-apisix-oauth
task testing:demo:keycloak-apisix-oauth:dry-run

# Opt-in live execution
task testing:demo:keycloak-apisix-oauth:live
```

Read DOT Anchors and Composition alongside `DOT-Repository/Compositions/keycloak-apisix-oauth/composition.yaml`.

# MARE PoC-derived coverage

The interoperability ledger pins the source revision used for every analysis, records the modeled interactions, identifies the executable fixtures, and keeps remaining acceptance gaps explicit. The current repository contains source-adapted evidence for all eleven PoCs; this is not a claim of unmodified partner-runtime acceptance.

| PoC | Composition focus | Fixture |
| --- | --- | --- |
| 1 | OIDC identity, OPA authorization, RIC forwarding, metrics feedback, and deny-list update | `DOT-Testing/composition/poc1-ric-aaa/` |

| PoC | Composition focus | Fixture |
|---|---|---|
| 2 | Repeated Flower trainers over gRPC/Protobuf and exposed BentoML inference | `DOT-Testing/composition/poc2-federated-inference/` |
| 3 | Five-stage contractless JSON insider-threat analytics dataflow | `DOT-Testing/composition/poc3-insider-threat-pipeline/` |
| 4 | Prometheus metrics, Elasticsearch logs, Grafana, and Kibana configuration | `DOT-Testing/composition/poc4-continuous-monitoring/` |
| 5 | Monitoring-to-analysis-to-assessment-to-mitigation security pipeline | `DOT-Testing/composition/poc5-security-pipeline/` |
| 6 | OpenAPI plus brokered AsyncAPI/MQTT, CACAO, OpenC2, and NWDAF remediation | `DOT-Testing/composition/poc6-closed-loop/` |
| 7 | Least-privilege packet capture, explicit CloudEvents mediation, and trust scoring | `DOT-Testing/composition/poc7-trust-scoring/` |
| 8 | cAdvisor/Prometheus observability and physical/twin integrity checks | `DOT-Testing/composition/poc8-ndt-observability/` and `poc8-twin-integrity/` |
| 9 | Amarisoft/Open5GS collection, InfluxDB storage, and data export | `DOT-Testing/composition/poc9-edge-ddos-collection/` |
| 10 | OIDC-protected capability exposure, SPARQL, OPA policy decisions, and observability | `DOT-Testing/composition/poc10-secure-capability-exposure/` |
| 11 | Reciprocal federation peers and local-core adapters with separate readiness edges | `DOT-Testing/composition/federation-peers/` |

The authoritative coverage and gap list is `DOT-Testing/interoperability/pocs.yaml`. The official project descriptions are available from the MARE PoC catalogue.

# Negative evidence is intentional

Rejected fixtures are part of the design evidence. They demonstrate that the model does not invent connectivity where the pinned source lacks a declared consumer input, does not treat stdout or a host path as a portable connection, and does not silently grant privileged runtime access.

Examples include:

- PoC7 rejects a collector-to-scorer wire when the collector declares no such Anchor or configuration input;
- PoC8 rejects an undeclared Prometheus-to-NDT handoff; and
- invalid conformance fixtures cover missing Contracts, selectors, ports, duplicate bindings, and incompatible Anchor pairs.

Failing closed is a successful test outcome when the available evidence cannot support the proposed composition.

## Adding or promoting a scenario

1. Pin the exact external source revision used as evidence.
2. Document what the source actually produces and consumes.
3. Add deployable definitions according to ownership: third-party software under `DOT-Repository/DOTs/external/`, MARE-owned software under `DOT-Repository/DOTs/internal/`, and only synthetic fixtures under `DOT-Repository/DOTs/testing/`. Record their evidence and purpose in the applicable test evidence ledger and select them by exact `repository://<uuid>@<version>` identity.
4. Define Contracts only for supported, observable interactions.
5. Use contractless Anchors only when no supported Contract standard applies.
6. Declare all service ports, configuration targets, privileges, and devices.
7. Add negative fixtures for tempting but unsupported connections.
8. Start under `DOT-Testing/composition/` with validate and plan assertions.
9. Add a lifecycle dry-run before any live execution.
10. Promote to `DOT-Testing/e2e/` only with isolated resources, explicit prerequisites, reliable cleanup, and externally observable assertions.
11. Record what remains unproved in the interoperability ledger.

## Interpreting results



A successful plan proves model consistency. A successful apply proves that the supported deployment adapters completed. Only an active scenario probe can prove that traffic, authentication, authorization, policy, or security behavior worked as intended.

## External DOT examples

The external-product examples demonstrate how an existing open-source workload can be described as a DOT without modifying that workload. The upstream project remains authoritative for the software, configuration, and interface Contract. The DOT descriptor selects a pinned artifact and declares the portable service surface that `dotx` can validate and deploy.

Upstream documentation is useful source evidence, but it is not acceptance evidence. DOT-Design therefore tests each example at separate levels:

1. direct `dotx validate`, using the reusable publication profile where applicable;
2. Contract-selector and DOT-to-DOT Anchor composition for reusable definitions;
3. offline inspection and deployment rendering;

4. networked rendering of the exact pinned Helm charts;
5. one real Linux lifecycle on Kubernetes/Helm or Docker; and
6. bounded repeated lifecycle stress with cleanup verification.

## Canonical external catalogue

The external catalogue currently has one root for each of these 17 products: APISIX, cAdvisor, Elasticsearch, Filebeat, Grafana, GraphDB, InfluxDB, Keycloak, Kibana, Loki, Eclipse Mosquitto, Prometheus node-exporter, OPA, OpenTelemetry Collector, Oxigraph, Prometheus, and Wireshark. PoC compositions reuse those identities and supply routes, realms, policies, scrape targets, data sources, credentials, files, and deployment selection as instance inputs. They do not create `pocN-*` copies of the product.

The following four products form the bounded release stress matrix. They are not the complete external catalogue.

## Release stress matrix

| DOT | Catalogue status | Mechanisms in the same DOT | Contract/Anchor surface | Live assertions |
|---|---|---|---|---|
| `Keycloak-DOT` | Stable reusable external DOT | Bitnami `HELM_CHART 25.2.0`; official Keycloak `DOCKER_COMPOSE`; official and PoC1 evidence `DOCKER_IMAGE` variants | Exact Keycloak 26.3.3 Admin REST OpenAPI plus an attributed OIDC endpoint projection; Admin realm, JWKS, and client-credentials provider Anchors | Exact StatefulSet binding and Compose `start-dev`; readiness; obtain an admin token and assert `GET /admin/realms/master` returns the master realm through both mechanisms |
| `Loki-DOT` | Runtime sample, not reusable | Grafana Community `HELM_CHART 18.15.1`; official Loki `DOCKER_COMPOSE` image pinned by digest | No fabricated Contract: the pinned release has HTTP documentation and internal multi-file protobuf APIs, but no supported self-contained machine-readable Contract for the exposed public HTTP surface | Exact single-binary StatefulSet binding and `/ready`; Compose `/ready` |

| DOT | Catalogue status | Mechanisms in the same DOT | Contract/Anchor surface | Live assertions |
|---|---|---|---|---|
| `Prometheus-DOT` | Stable reusable external DOT | Prometheus Community `HELM_CHART 29.37.0`; official Prometheus `DOCKER_COMPOSE` image pinned by digest; `DOCKER_IMAGE` for composition-owned scrape configuration | Exact Prometheus 3.15.0 OpenAPI 3.1 artifact for build information and PromQL; explicit contractless scrape-target requirement materialized through `prometheus.yml` | Exact server Deployment binding; readiness; assert `/api/v1/status/buildinfo` reports version `3.15.0` through both mechanisms |
| `Wireshark-DOT` | Runtime sample, not reusable | LinuxServer versioned `DOCKER_IMAGE`; equivalent `DOCKER_COMPOSE` image pinned by digest | No fabricated Contract: the image exposes a human web desktop, not a supported machine API | Container state/environment/volume/ports plus HTTPS for both mechanisms |

Each product is kept under `DOTs/external/` because it is owned outside MARE, with its supported deployment mechanisms in the same `dot.json`. Ownership does not imply interoperability maturity. Keycloak and Prometheus publish adjacent official Contract evidence and pass the strict reusable profile. Loki and Wireshark currently provide runtime evidence only; they must not be misrepresented as Contract-backed interface interoperability. Chart versions, Compose image indexes, and the direct Wireshark image tag are deliberately pinned. `latest` is convenient for exploration but unsuitable for release evidence because a repeated run could execute different bytes. The Wireshark OCI index currently contains Linux amd64 and arm64 manifests. Stable `0.2.0` retained a historical cross-built arm64 preview, but subsequent DOT-Design release packaging and automated acceptance intentionally target Linux amd64 only.

Authoritative upstream references:

- Keycloak 26.3.3 release assets, Keycloak Admin REST API, Bitnami Keycloak chart, and Keycloak containers
- Grafana Loki Helm documentation
- Prometheus 3.15.0 official OpenAPI 3.1 artifact, Prometheus Community chart, and Prometheus installation
- LinuxServer Wireshark image

## Run the examples

Run commands from the `DOT-Design` root with a `dotx` built from the checked-out `DOT-CLI` revision.

```
# Fully offline and non-mutating
task testing:dots:external:offline
```

```
# Downloads and renders the pinned charts, but does not deploy them
task testing:dots:external:dry-run

# Deploys each example once and cleans it up
task testing:dots:external:live

# Repeats the complete lifecycle three times by default
task testing:dots:external:stress
```

The live profiles require Linux, Docker, Kubernetes, Helm, `kubectl`, `curl`, `jq`, and network access to the pinned public artifacts. `dotx tool check` starts Minikube when the current environment has no selected cluster. Set `DOTX_E2E_SKIP_HELM=1` or `DOTX_E2E_SKIP_DOCKER=1` only for focused local diagnosis; a release acceptance run must not skip either runtime.

## What the stress profile adds

`DOTX_E2E_STRESS_REPEATS` accepts a bounded value from 1 through 5. Every cycle uses a new namespace, Docker runtime name, Compose project names, marker, and host-port range. It executes validation again before runtime mutation, selects each mechanism explicitly, verifies workload readiness and HTTP/HTTPS reachability, and destroys all resources. The runner then fails if the cycle namespace, Wireshark container, or any of the four Compose projects still exists. If a Kubernetes deployment or readiness check fails, the runner records pod, workload, event, and bounded container-log diagnostics before cleanup and skips only the assertions that are causally blocked by that failed workload. The root failure still fails the cycle; dependent timeout noise is not counted as independent evidence.

The resulting matrix contains eight cells:

- Keycloak: `HELM_CHART` and `DOCKER_COMPOSE`;
- Loki: `HELM_CHART` and `DOCKER_COMPOSE`;
- Prometheus: `HELM_CHART` and `DOCKER_COMPOSE`; and
- Wireshark: `DOCKER_IMAGE` and `DOCKER_COMPOSE`.

This is a runtime-mechanism matrix, not a reusable-interface count. Only the Keycloak and Prometheus cells carry official machine-readable Contracts and Contract-backed Anchors. A separate locked three-DOT composition wires those two providers to a headless consumer, resolves the exact official selectors, binds the Keycloak `{realm}` parameter, and asserts the resulting operation paths and environment materializers:

```
task testing:compose:keycloak-prometheus-openapi-planning
```

```
DOTX_E2E_STRESS_REPEATS=5 task testing:dots:external:stress
```

Evidence is written to `DOT-Testing/e2e/target/external-dots/`:

- one log per cycle;
- `summary.json`, including the DOT-CLI and DOT-Repository revisions; and
- `SHA256SUMS` covering the summary and logs.

The dedicated `External DOT examples` GitHub Actions workflow runs the non-mutating profile for relevant changes. Its weekly and manually dispatched Linux job runs the stress profile and uploads the evidence bundle. Push, schedule, and manual runs use distinct concurrency lanes, so requesting a live stress run cannot cancel the required push conformance result for the same revision.

## Recorded `0.2.0-rc.3` evidence

External DOT examples run 38708940363 completed two independent Linux lifecycle cycles against DOT-CLI `f8cf796afd985cc0514d42fb149ef67b6a1b7b53` and DOT-Repository `76585f6ec68a892f6cc6d48c9b2acdce6b08d48d`. Each cycle passed all 47 checks and cleaned up its resources. The complete eight-cell mechanism matrix was:

- Keycloak: Helm chart and Docker Compose;
- Loki: Helm chart and Docker Compose;
- Prometheus: Helm chart and Docker Compose; and
- Wireshark: Docker image and Docker Compose.

For both Keycloak mechanisms, the test invoked the exact Admin REST operation selected from the pinned official Contract. For both Prometheus mechanisms, it invoked the exact build-information operation selected from the pinned official Contract. The uploaded evidence contains `cycle-1.log`, `cycle-2.log`, `summary.json`, and `SHA256SUMS`; local checksum verification of the downloaded artifact passed for both cycle logs and `summary.json`. This evidence supersedes the earlier `rc.2` run because it exercises Keycloak-DOT `0.1.1` with the corrected Helm resource profile and pins the exact candidate CLI and repository revisions.

## Interpreting a pass

A pass means the pinned public artifacts satisfied the assertions executed in the disposable test environment. For Keycloak and Prometheus this includes a successful request to the exact operation selected by the DOT Anchor, in addition to readiness. It does not establish production sizing, availability, security hardening, backup/restore, or every upstream feature. In particular, the Wireshark example checks its web application surface; it does not grant packet-capture capabilities, host networking, or a CDI device. Those requirements must be declared and explicitly authorized in a separate acceptance profile.

When adding another external DOT, place it under `DOTs/external/` because of its ownership and provide an immutable runtime artifact. Use a vendor/project- official Contract whenever available. Otherwise either pin a runtime-extracted Contract, maintain a clearly identified derived Contract with live conformance, or expose an adapter-owned Contract. Every Contract asset declares its authority, source revision, license, and SHA-256. A runtime-only external DOT remains valid without a `contracts/` evidence directory but makes no interoperability claim. Add production claims only after a test explicitly proves them.

# Integration handoff

This handoff defines the interoperability boundary for a team implementing a catalogue, composer, Security Function Composer adapter, or other DOT consumer. It intentionally does not prescribe a service, database, API style, language, or framework.

## Authoritative inputs

The reference source of reusable definitions is `DOT-Repository`. Resolve a DOT by its immutable UUID and pinned version, then verify the returned `dot.json` matches both. Resolve referenced Contract payloads without copying their protocol, schema, media, or security facts into an Anchor.

Use these commands as executable reference behavior:

```
dotx repository inspect <dot-id>@<version> --repository DOT-Repository --json
dotx validate DOT-Repository
dotx compose validate composition.yaml
dotx compose inspect composition.yaml --json
```

An implementation may expose equivalent behavior through REST, gRPC, events, an SDK, or an embedded library. Its external contract should be versioned and tested independently from the DOT model.

## Disposition of legacy SFC requests

The original SFC issues 6 through 9 predate the stable repository-backed model. Their consumer needs are handled at the integration boundary without creating parallel sources of truth:

- **Parameter forms:** derive an SFC-facing form from exact service-scoped environment and configuration declarations plus authoritative Contract schemas. Composition validation rejects missing or ambiguous required inputs before runtime mutation. A duplicate root `params` model is not defined.
- **Skill search:** resolve stable `Skill` UUID references through the authoritative `DOT-Repository/Skills/` catalogue. Every referenced capability has one category there; `dotx repository list --skill <uuid-or-name>` and `--category <uuid-or-name>` expose the tested consumer projection without copying names or categories into every DOT as a second controlled vocabulary.
- **Catalogue API:** `DOT-Repository` and `dotx repository` define reference behavior. A catalogue team may expose equivalent queries through any versioned API; DOT-Design does not own or prescribe that service.
- **Low-barrier authoring:** a DOT author may hand-write one `dot.json`, place its folder under the appropriate recursive `DOT-Repository/DOTs/` category, and validate the whole filesystem catalogue with `dotx validate DOT-Repository`. There is no duplicate inventory file or wrapper script. This path requires no Maven, Jsonnet, MongoDB, or running service. `dot.json` remains the sole deployment definition; a parallel YAML format and inline copies of referenced entities

are intentionally not introduced.

These dispositions settle DOT-side model ownership. They do not replace joint SFC acceptance or prove a partner catalogue/search implementation.

# Required semantics

An interoperable consumer must:

- address DOTs, Contracts, and Anchors by stable UUID and pin DOT versions;
- resolve contract-backed selectors against the authoritative OpenAPI, AsyncAPI, Protocol Buffers, or GraphQL document;
- use `anchorBindings` to map a logical Anchor to a named service port;
- keep instance topology, endpoints, parameter values, credentials, and policy outside reusable DOT definitions;
- evaluate compatibility conservatively and preserve `UNKNOWN` as distinct from `COMPATIBLE`; and
- keep secret values out of DOTs, plans, persisted state, and logs.

Contractless Anchors are valid only when a supported Contract standard cannot describe the interaction. Their explicit direction, protocols, media types, and semantic types must be complementary before a wire is accepted.

# Acceptance evidence

The versioned consumer profile is under `DOT-Testing/acceptance/dot-anchors/v1/`. It resolves two catalogued test DOTs, verifies the committed composition lock, and plans 21 compatible wires across all supported Contract standards and versions.

```
task testing:anchors:acceptance
task testing:conformance
```

Passing the deterministic profile proves discovery and planning semantics. A joint live test must separately prove endpoint reachability, credential and token handling, authorization, policy enforcement, and application behavior.

# Future implementation freedom

The removed catalogue-service prototype is not part of the production architecture. A future team may replace it or implement discovery elsewhere in a different language or framework. Compatibility is judged by the stable DOT model and executable acceptance profile, not by reproducing the prototype's internal classes or former HTTP response shapes.

# Release notes

## Unreleased

Contract `standard` now accepts only `OpenAPI`, `AsyncAPI`, `Protobuf`, and `GraphQL`, generated from the Java domain enum's Jackson values. This is an intentional input-format change: uppercase values and case/whitespace aliases are rejected. Migrate Contract declarations and local asset wrappers, then explicitly regenerate and review affected composition locks with `dotx compose lock <composition.yaml>`. Existing locks fail closed; do not reuse saved runtime state against a changed graph. The `format` values remain `YAML`, `JSON`, `PROTOBUF`, and `GRAPHQL`. Use the pinned development CLI; previously released binaries retain the earlier vocabulary. Stable-tag docs remain unchanged.

`dotx compose inspect` replaces `compose plan` as the read-only command for resolving and explaining a composition. Flags, resolved JSON, locks and runtime state remain unchanged. Old invocations report the new command name. Use an installed development checkout; the original v0.2.5 binary retains the old name.

The official ecosystem presenter route now connects reuse across Docker and Kubernetes, secure multi-DOT policy enforcement and native stateless recovery. Shared preparation/preflight cover five lanes; the full clean-source rehearsal runs seven live paths twice and links cleanup-verified child evidence. The existing runtime workflow offers an explicit long-running full-demo dispatch.

Development CLI builds add `compose topology --status`: a locked-plan graph with scoped, bounded runtime observations and a distinct historical probe table. Matching saved execution inputs are required before runtime queries. Presenter pauses use this view without treating running workloads or a recent probe receipt as current API acceptance. Mermaid export remains plan-only; there is no portable-model or schema change, and no new release is implied.

Contracts now resolve only from their declaring DOT bundle or a complete digest-pinned URL declaration. The repository-level Contract catalogue and validator fallback have been removed; all remaining assets live in their owning bundles. Repository validation still rejects conflicting definitions of the same Contract UUID. The shared local/GitHub repository gate additionally validates every bundle in isolation and offline.

Getting started now begins with a checksummed Linux-amd64 CLI installation and a complete Prometheus Docker Compose lifecycle, with API/version verification, cleanup, and troubleshooting. Contributor toolchains and shared CI gates have their own guide. Antora includes the same command blocks executed by `testing:quickstart:dry-run` and `testing:quickstart:live`; the external-DOT workflow rehearses both, including port overrides and cleanup after failure. Development pages now export their own PDF alongside the latest stable PDF, matching the existing download link in the site header.

The Antora site now includes a guided demo for presentations and onboarding, with a tested five-minute read-only path, expected evidence, an optional live Contract-probe path, and explicit support boundaries.

`dotx compose apply --dry-run` no longer creates or changes the composition state/lock directory. Lifecycle rendering therefore works with a read-only cache while retaining the same fail-closed

planning and runtime checks.

## 0.2.4

`0.2.4` is a backward-compatible Linux-amd64 hardening release for the stable `dot.mare6g.eu/v1` single-host composition profile. It preserves the DOT, Contract, Anchor, composition-schema, repository-layout, deployment, and CLI interfaces published in `0.2.3`. The exact pinned revisions are DOT-CLI `e88daa83221ae5d49a1317b7412045ab2b78315f` and DOT-Repository `247606d4a55c5e85a822f1f82eca52a3574b5a04`.

DOT-CLI now exposes a local/GitHub `task ci:security` gate using the pinned stable OSV-Scanner release. It checks the module dependency graph and performs Go call analysis against the current OSV database. The network-backed scan is kept separate from deterministic `task ci` because vulnerability data changes independently of a source revision.

Standalone DOT-CLI packaging now emits a deterministic CycloneDX 1.6 SBOM for the exact Linux-amd64, CGO-disabled build profile. Release tests validate the SBOM identity, platform properties, dependency content, omitted timestamp and serial number, checksum coverage, and byte-for-byte equality across two package builds. Downloaded-binary acceptance retains the published SBOM as evidence.

Readable workflow badge publication now refreshes the complete status set on a coalesced 15-minute schedule, with manual dispatch available for immediate refresh. It no longer consumes one self-hosted-runner job for every completed workflow in the already large validation matrix.

Wasmtime lifecycle status and inventory checks now tolerate the short, bounded interval in which Linux procfs may expose an empty process command line during `exec`. A non-empty command line that does not match the recorded Wasmtime module identity is still rejected immediately, preventing lifecycle operations from acting on a reused PID.

Antora publication now deduplicates an exact stable tag from its identical release-commit worktree, closing the tag-push race while preserving immutable stable documentation. Current-profile documentation is guarded against the machine-readable acceptance profile: twenty-one wires, all four finite side-effect-free gRPC shapes, and MQTT, NATS, Kafka, and AMQP 0-9-1 delivery.

The complete pushed workflow matrix passed on `77d8f099e5f42430f1d30e8d01c27b2c534b8d49`. It includes 100% DOT-Domain instruction and branch coverage, 100% DOT-CLI aggregate and per-executable- package statement coverage, repository conformance, all twenty-one cross-standard wires, real-producer and real-broker active probes, Docker, Kubernetes, Helm, Vagrant, and WebAssembly lifecycle evidence, and versioned Antora publication. DOT-CLI `v0.2.4` additionally passed two-pass reproducible packaging and downloaded-binary acceptance.

The release publishes the versioned DOT-Domain JAR, DOT and Composition v1 JSON Schemas, JSON and XML CycloneDX SBOMs, the Linux-amd64 `dotx` binary, an exact source manifest, and checksums. Linux amd64 remains the only supported release target. DOT-Registry REST API work, native arm64, non-Linux targets, multi-host orchestration, and joint Security Function Composer or partner acceptance remain outside this release boundary.

# 0.2.3

`0.2.3` is a backward-compatible production-hardening release for the stable `dot.mare6g.eu/v1` single-host composition profile. It preserves the DOT, Contract, Anchor, composition-schema, repository-layout, and deployment interfaces published in `0.2.2` while broadening standards-authorized runtime evidence and strengthening deployment-configuration acceptance. The exact pinned revisions are DOT-CLI `b07ec3b568b59cb25631a9b51700a14ee124bb05` and DOT-Repository `247606d4a55c5e85a822f1f82eca52a3574b5a04`.

The cross-runtime configuration matrix now exercises every executable adapter through separate default-only and explicit-override paths. It verifies non-sensitive `defaultValue` and `defaultContent` materialization, explicit environment/file precedence, resolved service targeting, and redacted dry-run diagnostics.

Contract-authorized gRPC probes now cover unary, server-streaming, client-streaming, and bidirectional methods. All methods still require the Proto3 `NO_SIDE_EFFECTS` option and server reflection. Streaming fixtures must declare an exact finite response count; input and output messages are validated against the authoritative descriptors; and expectations explicitly select `ALL`, `ANY`, `FIRST`, or `LAST` response matching.

AsyncAPI delivery now supports NATS core messaging beside MQTT. NATS fixtures require the official channel binding and a concrete non-wildcard subject, then prove UUID-correlated subscribe-before-publish delivery through digest-pinned official NATS client and server images. The live standards fixture exercises all four gRPC shapes plus MQTT and NATS while retaining redacted state evidence.

AsyncAPI delivery also supports Kafka through the official 0.5.0 channel binding. `dotx` honors a binding-level `topic` override, rejects invalid Kafka topic names and MQTT-only QoS, waits for a correlation-scoped consumer-group assignment before publishing, and verifies the exact correlated record through the digest-pinned official Apache Kafka 4.3.1 JVM image. The ephemeral consumer group is removed; the record itself remains subject to the topic's broker retention policy. The canonical runtime-only Kafka DOT expands the locked standards composition to twenty wires and three real message brokers.

AsyncAPI delivery now also supports AMQP 0-9-1 through the official 0.3.0 binding. `dotx` validates exact exchange, routing-key, virtual-host, durability, auto-delete, and delivery-mode semantics; binds an exclusive auto-delete probe queue before publish; acknowledges and correlation-matches the delivered JSON; and obtains RabbitMQ credentials only through a secret reference. Queue-bound channels, headers exchanges, AMQPS, and AMQP 1.0 fail closed. The canonical digest-pinned RabbitMQ 4.3.6 runtime DOT expands the locked standards composition to twenty-one wires and four real message brokers.

The complete pushed workflow matrix passed on `79807aacebe7aab1112fe82104898ab4a0929db0`, including DOT-CLI CI, Contract active probes, compatibility matrix, coverage, Kubernetes and Helm runtime E2E, DOT repository conformance, and versioned Antora publication.

The release publishes the versioned domain JAR, DOT and Composition v1 schemas, CycloneDX JSON/XML SBOMs, Linux-amd64 `dotx` binary, immutable release manifest, and checksums. The tag workflow performs deterministic two-build packaging before publication, and post-publication

acceptance downloads and exercises the released binary.

Existing `0.2.0` through `0.2.2` definitions, compositions, and locks remain valid. Unsafe or unbounded gRPC, AsyncAPI transports other than MQTT, NATS, Kafka, and AMQP 0-9-1, AMQPS/AMQP 1.0, multi-scheme authentication, multi-host orchestration, automatic cross-platform gateway provisioning, general stateful-data migration, native arm64, non-Linux executables, DOT-Registry REST API work, and joint SFC/partner acceptance remain outside this Linux-amd64 release.

# 0.2.2

`0.2.2` is a backward-compatible extension of the stable `dot.mare6g.eu/v1` single-host composition profile. It retains the DOT, Contract, Anchor, composition-schema, repository-layout, and deployment interfaces from `0.2.1` while completing the standards-authorized active-probe profile. The exact pinned revisions are DOT-CLI `2dafb492bb657da4d6e6dfe3eab4e9ee4ee229c9` and DOT-Repository `31357e57ab1332e93de6afc25a2b0d67aafed857`.

GraphQL query probes validate documents, variables, and optional exact response expectations against the authoritative SDL. Mutations require verification and cleanup operations; GraphQL-over-SSE subscriptions require a separately triggered event plus cleanup. Unary gRPC probes require server reflection, protobuf-JSON request validation, and the Proto3 `NO_SIDE_EFFECTS` option. AsyncAPI MQTT probes require an explicit binding, concrete topic, JSON-Schema-valid correlated message, QoS, subscribe-before-publish ordering, bounded timeout, and exact non-retained delivery. All fixtures remain composition-owned and persisted evidence remains redacted.

The fifteen-wire deterministic matrix covers OpenAPI 3.0/3.1/3.2, AsyncAPI 3.0/3.1, Proto3, and GraphQL. A real GraphQL.js/gRPC producer and Eclipse Mosquitto broker provide live Docker acceptance. The complete pushed workflow matrix passed on `70b0e94760d60bc0fc37f887db0466796627829a`, including Contract active probes, coverage, compatibility, and Antora publication.

Existing `0.2.0` and `0.2.1` definitions and compositions remain valid; probe fields are optional and additive. Streaming or safety-unclassified gRPC, non-MQTT AsyncAPI transports, multi-scheme probe authentication, native arm64, non-Linux executables, multi-host orchestration, DOT-Registry REST API work, and joint SFC/partner acceptance remain outside this Linux-amd64 release.

# 0.2.1

`0.2.1` is a backward-compatible hardening release for the stable `dot.mare6g.eu/v1` single-host composition profile. It retains the `0.2.0` DOT, Contract, Anchor, composition-schema, repository, and deployment interfaces. The exact pinned revisions are DOT-CLI `82d7de45800f4e4a435ef08202d4f1e42c16eca8` and DOT-Repository `2fe0ecc81f9caea43c5bf6e0dc824c3e0faf34e3`.

The patch adds overflow-safe SemVer precedence, atomic WASM module-cache replacement and restart-profile verification, fail-closed empty Docker port forwarding, correct Minikube home-directory expansion, and child-process checks for detached Kubernetes forwarding. Repository commands now share one verified private-cache acquisition path. DOT-Domain enforces 100%

instruction and branch coverage, while DOT-CLI enforces 100% aggregate and per-executable-package statement coverage. The installer missing-`curl` negative test is hermetic and cannot fall through to host networking or `sudo`.

The complete 13-workflow matrix passed on the pushed hardening baseline `7c71e60393b1758d45e3558a15a2073450cb5cfc`, including coverage, Anchor conformance, and repository conformance. The tag workflow must also pass deterministic two-build packaging and checksum verification. Post-publication acceptance downloads and exercises the released binary rather than trusting only source-tree tests.

The release publishes the versioned domain JAR, DOT and Composition v1 schemas, CycloneDX JSON/XML SBOMs, release manifest, checksums, and `dotx` for Linux amd64. Linux amd64 is the only supported executable target. DOT-Registry REST API work, native arm64, non-Linux targets, and joint Security Function Composer/partner acceptance remain outside this release boundary.

## 0.2.0

`0.2.0` promotes the fully accepted `0.2.0-rc.8` candidate as the first stable release of the `dot.mare6g.eu/v1` composition profile. There are no behavioral changes from that candidate: the stable tag regenerates the versioned domain artifact, schemas, SBOMs, documentation, and Linux CLI assets from the same pinned DOT-CLI `4f415d43f08bb546d333ab61d6de272e92113903` and DOT-Repository `2fe0ecc81f9caea43c5bf6e0dc824c3e0faf34e3` revisions.

The release pipeline classifies `0.2.0` as stable and publishes it as the latest GitHub release. Before promotion, `0.2.0-rc.8` passed the full pushed commit matrix, reproducible release packaging, and post-publication Linux amd64 acceptance using the downloaded checksummed binary, including two repeated external-DOT lifecycle and cleanup cycles. Arm64 remains a cross-built preview until equivalent native evidence exists, and joint SFC acceptance remains a separate integration activity.

## 0.2.0-rc.8

`0.2.0-rc.8` supersedes `0.2.0-rc.7` with a current stable build and CI toolchain while retaining the accepted `dot.mare6g.eu/v1` composition, runtime, lifecycle, Contract, Skill-taxonomy, reusable-catalogue, and WASI profiles. Node.js 26 is now the explicit documentation and Contract-validation baseline; Go 1.27.2, Rust 1.98.1, Ruby 4.0, Java 27, and Maven 3.10.0 provide the remaining release toolchains. Direct Maven and Go dependencies and all third-party GitHub Actions were updated to compatible stable releases and immutable Action SHAs, with weekly Dependabot checks for Actions, Maven, Go, and the independently cloneable repositories. Documentation CI now rejects prerelease Ruby runtimes before Antora/PDF assembly.

Release publication now uses a tested, fail-closed SemVer channel classifier: prerelease versions such as `0.2.0-rc.8` are published as GitHub prereleases, while `0.2.0` is published as the latest stable release. The exact pinned revisions are DOT-CLI `4f415d43f08bb546d333ab61d6de272e92113903` and DOT-Repository `2fe0ecc81f9caea43c5bf6e0dc824c3e0faf34e3`.

# 0.2.0-rc.7

`0.2.0-rc.7` supersedes `0.2.0-rc.6` with two release-evidence hardening changes while retaining the same stable `dot.mare6g.eu/v1` profile and pinned DOT-CLI and DOT-Repository revisions. Source-adapted PoC harnesses now acquire private producer source through one guarded helper: it checks out the full pinned commit in a fresh temporary directory, verifies `HEAD` and the expected GitHub origin, retries bounded transient failures, and rejects partial, untrusted, missing-revision, or unsafe-destination results. Deterministic tests exercise those positive and negative boundaries without network access.

Contract conformance now installs the official Protocol Buffers 36.2 compiler for Linux amd64 or arm64 through the shared local bootstrap, verifies the download with an architecture-specific SHA-256, and uses that exact binary for both positive and deliberately invalid Protobuf fixtures. Local Taskfile runs and GitHub Actions no longer depend on different compiler setup paths. The exact pinned revisions remain DOT-CLI `b01f736393caa1a93d32e2c5593e032e854052ac` and DOT-Repository `d0b7fc1f3aac13a2b785482b07d5529f1d53404f`.

# 0.2.0-rc.6

`0.2.0-rc.6` supersedes `0.2.0-rc.5` with one shared, fail-closed bootstrap for the official WASI HTTP fixture. Local `task ci:live` and GitHub Actions now fetch and build the same exact Bytecode Alliance `sample-wasi-http-rust` revision before executing the same Wasmtime lifecycle assertions. The stable `dot.mare6g.eu/v1` composition profile, runtime adapters, lifecycle behavior, Contract model, Skill taxonomy, and reusable catalogue structure are otherwise unchanged. The exact pinned revisions remain DOT-CLI `b01f736393caa1a93d32e2c5593e032e854052ac` and DOT-Repository `d0b7fc1f3aac13a2b785482b07d5529f1d53404f`.

The bootstrap verifies the exact upstream URL and commit, rejects non-Git, wrong-origin, and modified tracked source caches, and rebuilds the `.wasm` component rather than trusting an ignored build artifact. The GitHub workflow no longer contains a separate checkout/build implementation; it invokes the same `ci:wasm-runtime` Taskfile gate available locally. Documentation records the Rust 1.98.1, `wasm32-wasip2`, and first-build network prerequisites and the optional `WASI_HTTP_COMPONENT` override for a deliberately supplied component.

# 0.2.0-rc.5

`0.2.0-rc.5` supersedes `0.2.0-rc.4` with one fail-closed representation for complete Contract payloads. The stable `dot.mare6g.eu/v1` composition profile, runtime adapters, lifecycle behavior, Skill taxonomy, and reusable catalogue structure are unchanged. The exact pinned revisions are DOT-CLI `b01f736393caa1a93d32e2c5593e032e854052ac` and DOT-Repository `d0b7fc1f3aac13a2b785482b07d5529f1d53404f`.

Complete Contract declarations now carry a structured `payload` object:

```
"payload": {
  "url": "https://vendor.example/contracts/service-openapi.yaml",
  "sha256": "0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef"
```

```
}
```

The URL must be absolute HTTPS without user information or a fragment, and the digest must be exactly 64 lowercase hexadecimal characters. Raw embedded Contract documents and the former flat `payloadUrl`/`payloadSha256` fields now fail schema and `dotx validate` checks. UUID-only Contract references remain a migration-compatible form for verified repository-local wrapper definitions.

`dotx validate` downloads the authoritative bytes, verifies their digest, and uses `MARE_READ_ONLY_TOKEN` or `GITHUB_TOKEN` only for `raw.githubusercontent.com`. Authorization is removed before following a cross-host redirect. Unit and repository tests cover malformed URLs and digests, unavailable assets, digest mismatch, authentication scoping, redirect credential leakage, inline payload rejection, flat-field rejection, and the official external Contract composition. The standalone DOT-Repository gate also runs the pinned validator directly for every catalogue revision.

GitHub Actions and local pre-push verification now share root `ci:*` Taskfile gates. `task ci:deterministic` covers the deterministic non-live release boundary; `task ci:live` covers the mutable Linux runtime workflows and requires the same official WASI HTTP component exercised by Actions, preventing a reduced local test path from being mistaken for workflow parity.

# 0.2.0-rc.4

`0.2.0-rc.4` retains the runtime and lifecycle profile accepted in `0.2.0-rc.3` and completes the closed SFC-requirement audit with repository-backed Skill discovery. `DOT-Repository/Skills/` now contains an authoritative asset for every capability UUID referenced by the catalogue and links each capability to one of the five controlled MARE categories.

`dotx validate DOT-Repository` resolves all 67 DOT definitions against that catalogue and rejects missing capabilities, category assets used as capabilities, duplicate references, and invalid category links. `dotx repository list --skill <uuid-or-name>` and `--category <uuid-or-name>` provide resolved, machine-readable discovery without copying category fields into every `dot.json`. The exact published revisions were DOT-CLI `3069f907bfd2056b16b92800bb5b98552c28832d` and DOT-Repository `3117a18da3004c57f4edf3714494fd3954225ae7`.

The audit also reconfirmed the existing DOT-side dispositions for stable Anchor identity and semantics, authoritative payload Contracts, service-scoped pre-deployment configuration validation, and single-file `dot.json` authoring. The separate DOT-Registry REST API remains outside this candidate, and joint SFC/partner acceptance remains a manual gate.

External products now have one canonical reusable DOT root rather than PoC-specific identities. Seventeen product roots plus two shared external test fixtures replace the former duplicated Keycloak, OPA, Prometheus, Elasticsearch, and APISIX definitions. PoC realms, policies, routes, scrape targets, data sources, secrets, and files remain composition-owned inputs; unavoidable source adaptations are named deployment variants under the same product UUID. Repository validation rejects any new deployable definition below `DOTs/external/pocs/`.

Composition lifecycle commands preserve the exact deployment ID rather than reducing it to a manifest type. The `--deployment` selector prefers an exact ID and accepts a manifest type or alias only when it identifies one deployment, so source-adapted and upstream deployments of the same

type cannot be selected silently.

Catalogue placement now records ownership rather than Contract maturity: third-party software is under `DOTs/external/`, MARE-owned/developed software is under `DOTs/internal/`, and only synthetic fixtures remain under `DOTs/testing/`. Every DOT bundle includes a local quickstart `README.md`. External/internal bundles opt into strict reusable validation by publishing a `contracts/` evidence directory whose assets identify their authority as upstream-official, runtime-extracted, DOT-ecosystem-derived, or adapter-owned, along with source revision, license, and SHA-256. Loki and Wireshark are now correctly classified as external runtime-only definitions without fabricated vendor Contracts.

## 0.2.0-rc.3

`0.2.0-rc.3` supersedes `0.2.0-rc.2` after post-publication, exact-pin lifecycle acceptance found that the Keycloak Helm workload could not reliably complete Quarkus augmentation within its declared 1536-MiB container limit. Three independent real-cluster runs retained pod and event diagnostics showing `OOMKilled` exit 137; the runner had 62 GiB of host memory and Minikube exposed 16.8 GB and 20 CPUs, isolating the defect to the DOT resource declaration.

Keycloak-DOT `0.1.1` declares a 2-GiB memory request and 4-GiB limit while retaining the same official Keycloak 26.3.3 Contract, Anchor selectors, chart, image, service, and Docker Compose mechanism. The external-DOT render suite now inspects the real upstream chart output and rejects any mismatch in the Keycloak container's CPU or memory requests and limits before live deployment. The official external-Contract composition and content-addressed lock resolve the versioned `0.1.1` definition.

External DOT examples run 38708940363 passed both the non-mutating chart conformance job and two complete Linux lifecycle cycles. Each cycle passed all 47 checks with zero failures and cleanup verification across Keycloak, Loki, Prometheus, and Wireshark's eight deployment-mechanism cells. The downloaded, checksum-verified evidence pins DOT-CLI `f8cf796afd985cc0514d42fb149ef67b6a1b7b53` and DOT-Repository `76585f6ec68a892f6cc6d48c9b2acdce6b08d48d`.

This candidate also adds a post-publication workflow that downloads the actual checksummed Linux amd64 `dotx` release asset, verifies its release manifest and source revisions, then runs repository validation, Anchor acceptance, Composition v1 acceptance, and two complete external-DOT lifecycle cycles. Maven dependency submission is now an explicit self-hosted workflow; the billing-blocked GitHub-managed automatic submission feature is disabled. Linux amd64 is the release-accepted executable target. The cross-built arm64 asset remains preview-only until equivalent native arm64 evidence exists. Joint SFC/partner acceptance remains a separate manual gate and is not claimed by this candidate.

## 0.2.0-rc.2

`0.2.0-rc.2` supersedes the first candidate with the fixes and acceptance evidence completed after `0.2.0-rc.1` was tagged. The stable serialized composition API remains `dot.mare6g.eu/v1`.

## Changes since rc.1

- Exact service-to-workload selection and configuration isolation for multi-service Docker Compose, Kubernetes-manifest, and Helm deployments.
- Exact official, digest-pinned OpenAPI Contracts and Contract-backed Anchors for reusable Keycloak and Prometheus external DOTs, with every Helm and Docker Compose deployment binding the same interface.
- `dotx validate --profile reusable` now enforces declared HTTPS Contract provenance, payload integrity, selector resolution, Contract use, and per- deployment Anchor binding. Review establishes that the pinned source is the official upstream artifact. `dotx validate <repository-root>` recursively discovers every `dot.json`; the later rc.4 ownership migration derives strict reusable validation from adjacent Contract evidence rather than category path, and is the single DOT-Repository GitHub Actions gate; no duplicate inventory or shell wrapper is required.
- Same-DOT alternative deployment mechanisms for the Keycloak, Loki, Prometheus, and Wireshark external-product examples. Loki and Wireshark were initially retained as runtime-tested samples; rc.4 reclassifies them as external runtime-only DOTs without claiming Contract-backed interoperability.
- A locked Keycloak/Prometheus/consumer composition proves that both stable external Anchors resolve and materialize the exact official operations.
- Failed Kubernetes/Helm workloads now retain pod, event, workload, and bounded log diagnostics before cleanup; causally blocked endpoint assertions are skipped rather than counted as independent failures.
- Repeated hosted-Linux acceptance run 38600327954 covering all eight external-DOT mechanism cells: two complete lifecycle cycles, 46 checks per cycle, zero failures, verified cleanup, and four successful official- Contract operation checks per cycle. The checksummed evidence pins DOT-CLI `848b2d56e9e416ae87fa46d221c9649041b64541` and DOT-Repository `b681c937422fa3db13ad16587cf271700eff5d79`.
- Portable evidence checksums and retained GitHub Actions artifacts.
- Separate workflow concurrency lanes for push conformance and manually requested live stress, preventing one evidence class from canceling the other.

Linux amd64 is the release-accepted executable target. The published arm64 executable is a cross-built preview and is not a supported final-release target until equivalent native arm64 evidence exists. Producer/partner acceptance, non-Linux executables, multi-host orchestration, general stateful migration, and automatic provisioning of cross-platform gateways remain outside this candidate. GraphQL, gRPC, and event-driven runtime probes beyond the documented safe profile remain standards-led roadmap work rather than an `rc.2` blocker.

The candidate assets and verification procedure are the same classes described for `rc.1` below, with filenames versioned as `0.2.0-rc.2`. The tag workflow rebuilds the Java artifact, both schemas, SBOMs, and Linux `dotx` binaries from the immutable revisions recorded in `release-manifest.json`, then verifies reproducibility and `SHA256SUMS` before publication.

# 0.2.0-rc.1

`0.2.0-rc.1` is the first integrated release candidate for the stable `dot.mare6g.eu/v1` composition profile. It replaces the narrower 0.1.0 artifact set with a traceable domain model, both machine-readable schemas, and version-stamped `dotx` binaries built from the exact DOT-CLI revision pinned by this repository.

## Highlights

- Stable, repository-backed Composition v1 lifecycle, probe, and upgrade commands.
- Contract-driven Anchors for OpenAPI 3.0/3.1/3.2, AsyncAPI 3.0/3.1, Protobuf/gRPC, GraphQL, and declared contractless interactions.
- Immutable repository resolution, content-addressed locks, resumable lifecycle operations, and last-known-good upgrade rollback.
- Managed Docker image, Docker Compose, Kubernetes manifest, Helm chart, Vagrant/libvirt, and Wasmtime/WASI profiles.
- Explicit cross-platform networking through externally owned gateway bindings with pinned host, port, ownership, and no-cleanup semantics.
- Service-scoped configuration, secrets, OCI requirements, CDI devices, and exact native-workload bindings.

## Assets and verification

The candidate includes the DOT-Domain JAR, DOT and Composition v1 schemas, CycloneDX SBOMs, and `dotx` binaries for Linux amd64 and arm64. `release-manifest.json` records the immutable DOT-Design, DOT-CLI, and DOT-Repository revisions. The repository revision pins its `DOTs/`, `Skills/`, and `Schema/` assets together. Verify the download set with:

```
sha256sum --check SHA256SUMS
```

The tag workflow publishes only after the Java build, generated-schema drift check, DOT-CLI tests, repository test, Composition v1 acceptance suite, deterministic packaging comparison, manifest validation, and checksum verification pass.

## Compatibility and boundaries

The serialized composition API is `dot.mare6g.eu/v1`; breaking serialized changes require a new API version. The experimental local catalogue/registry prototype from the 0.1.0 era is not included: DOT-Repository is authoritative.

This candidate does not claim multi-host orchestration, automatic provisioning of cross-platform gateways/tunnels/brokers, general stateful data migration, arbitrary privileged workloads, non-Linux `dotx` executables, or producer/partner acceptance. The current CLI includes Linux process controls for its managed WASI runtime, so this candidate publishes Linux amd64 and arm64 binaries only. Closed legacy SFC issues record DOT-side disposition; joint consumer acceptance

remains a separate manual gate.

See the repository-root `RELEASE_NOTES.md` included with the source for the complete candidate notes and upgrade guidance.

# DOT-Domain

## Domain model

`DOT-Domain` is the persistence-neutral Java library that defines the portable DOT aggregate and its validation rules. It contains no application, web, or storage-adapter layer.

### Aggregate overview

Read the diagram from `Dot` down to a deployment's logical services. It shows the main relationships without the configuration and validation details below.

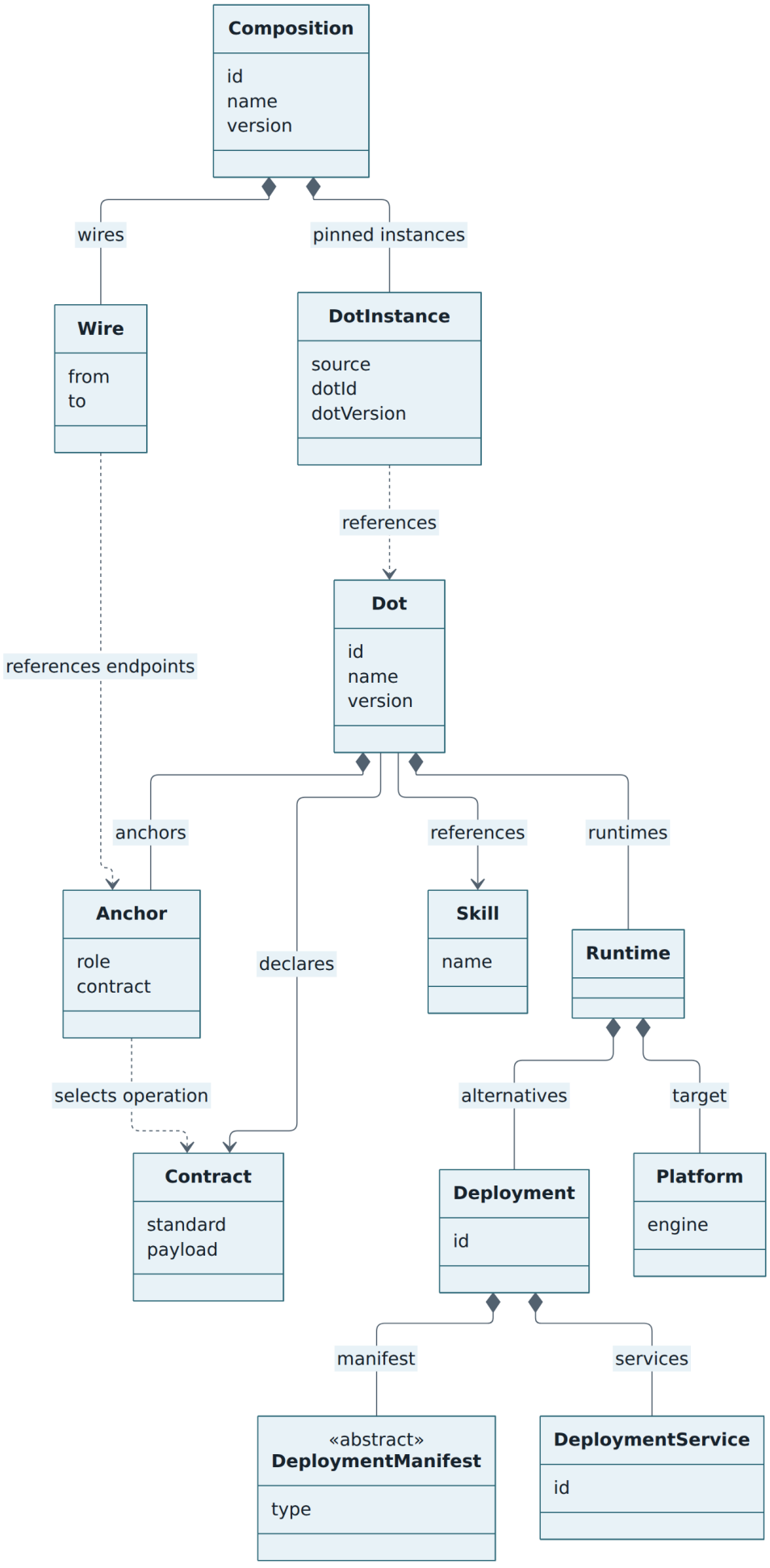



*Figure 9. Simplified domain model: reusable DOT definitions and a separate Composition*

Diamonds indicate embedded values; arrows indicate declarations or references. The dashed arrow applies to Contract-backed Anchors. Collection sizes, configuration, ports, binding helpers and audit metadata are omitted for readability; see the full generated class diagram for implementation detail. `Composition` is a separate DOT-Domain aggregate: its instances reference DOTs and its Wires reference their Anchors. Neither is embedded in the reusable Java `Dot` aggregate.

| Type | Responsibility |
|---|---|
| `Dot` | Aggregate root: identity, version, description, classification groups, metadata, Skills, Contracts, Anchors, and Runtimes. |
| `Skill` | Standalone human-readable capability description. |
| `Composition` | Separate aggregate: pinned DOT instances, Anchor Wires, exposed Anchors, deployment inputs, Secret references and active probe expectations. |
| `Contract` | Standalone authoritative OpenAPI, AsyncAPI, Protobuf, or GraphQL document. |
| `Anchor` | Embedded stable logical point through which the DOT participates in composition. |
| `Runtime` | A target `Platform` plus one or more compatible deployment definitions. |
| `Deployment` | Stable deployment ID, one polymorphic manifest, and the logical services it materializes. |
| `DeploymentService` | Service identity, host, named ports, Contract references, Anchor bindings, and its supported configuration surface. |

`Dot`, `Skill`, `Contract`, and `Composition` are independently identified domain entities. Other types are embedded values owned by these aggregates. The module carries no database or ORM annotations; persistence is an adapter concern.

## One model authority

DOT-Domain defines **all** portable models, including Composition, not just `Dot`. The existing Victools build generates `Schema/schema.json` for DOTs, `Schema/composition-v1.schema.json` for Composition v1, and standalone `Schema/models/*.schema.json` files for every public domain data class and enum. Jackson reference views remain intentional: a Skill reference containing only `id` is not the full standalone Skill entity.

The CLI consumes generated Go input types and embeds the generated Composition schema. It does not require Java at runtime. From DOT-Design, regenerate with:

```
mvn -pl DOT-Domain process-test-classes
task cli:domain:generate
```

Never edit generated files manually. `task testing:composition:model` checks generation drift and v1 compatibility after the Contract validator bootstrap.

`Composition` references reusable DOTs; it neither inherits from `Dot` nor copies its runtimes or Contracts. Resolved execution plans, lock digests, private lifecycle state, endpoint allocation and runtime policy enforcement remain CLI/executor concerns. Standard Jakarta constraints and Jackson shape annotations describe fields; model-owned JSON Schema annotations express conditional input rules such as exactly one literal value or Secret reference. Cross-DOT resolution, secret authorization and ownership checks still run in the CLI after schema validation.

This migration preserves `dot.mare6g.eu/v1`, existing documents and reviewed locks. Composition `version` and `dotVersion` retain the existing v1 string semantics; they have not silently become stricter Java `Version` fields.

## Identity, versioning, and references

`Dot`, `Anchor`, and `Composition` identities remain UUIDs. Skills and Contracts use stable, case-sensitive textual IDs such as `Metrics-Scraping` and `Prometheus-3.15.0-OpenAPI`. IDs start with an ASCII letter or digit and contain only letters, digits, dots, underscores and hyphens. Existing UUID-shaped IDs remain valid unchanged. IDs are not display names, URLs, or automatically derived from mutable fields; renaming one requires updating every reference. Equality for `Dot`, `Skill`, `Contract`, and `Composition` is ID-based so changes to mutable metadata or collections do not destabilize sets or persistence identity.

`Version` accepts Semantic Versioning 2.0.0 values and the special `latest` selector. It serializes as a string and implements SemVer precedence; `latest` sorts after concrete versions. A new `Dot` or `Contract` defaults to `0.0.0`, while `Platform.version` defaults to `latest` so no platform-specific release is implied.

JSON serializes `Dot.skills` and `DeploymentService.contracts` as ID-only references. Root `Dot.contracts` carries authoritative Contract declarations: identity, standard, format, and a structured `payload` with an HTTPS `url` and `sha256` pin. Raw API documents are not embedded in `dot.json`. Temporary root ID-only entries must resolve to verified assets in that DOT's own `contracts/` directory; repository-level and sibling-DOT fallback are not allowed. An Anchor references the root declaration by Contract ID and selector.

`dotx validate` enforces catalogue identity uniqueness, including DOT `(id, version)` and `(name, version)`, and rejects conflicting documents for one Contract ID across bundles. These catalogue rules are not database indexes or persistence requirements enforced by the Java entities.

## Contracts

Contract `standard` accepts only the canonical JSON spellings `OpenAPI`, `AsyncAPI`, `Protobuf`, and `GraphQL`. Uppercase enum names, alternative casing, and surrounding whitespace are rejected. The Java enum constants remain uppercase; Jackson serialization and the generated JSON Schemas use their canonical values. `DataFormat` is a separate enum and is unchanged.

| Standard | Payload format | Anchor selector |
|---|---|---|
| `OpenAPI` | `YAML` or `JSON` | RFC 6901 URI-fragment JSON Pointer to an Operation Object |

| Standard | Payload format | Anchor selector |
|---|---|---|
| AsyncAPI | YAML or JSON | RFC 6901 URI-fragment JSON Pointer to an Operation Object |
| Protobuf | PROTOBUF | Fully qualified service method |
| GraphQL | GRAPHQL | Root-field schema coordinate |

`Contract.isFormatCompatibleWithStandard()` rejects mismatched standard/format pairs. Payload parsing and selected-element existence are resolver concerns; the domain model validates selector syntax and references within the aggregate.

## Anchors and bindings

A contract-backed Anchor contains:

- a stable UUID and display name;
- a `PROVIDES` or `REQUIRES` role;
- optional semantic-type URIs; and
- a Contract ID plus standard-specific selector.

It omits direction, protocol, and media type because those facts come from the selected Contract operation. A contractless Anchor must declare `direction`, at least one protocol, and at least one media type explicitly.

`DeploymentService.anchorBindings` maps an Anchor UUID to a named `ServicePort` on that same service. An Anchor contains no host, numeric port, credential, placement, or live policy state.

The class-level `@ValidAnchor` constraint delegates cross-reference validation to `AnchorValidator`. It rejects:

- duplicate Anchor UUIDs within one DOT;
- Contract references not declared by the same DOT;
- selectors invalid for the referenced Contract standard;
- bindings to undeclared Anchors;
- bindings to port names absent from that service; and
- duplicate bindings of one Anchor within the same service.

See DOT Anchors and Composition for authoring and wiring examples.

## Runtime two-layer model

`Runtime` separates infrastructure from deployment mechanism:

| Enum | Used by | Answers |
| --- | --- | --- |
| `PlatformEngine` | `Platform.engine` | Where the workload runs: `DOCKER`, `KUBERNETES`, `VAGRANT`, or `WEBASSEMBLY`. |
| `DeploymentManifestType` | `DeploymentManifest.type` | How it is deployed: Docker image, Docker Compose, Kubernetes manifest, Helm chart, Vagrantfile, or WASM module. |

Every `DeploymentManifestType` declares its required `PlatformEngine`. `Runtime.isDeploymentPlatformAligned()` validates every deployment against the runtime platform. `Runtime.isDeploymentIdsUnique()` enforces stable deployment selection within that runtime.

`Deployment.services` is a set. `DeploymentService` uses ID-based equality, so logical service IDs remain unique within the deployment.

## Deployment manifests

`DeploymentManifest` is polymorphic JSON dispatched by its existing `type` property.

| Type | Java subtype | Important fields |
| --- | --- | --- |
| `DOCKER_IMAGE` | `DockerImage` | Image reference and ordered dotenv payloads |
| `DOCKER_COMPOSE` | `DockerCompose` | Complete YAML/JSON payload and ordered dotenv payloads |
| `KUBERNETES_MANIFEST` | `KubernetesManifest` | Complete YAML/JSON manifest payload |
| `HELM_CHART` | `HelmChart` | Repository, chart or OCI reference, version, and ordered values payloads |
| `VAGRANTFILE` | `Vagrantfile` | Complete Ruby DSL payload |
| `WASM_MODULE` | `WebAssemblyModule` | Relative/file or HTTPS module URI, WASI execution world, command arguments, and required SHA-256 integrity pin |

Docker Compose and Kubernetes manifest payloads accept YAML or JSON only. Helm values also preserve YAML/JSON order: later entries override earlier entries as with repeated Helm `-f` arguments.

Runtime instance concerns do not belong in a manifest. For example, a Helm release name and namespace are selected by deployment tooling, not embedded in `HelmChart`.

## Service configuration declarations

Every `DeploymentService` owns one `DeploymentConfig` containing:

- environment variables, including default, required, and sensitive metadata;
- configuration-file paths, content type, defaults, and sensitivity;
- logical volume mounts;

- logical CNCF CDI device requirements; and
- optional OCI capability and host-namespace requirements.

These are supported inputs and requirements, not applied runtime state. Operator values, host paths, concrete CDI device names, and permission grants belong to deployment or composition input.

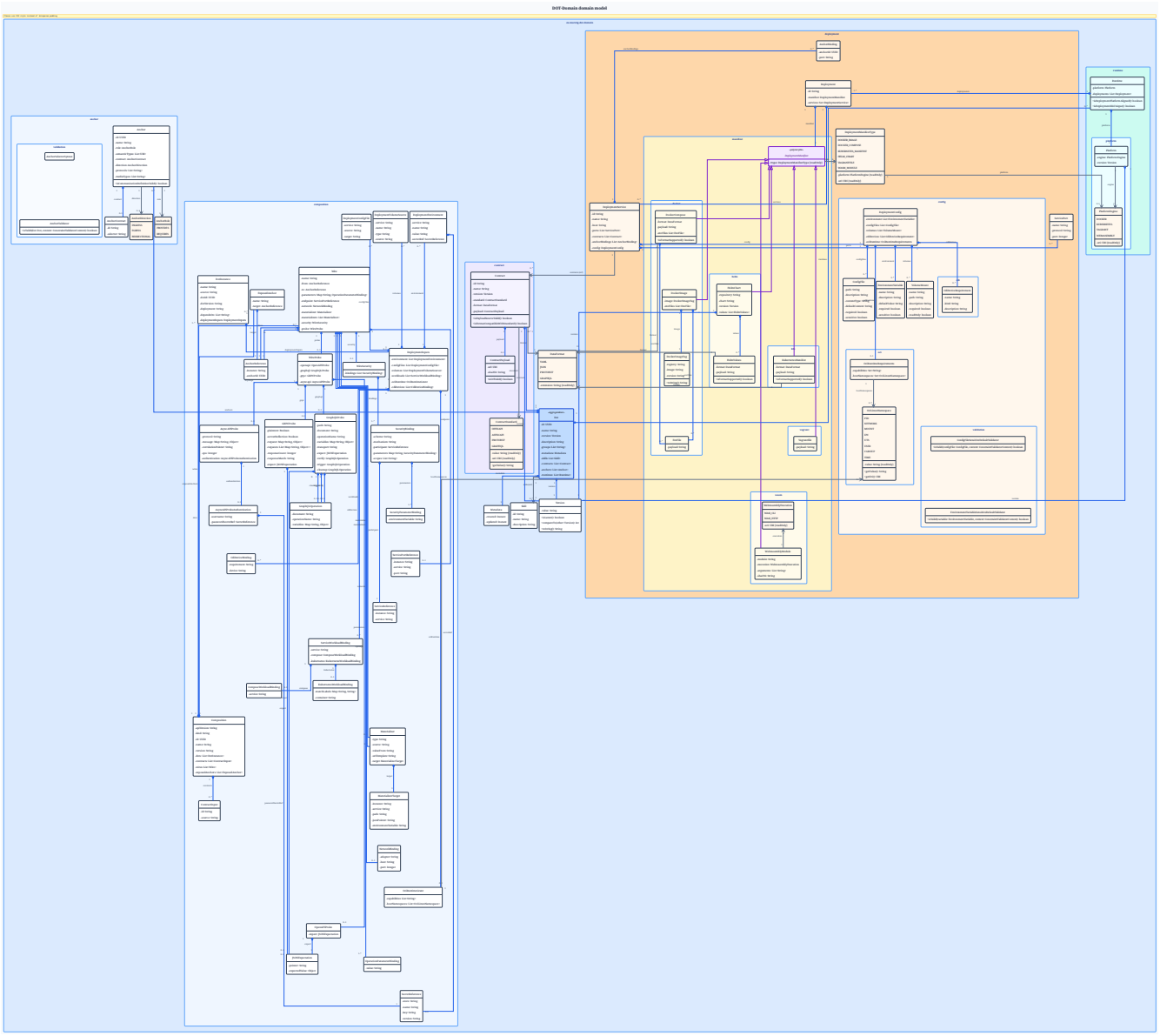
OCI requirements use standard `CAP_*` capability names and typed Linux namespaces. The model deliberately has no unrestricted `privileged` flag.

A CDI requirement declares a stable logical name and a `vendor.example/class` kind. A composition binds it to one exact `vendor.example/class=device` identifier. CDI installation, allocation, and authorization remain operator responsibilities.

## Serialization and persistence adapters

- Collections have initialized builder defaults and do not become `null` when constructed through Lombok builders.
- `Version` is JSON-serialized as a normalized string.
- Persistence adapters must preserve UUID, textual identifier and version representations.
- Skill and service-level Contract references expose only `id`; root `Dot.contracts` contains authoritative declarations or verified local transition references, as described above.
- Domain values carry no database or ORM annotations.

## Class diagram



The diagram is generated from current domain sources during the DOT-Domain build. Open it separately for the full-resolution view.

## Build and test

```
# Unit tests only; no service or database required
mvn -pl DOT-Domain test

# Regenerate schema and diagram through the reactor lifecycle
mvn -pl DOT-Domain clean install
```

Representative tests cover version precedence, reference serialization, manifest polymorphism, runtime/manifest alignment, deployment configuration, Anchor rules, and entity equality.

## Related pages

- Architecture
- DOT Anchors and Composition

- Integration handoff

# DOT-CLI

## CLI overview

`dotx` is the Go command-line application provided by the independent DOT-CLI repository. It operates on DOT definitions and compositions; it is not a catalogue server or a replacement for Docker, Kubernetes, Helm, Vagrant or Wasmtime.

### Start with a working DOT

Follow Getting started for checksummed Linux-amd64 installation and the Prometheus Docker Compose quickstart: validate, deploy, verify an API response, inspect status and clean up. Running the released binary does not require a Java or Go development toolchain.

Use the guided demo to see a topology and a multi-DOT composition. The MARE PoC walkthroughs include a presenter pause for inspecting verified live results before cleanup.

### Find the right command

| Task | Command family |
|---|---|
| Check a definition, bundle or complete catalogue | `dotx validate` |
| Read a DOT's declared interfaces and deployment choices | `dotx inspect` |
| Discover and inspect catalogue entries | `dotx repository` |
| Operate one selected DOT deployment | `dotx deploy`, `status`, `start`, `stop`, `destroy`, `ps` |
| Validate, lock, plan and display a multi-DOT graph | `dotx compose validate`, `lock`, `plan`, `topology` |
| Operate and verify a composition | `dotx compose apply`, `status`, `probe`, `destroy` |
| Relate a locked graph to observed runtimes and historical probes | `dotx compose topology --status` (use the applied run's cache/configuration) |
| Check deployment tools and runtime prerequisites | `dotx tool` |

Ask the installed binary for its exact options:

```
dotx version
dotx help
```

```
dotx compose --help
```

For reusable catalogue examples and explicit deployment selection, see Deployments and examples. For Anchor selectors, Wires, native Secret references, locks and lifecycle safety, use the composition guide.

Validation and planning are not live acceptance. Active probes run only for supported, explicitly configured Contract operations and transports. The walkthroughs state which endpoints are actually exercised and distinguish real producers, source-adapted integrations and synthetic fixtures.

### Versions and model authority

The CLI release version is independent of DOT-Design. Release installation instructions use the published binary; development demos may require the exact source revision pinned by the DOT-Design checkout, even when its version label matches a release. See Development and testing for source installation, Bash completion, local CI-equivalent Tasks and prerequisites.

All portable models belong to DOT-Domain. The CLI consumes its generated JSON Schemas and Go bindings; it does not define a competing DOT or Composition input model. Resolved execution plans, lock fingerprints and private runtime state are executor concerns. DOT definitions and reusable composition inputs live in DOT-Repository.

## Installation and first DOT

## Deployments and examples

## Compositions and lifecycle

## Topology and guided demo

## Development and testing